FACULTY OF COMPUTING AND INFORMATION

TECHNOLOGY Programme: (Group: 14)

Assignment

BMSE2003 SOFTWARE TESTING

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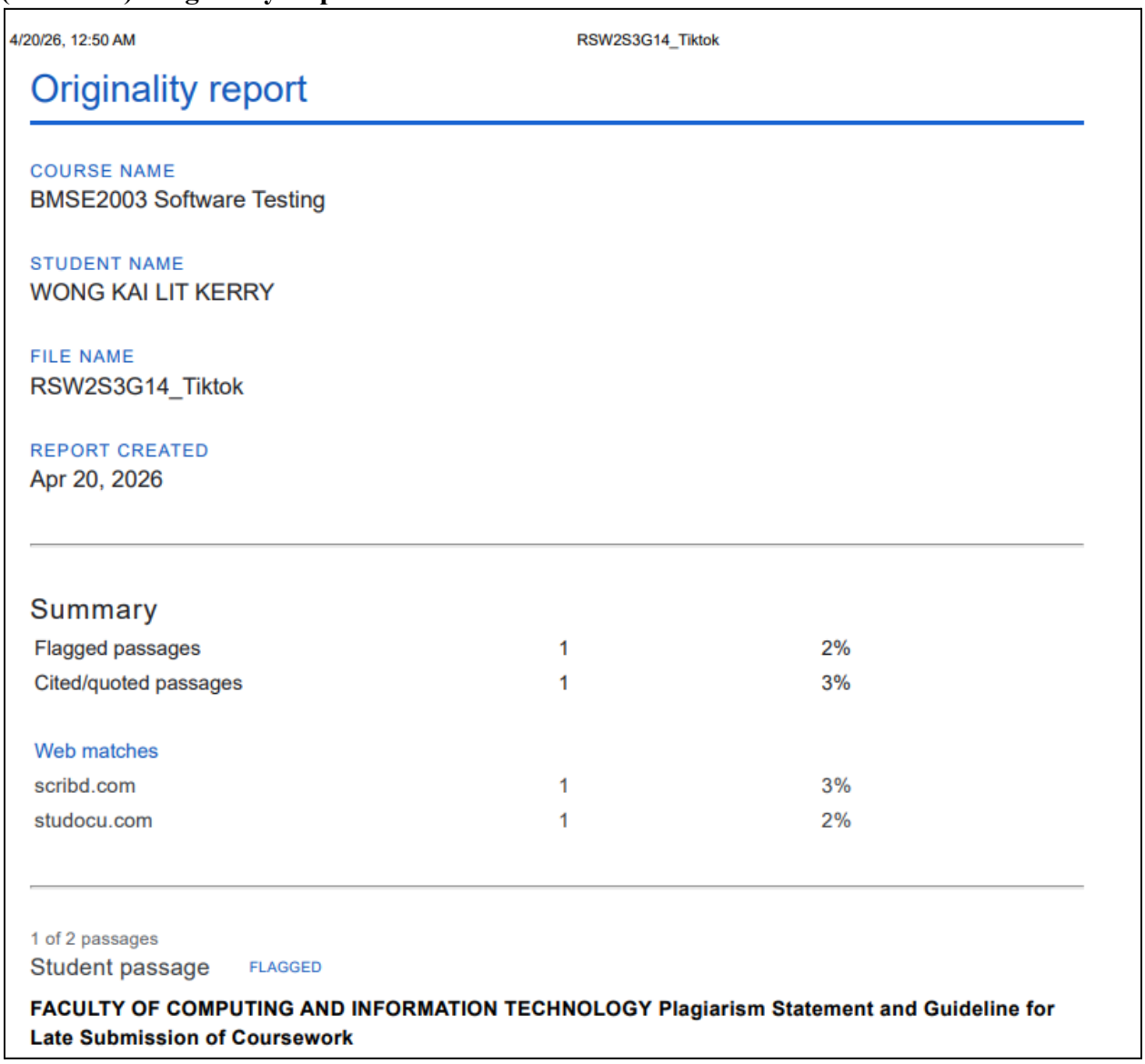
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(Test Plan) Originality Report



4/20/26, 12:50 AM RSW2S3G14_Tiktok

Originality report

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BMSE2003 Software Testing

STUDENT NAME
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REPORT CREATED
Apr 20, 2026

Summary

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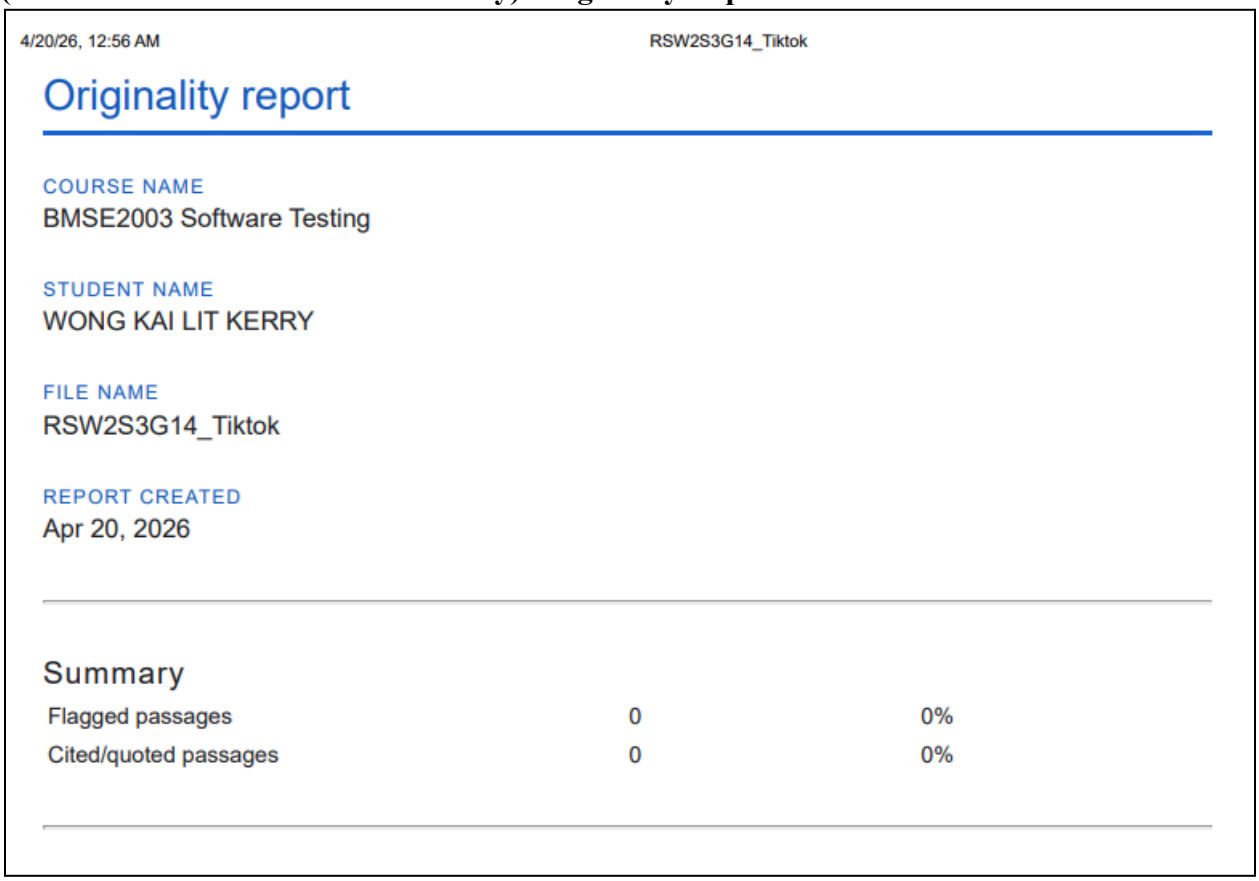
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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY Plagiarism Statement and Guideline for Late Submission of Coursework

(Test Case - Test Execution Summary) Originality Report



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BMSE2003 Software Testing - Assignment Rubrics

1 Tutorial Group: G14

Student Names: 1. Ryan Julian Rajesh Modules: User Profile Management
2. Natalie Koa Hao Yee Modules: Video Comment
3. Lester Liau Jiun Lang Modules: TikTok Shop Seller Center
4. Wong Kai Lit Kerry Modules: Messaging
5. Tiong Chang Vui Modules: Video Creation

CLO2: Adopt software test documentation standard in the development of test plan, test cases, and test report. (A3, PLO4)
CLO3: Utilise testing tools in executing a software testing process for the selected test object. (C3, PLO6)

Criteria / Area	Excellent [9-10]	Good [7-8]	Average [5-6]	Fair [3-4]	Poor [2-0]	Score				
						1	2	3	4	5
Test Plan (Clarity and Completeness)	Test plan is detailed, clear, and covers all sections (test item, scope, risks, staffing, schedule etc). All objectives align perfectly.	Test plan is mostly clear and includes major sections. A few details might be missing or partially aligned.	Test plan is average, missing key sections or details, with average alignment to the objectives.	Test plan is basic, missing key sections or details, with minimal alignment to the objectives.	Test plan is incomplete, vague, or fails to cover major sections like scope or objectives.					
Application of Testing Techniques	Choosen testing techniques are applied accurately and effectively align with the assignment.	Choosen testing techniques are applied well but with minor errors or omissions in execution.	Choosen testing techniques are applied average with minor errors or omissions in execution.	Choosen testing techniques are applied but lack depth, accuracy, or proper alignment with the assignment.	Choosen testing techniques are incorrect, incomplete, or not applied at all.					
Test Case Design Using Tools	Test cases are comprehensive, created using appropriate tools with detailed steps, test data, and expected results.	Test cases are detailed and created using tools, with minor gaps in clarity or coverage of scenarios.	Test cases are average and created using tools, with gaps in clarity or coverage of scenarios.	Test cases are incomplete, lack variety, or are unclear. Minimal use of testing tools for creation.	Test cases are poorly written, irrelevant, or not created using appropriate tools.					
Test Execution Using Testing Tools	Test execution is conducted using relevant tools. Execution logs, screenshots, or results are provided. All test cases are executed accurately.	Test execution is conducted using tools, but some cases are incomplete or lack evidence.	Test execution is averagely conducted using tools and most cases are incomplete or lack evidence.	Test execution is partially conducted or lacks proper use of testing tools. Results are inconsistent or incomplete.	Test execution is not conducted, or no use of testing tools is evident.					
Test Execution Summary Report	Report is highly detailed, professional, and includes comprehensive results, defect summary, and observations.	Report is clear and professional but missing minor details like observations or defect patterns.	Report is average and missing minor details like observations or defect patterns.	Report is basic, lacks important details, or is unclear in presentation.	Report is incomplete, vague, or does not summarize the test execution effectively.					
Test artifacts based on ISO/IEC/IEEE 29119-3 standard	Excellent-developed test artifacts (test plan, test case specification, test execution summary report) on its objectives, correct and complete information provided based on the given standard without any error.	Good-developed test artifacts (test plan, test case specification, test execution summary report) on its objectives, correct and complete information provided based on the given standard with minor errors.	Shows average attempt to develop test artifacts (test plan, test case specification, test execution summary report) on its objectives, correct and complete information provided based on the given standard with severe errors.	Shows fair attempt to develop test artifacts (test plan, test case specification, test execution summary report) on its objectives, correct and complete information provided based on the given standard with major errors.	Inadequate attempt to develop test artifacts (test plan, test case specification, test execution summary report) on its objectives, correct and complete information provided based on the given standard with incorrect/incomplete information or errors.					
Team Collaboration and Communication	Demonstrates excellent team collaboration, communication, and empathy throughout the testing process.	Demonstrates good interpersonal skills but with minor gaps in teamwork or communication.	Demonstrates average interpersonal skills with gaps in teamwork or communication.	Shows minimal interpersonal skills with limited evidence of collaboration or effective communication.	Fails to demonstrate interpersonal skills, with no evidence of teamwork or effectve collaboration					
Presentation and Documentation	Work is highly professional, well organized, and free of errors. Documentation is clear and visually appealing. Consists of more than 10 related references.	Work is professional and organized, with minor formatting or clarity issues. Consists of 8 to 10 related references.	Work is average organized with minor formatting or clarity issues. Consists of 5 to 7 related references.	Work is somewhat professional but has noticeable errors, poor formatting, or unclear documentation. Consists of below 5 related references.	Work is unprofessional, disorganized, and contains numerous errors or lacks clarity. No related reference.					
Accessing Digital Skills in Testing	Demonstrates excellent use of testing tools, platforms, and technologies. Skills in automation, reporting, or collaboration are evident and effective.	Demonstrates good use of testing tools and technologies, with minor gaps in proficiency or integration	Demonstrates average use of testing tools and technologies, with major gaps in proficiency or integration	Demonstrates minimal use of tools or technologies, with limited effectiveness or integration.	Fails to demonstrate proficiency in using testing tools or integrating digital skills into the process.					
Collaboration and Communication Using Digital Platforms	Demonstrates excellent use of digital communication and collaboration platforms for team communication, sharing testing artifacts in real time, and conducting virtual presentations.	Demonstrates good use of digital communication and collaboration platforms for team communication, sharing testing artifacts in real time, and conducting virtual presentations with minor gaps.	Demonstrates average use of digital communication and collaboration platforms for team communication, sharing testing artifacts in real time, and conducting virtual presentations with major gaps.	Shows minimal use of digital communication and collaboration platforms.	Fails to demonstrate proficiency in using digital communication and collaboration platforms.					

TOTAL (100 marks)						
Comments:						

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1.Test Plan

1.1 Overview

The purpose of this Test Plan is to define the scope, approach, resources, and schedule for the software testing activities of the Tiktok web portal. This document outlines the strategy for verifying the application's functionality, focusing on identifying defects and ensuring the system meets user requirements. This testing project is conducted as a Black-Box Testing initiative, meaning the test team will validate the external behavior of the application without access to the internal source code or development database consequently, testing will be performed in the Web Browsers. The primary objectives of this test plan are to validate the functional correctness of five key modules which are User Profile Management, Video Comment, TikTok Shop, Video Creation, and Messaging. While executing manual test cases to ensure stability and reliability. Furthermore, all testing activities, defects, and results will be documented in accordance with the ISO/IEC/IEEE 29119-3 standard for software test documentation.

1.2 References

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1.3 Context of Testing

1.3.1 Project(s)

The project object that we have chosen is TikTok, it is a short-form video social networking platform developed by the Chinese company ByteDance and its global launch in 2017. ByteDance was founded by Zhang Yiming and later acquired by musical.ly in 2018. Nowadays, TikTok has grown into one of the biggest social media applications platforms that has over one billion active users.

The TikTok platform's first core function is to let the users create, share, and watch short videos created by content creators. Normally, the video duration will be around 15 seconds to ten minutes only for some special cases can be up to 1 hour. The second core functionality is using a very complex content discovery engine which is the "For You" page that utilizes the machine learning algorithms to predict users favorite content then give the personalized feed videos based on individual user behavior, interactions, and preferences.

Furthermore, TikTok also has comprehensive creative tools, to let users record multi-segment videos, use filters and visual effects during editing the video, and add music from an extensive licensed library. The application also integrates social media platform elements such as following other friends or creators, liking video and commenting on the video content, private messaging, live streaming capabilities, and finally TikTok also integrated e-commerce functionalities that able users to buy items or products directly from the list that are created by TikTok sellers through the app.

Tiktok became very successful because these features are enabled by the organizational structure. Which includes many sections such as the Product Development Team, the Global Operations Team, and the Trust and Safety Division. The current CEO of TikTok is Shou Zi Chew, this person is a person who reports to the senior leadership team at ByteDance. Furthermore, the organizational structure is relevant to the testing of the web application because the decisions made by these sections, such as the changes to the algorithm and the security updates, are what drive the web application for the end users.

1.3.2 Test Item(s)

Module	Test Item 1	Test Item 2
1. User Profile Management	Login Account Verifying authentication using registered credentials and handling incorrect inputs.	Edit Profile Verifying input validation for username and bio fields when updating profile information.
2. Video Comment	Post Comment Testing the ability to post text comments and verifying rate-limiting constraints.	Reply to Comment Testing nested comment creation and enforcing character length boundaries.
3. TikTok Shop Seller Center	Create Product Listing Verifying sellers can create a new product listing with valid details.	Create Promo Code Verifying sellers can create promo codes with conditional logic and minimum spend requirements.
4. Video Creation	Upload Video Verifying input validation by testing supported and unsupported file formats.	Schedule Video Testing the ability to schedule videos for future posting, including editing scheduled posts and preventing past date selection.
5. Messaging	Send Message Verifying the delivery of direct text messages and handling of empty/null inputs and messages containing only spaces.	Manage Message Requests Testing the ability to accept/decline message requests from non-followers and verifying message limits before acceptance.

1.3.3 Test Scope

For the testing TikTok web application, since we don't have any access to the internal source code or database structure we decided to use the black block testing approach for this project only. After that, we will use the user interface to observe inputs and outputs for all test validation. So this test plan will focus on three testing types only which is Functional Testing, User Interface Testing and Error Testing.

Functional Testing

This testing is using the input data and examining the output to verify the functions performed the task correctly and Functional testing is testing that can ensure the functionality of the object is really implemented in the system. For this project this testing will validate the User Profile Management Module to test whether users can successfully log in their account and manage their account or not. For the Video Comment Module will test the post and reply comment function. For Tiktok Shop Seller Center Module will test the create product listing and promo code function in the system. The Video Creation Module will test the upload and schedule video function in the system. Lastly, the Message Module will test the send message and manage message request function.

User Interface Functional Testing

This testing is focused on the UI elements testing of the TikTok web application by ensuring that all buttons are clickable, inputting the values into the input fields are either accepted or rejected, all navigation links are redirected to the corresponding page, and clickable and interactive elements respond appropriately whenever clicked on. Testing will be done on the functionality of various interface elements of the TikTok web application in all five modules, such as input fields allow text input, clicking buttons activates the desired actions, the comment block is loaded, and when the message function is selected, the desired conversation is also loaded. In the User Profile Management Module, the login button should be a form submission, and the profile edit button should call the change user name function. For the Video Comment Module, the comment field should only contain texts, the post button should be active only after posting a text, while the reply should lead to the opening of the right comment thread. Considering this, in the TikTok Shop Seller Center, a value will be added to ensure that all the forms related to the product

creation correctly submit data, all the dropdown menus contain all options, and the ‘Save’ and ‘Cancel’ buttons work as expected. In “Video creation module”, the upload button should open a file browser, the progress bar should be updated in real-time when a file is being uploaded, and the date-picker should correctly pick and confirm dates. For the Messaging Module, we will be testing that sending text is only enabled when there is a text in the text area and clicking on a message request accepts or declines the message request. Lastly, all UI elements that fail to function as expected will be identified, including non-actionable buttons, broken links, and input fields that fail to accept valid inputs.

Error Handling Testing

This testing type is to verify that the system is able to handle wrong input with correct approach and check the system when an error occurs will the error will easily bring it to crash or not. Which means this Error Testing contains both negative testing and security testing. For this project, error testing will verify and validate the system such as will the login fails with incorrect input, will the system detect when duplicate name happen and reject the users with a good output message, will the empty or over long comment is able to post or not, will the product listings cant create when missing compulsory input fields data or wrong prices format, will the unsupported file format will be rejected by system or not, will the past date is can’t selected for scheduling, will the empty message sendable or not, and will the system limit the user only can send one message to non followers.

Ultimately, these three testing types can ensure the Tiktok web application functions correctly, provide a great user experience to the users, and ensure the system is stable and reliable when users input wrong data or unexpected user behavior.

1.4 Risk Register

The following table identifies potential risks associated with the testing of the TikTok web application. It outlines the project risk and product risk of the Tiktok web application.

Project Risk

Time Constraints

Explanation:

Having a fixed deadline in Week 13, means that any delays in setup or test case development may cause the team to be unable to execute all planned test cases.

What is the Risk:

Setting up Selenium IDE and configuring browsers can be technically challenging, especially with varying system configurations across team members.

How to Manage:

Strictly follow the schedule defined in Section 1.8. The Test Manager should monitor progress weekly to ensure testing stays on track and prioritize critical test cases if delays occur.

Test Environment Issues

Explanation:

Setting up Selenium IDE and configuring browsers can be technically challenging, especially with varying system configurations across team members.

What is the Risk:

A non-functional test environment will become an obstacle that can cause test execution to waste valuable time, and can cause significant delays to the project timeline.

How to Manage:

The Test Lead should configure the environment early in Week 10 and provide a step-by-step setup guide with screenshots to all testers. Any issues will be documented and shared with the team.

Resource Unavailability

Explanation:

Team members may get sick, have holidays, or face other personal commitments during the testing phase.

What is the Risk:

Each member is responsible for a specific module, which means if someone is unavailable, those modules may not be tested thoroughly thus creating gaps in test coverage.

How to Manage:

Put all test cases and project files stored into Google Drive where everyone can access them. This allows other members to take over pending tasks if necessary.

Product Risks

Functional Failure in TikTok Shop

Explanation:

The cart system must calculate prices correctly when users update quantities or apply promo codes.

What is the Risk:

Incorrect pricing could lead to poor user experience, financial discrepancies, and loss of trust in the TikTok Shop feature, damaging the platform's reputation.

How to Manage:

Design thorough test cases using Boundary Value Analysis to test calculation logic and maximum quantity limits. Use Equivalence Partitioning to test different types of promo codes (valid, expired, below minimum spend).

Security Flaws in Messaging Module

Explanation:

The messaging system needs to filter out malicious links and inappropriate content to protect users.

What is the Risk:

If harmful content slips through, it could expose users (especially younger audiences) to phishing attempts or inappropriate material, putting them at risk will damage the company's reputation.

How to Manage:

Include specific negative test cases that attempt to send blocked URLs, special characters, and inappropriate text to verify the system's text-filtering and security sanitization mechanisms work as intended.

Performance Bottlenecks in User Profile

Explanation:

Login and authentication processes need to be responsive for good user experience.

What is the Risk:

Slow loading times or timeouts during login can frustrate users, lead to abandoned login attempts, and create a negative first impression of the web application's reliability.

How to Manage:

Testers will monitor and record response times during execution of all profile management test cases. Any performance degradation will be logged and reported to identify patterns or recurring issues.

1.5 Test Strategy

1.5.1 Testing Techniques

Our team will apply multiple Dynamic Black-Box testing techniques which are Cause-Effect Graphing (CEG) and Decision Tables. Since the internal source code is not accessible, test cases will be designed based on the web application's external specifications and user interface constraints which is in turn Black-Box Testing.

Cause-Effect Graphing (CEG) technique is used for the specification of logical relationship between input condition 'causes' and outcome 'effects'. In CEG, 'cause' is defined as a particular condition, which can either be true or false, whereas 'effect' is defined as the outcome of 'cause', which must be, and can either be, true or false. In CEG, the logical relationship gets specified between 'cause' and 'effect', and different combinations of 'cause' resulting in a particular 'effect' get established, along with constraints, if required, i.e., 'mutual exclusiveness', etc. The project makes use of 'Cause-Effect Graphing' for different modules of the application, which involve complex logic, i.e., 'login account', 'create promo code', 'schedule video', 'upload video', 'send message', 'manage message requests', etc. Logic of 'cause and effect' gets used for contemplating different combinations of 'cause' resulting in a particular 'effect'.

Then, "**Decision Tables** are constructed from these cause-effect graphs, which systematically list all test cases." A decision table is "a complete set of possible combinations of cause values, 'true or false,' and their effects, making it transparent and ensuring that none are omitted." Once the decision table is prepared for the various modules that are being tested through the CEGs, the various decision tables included are: "the login decision table for valid login, invalid password, and unregistered email. The promo code decision table for valid promo code, invalid discount, invalid minimum spend, and invalid date range. The schedule video decision table for valid future times, early times, far future dates, past dates, and edit after scheduling. The upload video decision table for valid file, file too large, and file type. The send message decision table for valid messages, empty messages, spaces only messages, and messages over length, and message requests decision table for first message to requests, second message sent, accept request, and normal messaging after accept." The use of decision tables "ensures that all test cases are justified based on a unique combination of input conditions, ensuring completeness in testing."

Module	Test Item	Technique(s) Used
Module 1: User Profile Management	Login Account	Cause-Effect + Decision Table
	Edit Profile	Cause-Effect + Decision Table
Module 2: Video Comment	Post Comment	Cause-Effect + Decision Table
	Reply to Comment	Cause-Effect + Decision Table
Module 3: TikTok Shop Seller Center	Create Product Listing	Cause-Effect + Decision Table
	Create Promo Code	Cause-Effect + Decision Table
Module 4: Video Creation	Upload Video	Cause-Effect + Decision Table
	Schedule Video	Cause-Effect + Decision Table
Module 5: Messaging	Send Message	Cause-Effect + Decision Table
	Manage Message Requests	Cause-Effect + Decision Table

Module 1: User Profile Management

Test Item: Login Account

Cause-Effect Graph
Causes (Conditions):
C1: Username exists in the system
C2: Password matches the username password

Effects (Results):
E1: Login successful – User redirected to home feed
E2: Error message: "Username or Password doesn't match our records.Try again"
E3: Error message: ""Invalid Username or password you have 4 attempts left.Please try again""

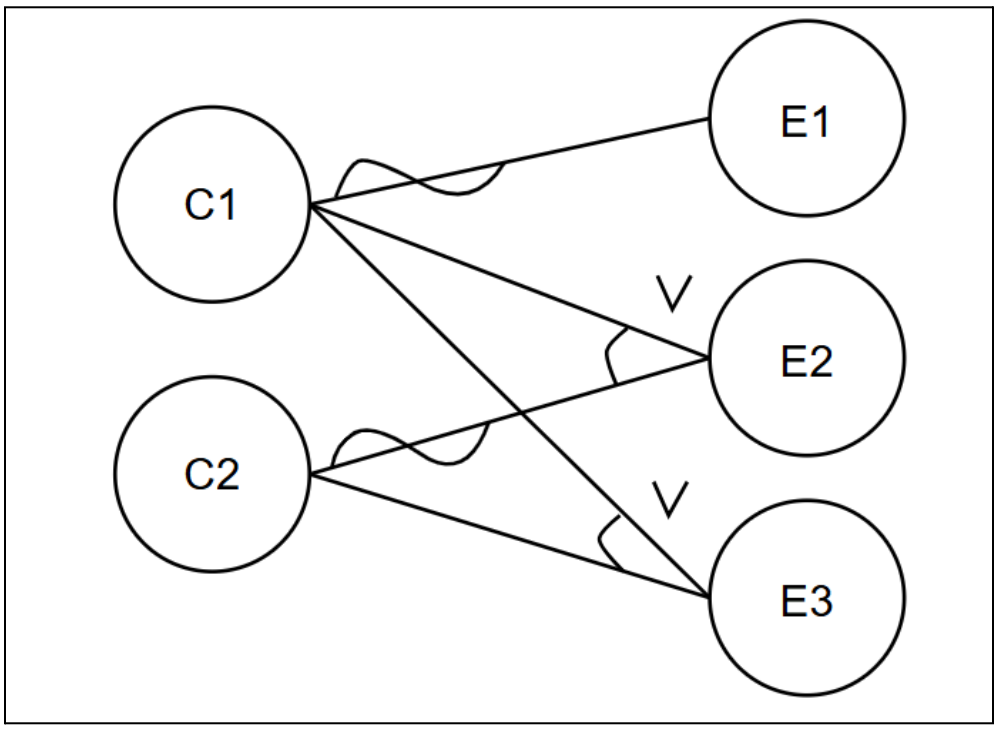


Figure 1: Cause-Effect Graph for Login Account

Decision Table

Cause	**TC1**	**TC2**	**TC3**
C1	T	T	F
C2	T	F	-
Effect			
E1	T	F	F
E2	F	T	F
E3	F	F	T

Test Item 2: Edit Profile

Cause-Effect Graph
Causes (Conditions):
C1: Username length is less than 2 characters
C2: Name length is more than 30 characters
C3: Bio length is more than 80 characters

Effects (Results):
E1: Error message: "Include at least 2 characters in your username"
E2: Error message: "Maximum 30 characters"
E3: Error message: "Maximum 80 characters"
E4: Profile updated successfully

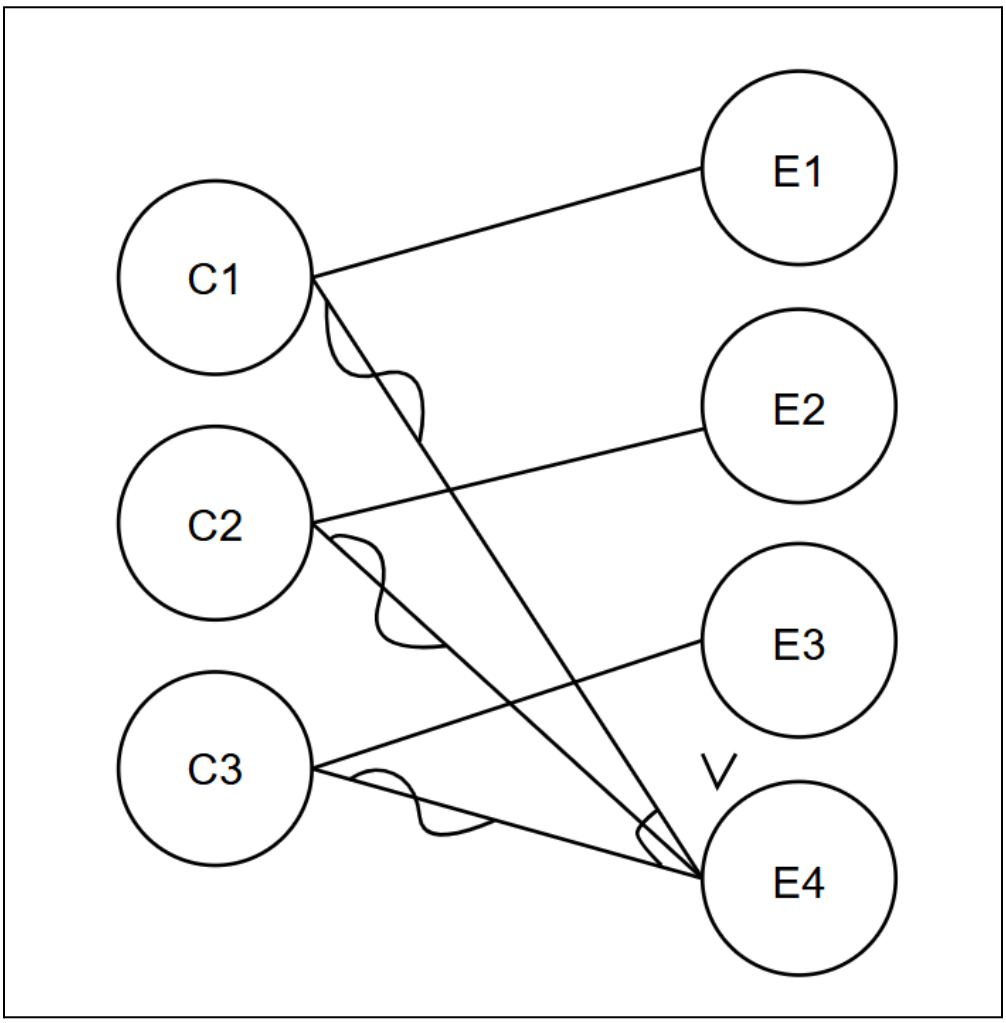


Figure 2: Cause-Effect Graph for Edit Profile

Decision Table

Cause	**TC1**	**TC2**	**TC3**	**TC4**
C1	T	-	-	T
C2	-	T	-	T
C3	-	-	T	T
Effect				
E1	T	F	F	F
E2	F	T	F	F
E3	F	F	T	F
E4	F	F	F	T

User Profile Management: Selected Test Cases

Test Items	Test Case #	Test Cases Selected
Login Account	UserProfileManagement_Login Account_ TC001	TC1
	UserProfileManagement_Login Account_ TC003	TC2
	UserProfileManagement_Login Account_ TC003	TC3
Edit Profile	UserProfileManagement_EditProfile_ TC004	TC1
	UserProfileManagement_EditProfile_ TC005	TC3

Module 2: Video Comment

Test item 1: Post Comment

Cause-Effect Graph
Causes (Conditions):
C1: Comment is not empty.
C2: Comment does not contain only spaces.
C3: Comment length less than or equal to 2200 characters.

Effects (Results):
E1: Comment post successfully.
E2: Post button grayed.
E3: Error: "Couldn't post comment. Try Again".
E4: Error: "Character limit reached"

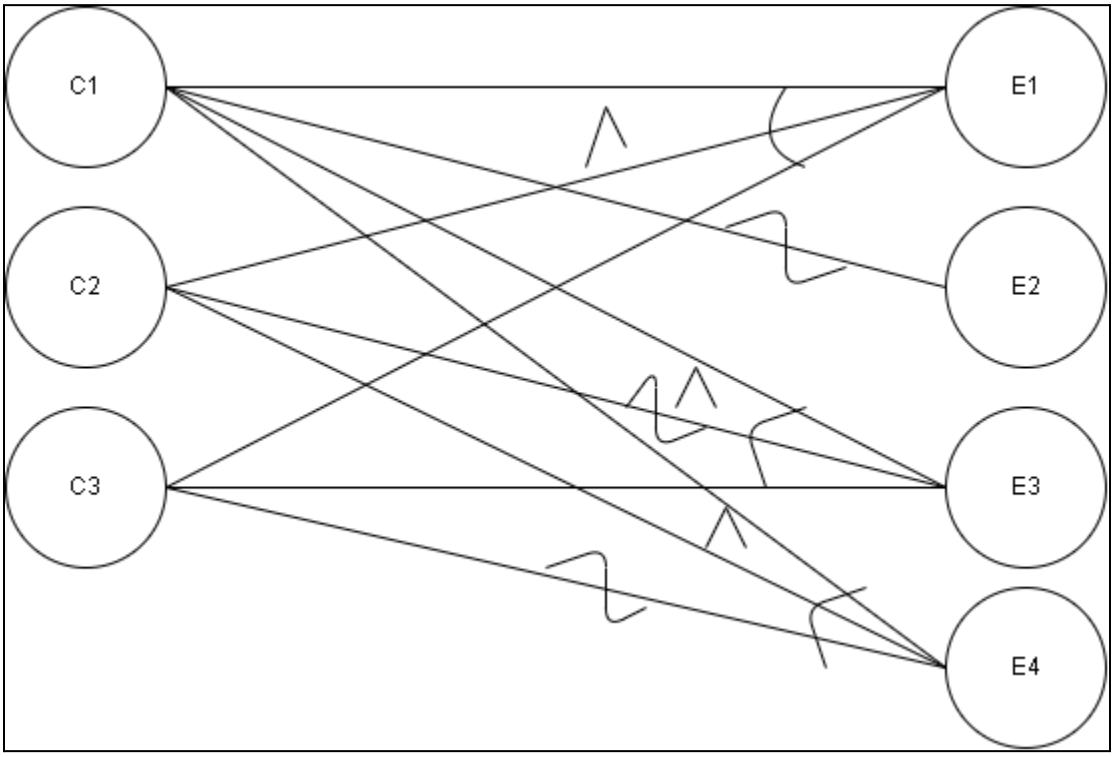


Figure 3: Cause-Effect Graph for Post Comment

Decision Table

Cause	**TC1**	**TC2**	**TC3**	**TC4**
C1	T	F	T	T
C2	T	-	F	T
C3	T	-	T	F
Effect				
E1	T	-	-	-
E2	-	T	-	-
E3	-	-	T	-
E4	-	-	-	T

Test Item 2: Reply to comment

Cause-Effect Graph
C1: Original comment exists (to reply to)
C2: Reply text is not empty
C3: Reply text does not contain only spaces
C4: Reply length is less than or equal to 2200 characters
C5: Reply contains @mention of another user
C6: Mentioned users privacy settings allow mentions. (recipient side)
Effects
E1: Reply field appears
E2: Reply posts successfully
E3: Post button grayed out
E4: Error: "Couldn't post comment. Try Again".
E5: Error: “Character limit reached”
E6: @username becomes clickable link
E7: Mentioned user receives notification
E8: Error: “Can’t mention this person due to their privacy settings”

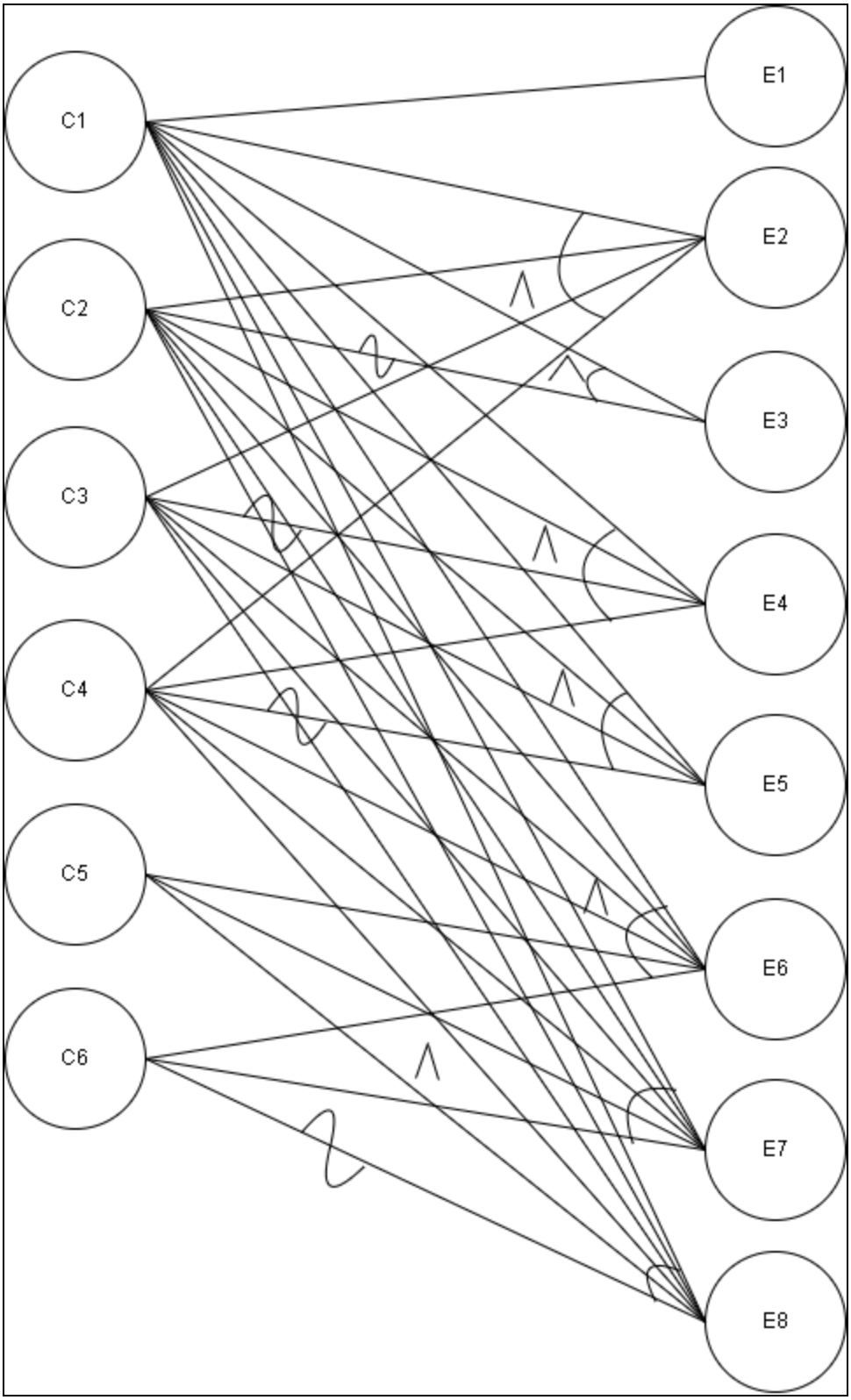


Figure 4: Cause-Effect Graph for Reply to Comment

Decision Table

Cause	TC1	TC2	TC3	TC4	TC5	TC6	TC7	TC8
C1	T	T	T	T	T	T	T	T
C2	-	T	F	T	T	T	T	T
C3	-	T	-	F	T	T	T	T
C4	-	T	-	T	F	T	T	T
C5	-	-	-	-	-	T	T	T
C6	-	-	-	-	-	-	T	F
Effect								
E1	T	-	-	-	-	-	-	-
E2	-	T	-	-	-	-	-	-
E3	-	-	T	-	-	-	-	-
E4	-	-	-	T	-	-	-	-
E5	-	-	-	-	T	-	-	-
E6	-	-	-	-	-	T	-	-
E7	-	-	-	-	-	-	T	-
E8	-	-	-	-	-	-	-	T

Video Comment: Selected Test Cases

Test Items	Test Case #	Test Cases Selected
Post Comment	VideoComment_PostComment_TC001	TC1
	VideoComment_PostComment_TC003	TC3
	VideoComment_PostComment_TC004	TC4
Reply to Comment	VideoComment_ReplytoComment_TC006	TC6
	VideoComment_ReplytoComment_TC008	TC8

Module 3: TikTok Shop Seller Center

Test Item 1: Create Product Listing

Cause-Effect Graph

Causes (Conditions):
C1: Product name is not empty.
C2: Price is a valid number format.
C3: Product Quantity is not empty.
C4: Product images are not empty.

Effects (Results):
E1: Product listing is created successfully.
E2: Error Message: "Product name is required."
E3: Error Message: "Price must be a valid number."
E4: Error Message: “Missing quantity”
E5: Error Message: "Product images are required."

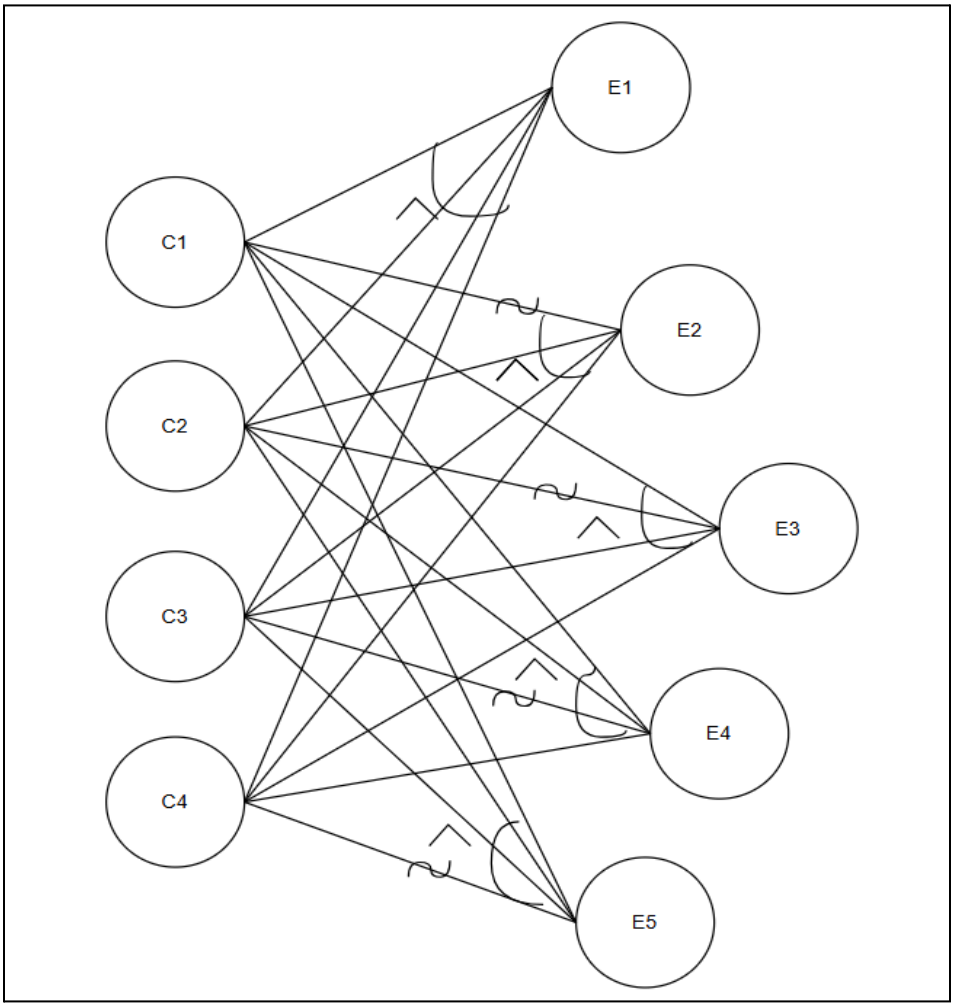


Figure 5: Cause-Effect Graph for Create Product Listing

Decision Table

Cause	**TC1**	**TC2**	**TC3**	**TC4**	**TC5**
C1	T	F	T	T	T
C2	T	T	F	T	T
C3	T	T	T	F	T
C4 (	T	T	T	T	F
Effect					
E1	T	-	-	-	-
E2	-	T	-	-	-
E3	-	-	T	-	-
E4	-	-	-	T	-
E5	-	-	-	-	T

Test Item 2: Create Promo Code

Cause-Effect Graph
Causes (Conditions):
C1: Promo code name is not empty
C2: Minimum spend value is valid
C3: Discount type is valid (percentage or fixed amount).
C4: End date is after the start date and not exceed 365 days.

Effects (Results):
E1: Promo code created successfully.
E2: Error Message: "Promo code name is required."
E3: Error Message: "Invalid minimum spend”
E4: Error Message: "Invalid discount amount."
E5: Error Message: "End date should not be more than 365 days."

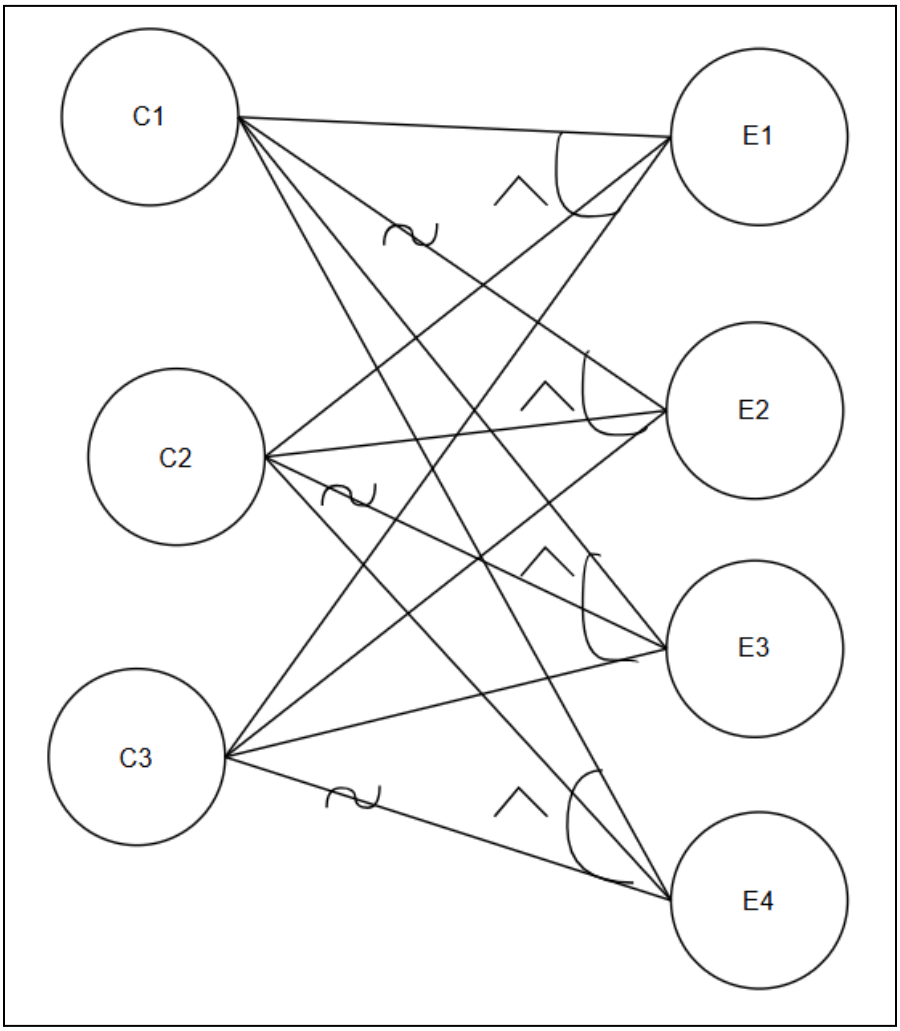


Figure 6: Cause-Effect Graph for Create Promo Code

Cause	**TC1**	**TC2**	**TC3**	**TC4**
C1	T	F	T	T
C2	T	T	F	T
C3	T	T	T	F
Effect				
E1	T	-	-	-
E2	-	T	-	-
E3	-	-	T	-
E4	-	-	-	T

Module 4: Video Creation

Test Item 1: Upload Video
Cause-Effect Graph
Causes (Conditions):
C1: File format is supported (MP4/MOV)
C2: File size is within limit (1 MB ≤ size ≤ 30,000 MB)

Effects (Results):
E1: Upload successful – video processed
E2: Upload failed – error message: "Unsupported file format"
E3: Upload failed – error message: "File too large (max 30GB)"

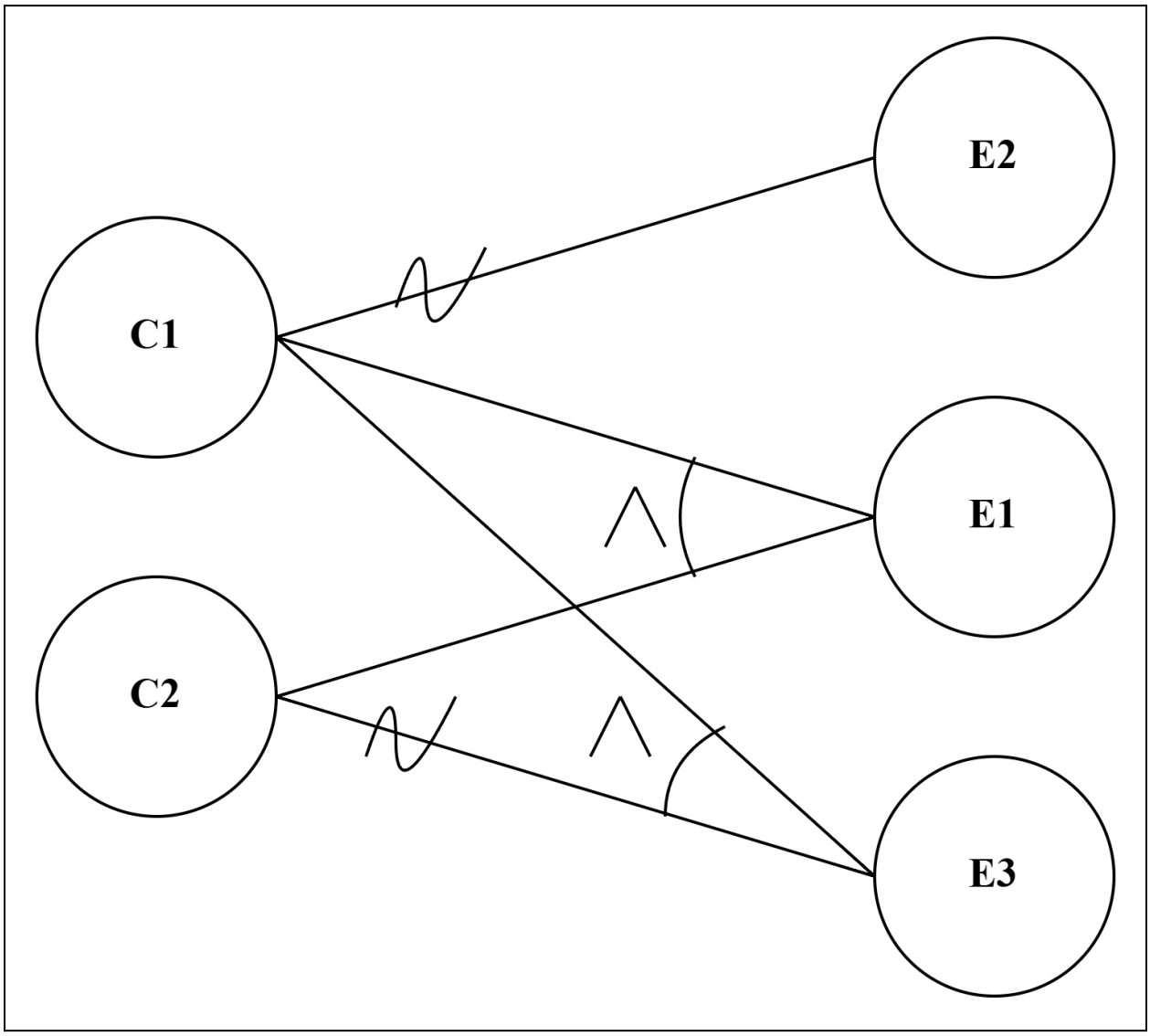


Figure 7: Cause-Effect Graph for Upload Video

Decision Table

Cause	**TC1**	**TC2**	**TC3**
C1	T	T	F
C2	T	F	–
Effect			
E1	T	-	-
E2	-	-	T
E3	-	T	-

Test Item 2: Schedule Video

Cause-Effect Graph

Causes (Conditions):
C1: Selected time is more than 15 minutes in advance
C2: Selected date is less than 30 days in advance
C3: selected date in the future
C4: User does not attempts to edit after scheduling

Effects (Results):
E1: Video scheduled successfully
E2: Error message: "Cannot edit after scheduling"
E3: Warning: "Schedule at least 15 minutes in advance"
E4: Warning: "Posts can be scheduled up to 30 days ahead"
E5: Date selection prevented become grayed out and unclickable

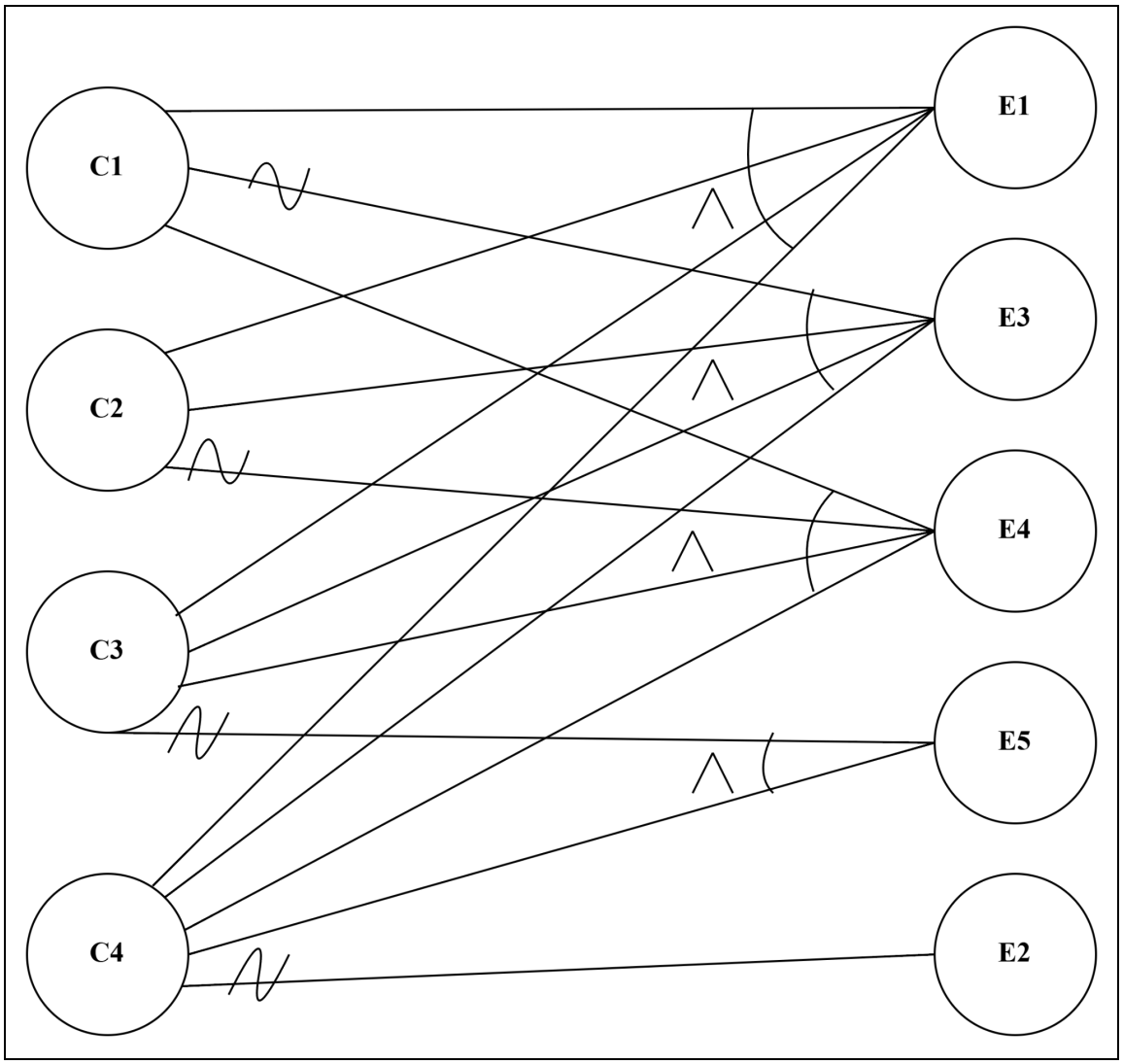


Figure 8: Cause-Effect Graph for Schedule Video

Decision Table

Cause	**TC1**	**TC2**	**TC3**	**TC4**	**TC5**
C1	T	F	T	-	-
C2	T	T	F	-	-
C3	T	T	T	F	-
C4	T	T	T	T	F
Effect					
E1	T	-	-	-	-
E2	-	-	-	-	T
E3	-	T	-	-	-
E4	-	-	T	-	-
E5	-	-	-	T	-

Video Creation: Selected Test Cases

Test Items	Test Case #	Test Cases Selected
Upload Video	VideoCreation_UploadVideo_TC001	TC1
	VideoCreation_UploadVideo_TC002	TC3
Schedule Video	VideoCreation_ScheduleVideo_TC003	TC1
	VideoCreation_ScheduleVideo_TC004	TC2
	VideoCreation_ScheduleVideo_TC005	TC3

Module 5: Messaging

Test Item 1: Send Message

Cause-Effect Graph

Causes (Conditions):
C1: Message is not empty
C2: Message is not contains only spaces
C3: Message length < 6000 characters

Effects (Results):
E1: Message sent successfully
E2: Unable to add more character
E3: Message unable to sent

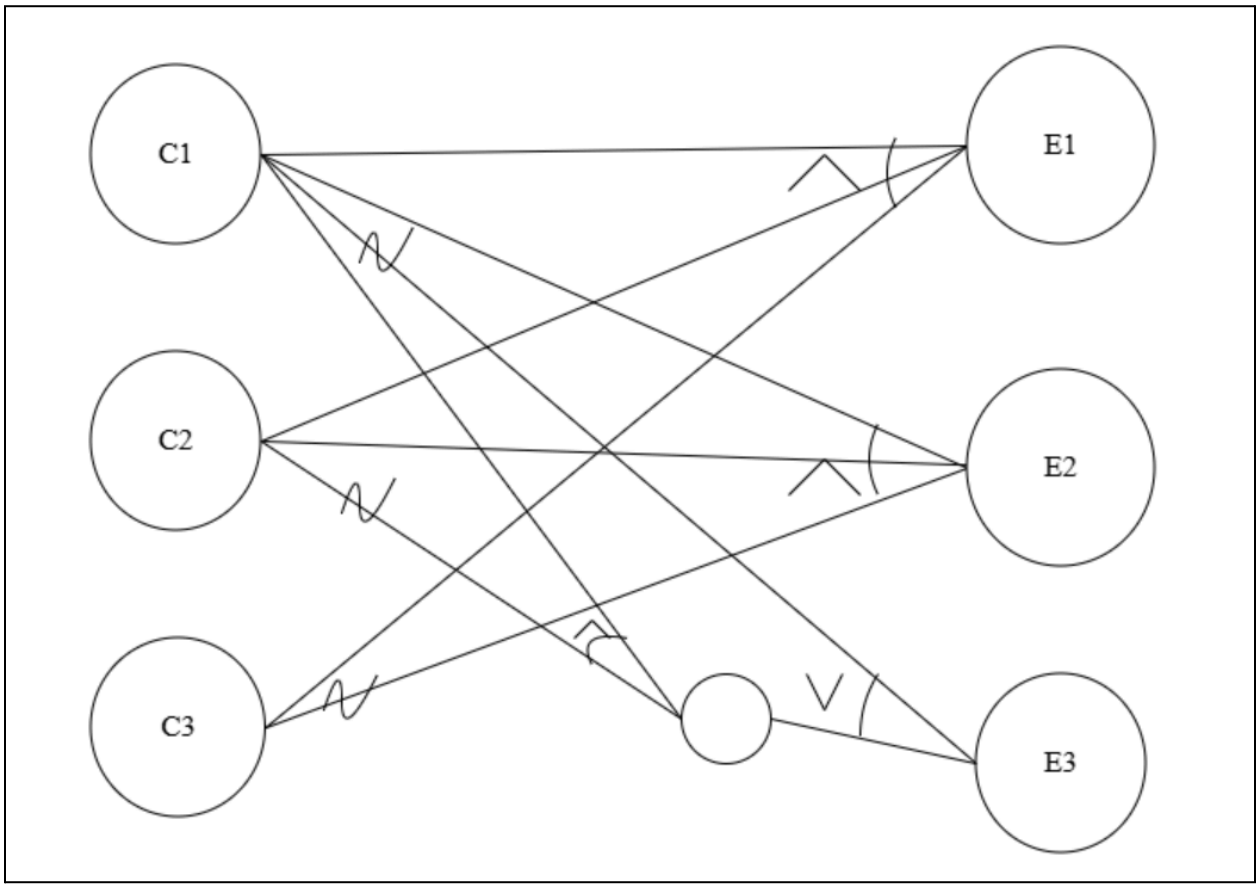


Figure 9: Cause-Effcet Graph for Send Message

Decision table

Cause	**TC1**	**TC2**	**TC3**	**TC4**
C1	T	T	F	T
C2	T	T	-	F
C3	T	F	-	-
Effect				
E1	T	-	-	-
E2	-	T	-	-
E3	-	-	T	T

Test Item 2: Manage Message Requests

Cause-Effect Graph

Causes (Conditions):
C1: Sender is a non-follower
C2: First message from non-follower
C3: Recipient accepts request from non-follower
C4: Second message from non-follower

Effects (Results):
E1: Message goes to requests folder
E2: Second message blocked
E3: Message moves to main inbox
E4: Normal messaging enabled

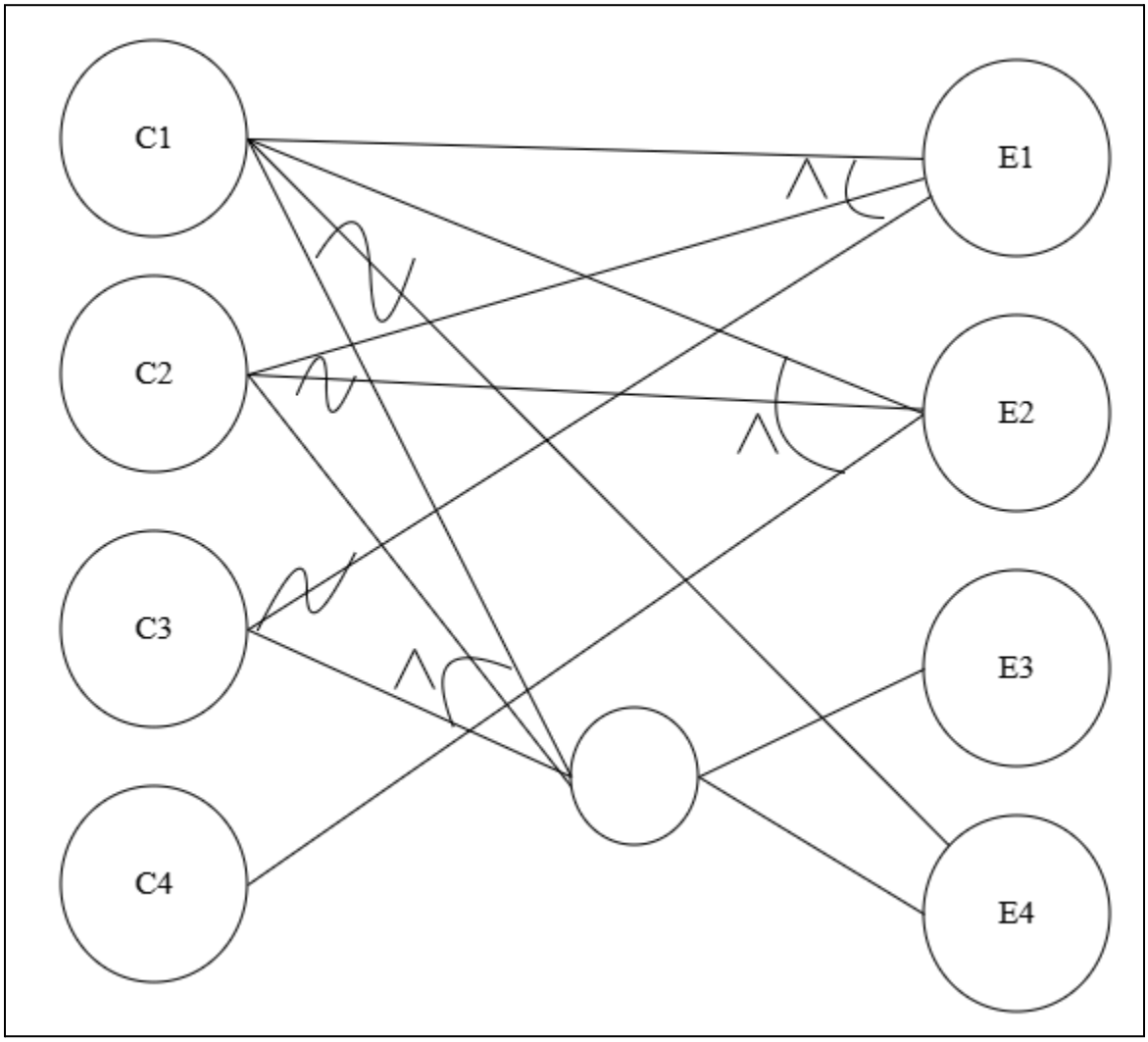


Figure 10: Cause-Effect Graph for Manage Message Requests

Decision Table

Cause	TC1	TC2	TC3	TC4
C1	T	T	T	F
C2	T	F	T	-
C3	F	-	T	-
C4	-	T	-	-
Effect				
E1	T	-	-	-
E2	-	T	-	-
E3	-	-	T	-
E4	-	-	T	T

Messaging: Selected Test Cases

Test Items	Test Case #	Test Cases Selected
Send Message	Messaging_SendMessage_TC001	TC1
	Messaging_SendMessage_TC002	TC3
	Messaging_SendMessage_TC003	TC4
Manage Message Requests	Messaging_ManageMessageRequests_TC004	TC3
	Messaging_ManageMessageRequests_TC005	TC2

1.5.2 Test Entry Criteria and Test Exit Criteria

Test Entry Criteria:

- Test plan document is completed and approved
- System requirements and design documentation are available to the testing team
- TikTok web application build is delivered and accessible through target browsers
- Test environment (hardware, operating system, and browsers) is fully configured
- Required testing tools (e.g., Selenium IDE) are installed and operational
- Test accounts and required test data are prepared
- All test cases for the five modules have been designed and reviewed

These criteria ensure that testing can be performed effectively and efficiently from the start. Without proper documentation, testers cannot determine expected results. Without a stable system and ready environment, test execution cannot proceed smoothly.

Test Exit Criteria:

- All planned test cases across the five modules have been executed at least once
- At least 95% of test cases have passed
- No critical or high-priority defects remain unresolved
- All detected defects have been logged and documented
- Test execution results are recorded and verified
- Test summary report is completed and reviewed by the team
- Final testing documentation is prepared for submission

These criteria ensure that the TikTok web application has been thoroughly tested and meets the required quality standards before the project is considered complete.

1.5.3 Test Environment

Component	Description
Hardware	**Primary Workstation:** - PC/Laptop with at least 8GB RAM, modern multi-core processor running Windows 10/11 or macOS. **Secondary Device:** - An alternative PC/Mac to ensure cross-browser compatibility.
Software	**Target System:** - TikTok Web Application (accessed via Google Chrome v120+ / Safari / Edge). **Automation Framework:** - Selenium WebDriver (Python) for automating UI-based web tests **Additional Tools:** - Google Chrome v120+ (primary test browser). - Microsoft Edge (secondary browser for compatibility checks).
Operating System	**Windows 10/11**
Automated Tools	**Test Automation:** - Selenium WebDriver with Python (selenium package) **Evidence Capture:** - Windows Snipping Tool for additional manual screenshots. **Defect Tracking:** - Google Sheets (shared among team members).
Network	Stable broadband or 4G/5G mobile connection.

1.5.3.1 Test Data

Module 1 - User Profile Management

Test data	Data Type	Size	Descriptions
Username(login)	String	5 - 24 character	**Valid** – “registered_user” Username exists in TikTok system
	String	5 - 24 characters	**Invalid** – “ghost_user_999” Username not registered in system
Password	String	8 + characters	**Valid** – “CorrectPass123” Correct password for registered_user
	String	8 + characters	**Invalid** – “dummy123” Password does not match the registered username
Username(profile)	String	1 character	**Invalid** – “a” Username less than 2 chars
Bio	String	=< 80 characters	**Valid** – “Short Bio” Used as baseline for username error test

Module 2 - Video Comment

Test data	Data Type	Size	Descriptions
Comment Text	String	1 - 2200 characters	Valid: **“Great Video!”**
		0 characters	Invalid: **Empty**
		≥ 1 space character	Invalid: **“ ” (Enter spaces)**
		> 2000 character	Invalid: **“HIHIHIHIHIHIHI HIHIHIHIHIHIHI HIHIHIHIHIHIHI

			HIHIHIHIHIHIHI HIHIHIHIHIHIHI HIHIHIHIHIHIHI

			HIHIHIHIHIHIHI HIHIHIHIHIHIHI HIHIHIHIHIHIHI

			HIHIHIHIHIHIHI HIH" (Exceeds limit)
Reply to comment	TikTok account	-	Valid: **Account used to send @mention replies (User A)**
			Valid: **privacy setting mentions set to "Everyone" (User B)**
			Invalid: **Account privacy settings mention set to "No**

			one”. (User C)

Module 3 - TikTok Shop Seller Center

Test data	Data Type	Size	Descriptions
Product Name	String	25 - 100 characters	Valid: **Enter a product name with valid size between 25 to 100 characters**
		0	Invalid: **Product name not entered**
Price	Decimal	RM 0.01 - RM 9999	Valid: **Enter a valid numeric product price**
	String	N/A	Invalid: **Enter a non-numeric value**
Quantity	Integer	1-1000	Valid: **Enter a valid product stock quantity**
		Null	Invalid: **Quantity not provided**
Schedule Time & Date	Date/Time	-	Valid: **27/04/2026 5:00 AM - 28/04/2026 5:00 PM**
			Invalid: **27/04/2026 5:00 AM - 30/04/2027 5:00 PM** (365 days from 27 April 2026 – exceeds 365-day maximum schedule window)

Module 4 - Video Creation

Test data	Data Type	Size	Descriptions
Video File	MP4	50 MB	Valid: **video.mp4**
	PDF	2 MB	Invalid: **video.pdf** (TikTok Web only accepts MP4/MOV video files)
Schedule Time & Date	Date/Time	-	Valid: **12/04/2026 12:45 PM**
			Invalid: **12/04/2026 12:40 PM** (5 minutes from current time, e.g., if current time is 14:00, fails minimum 15-minute advance)
			Invalid: **12/05/2026 12:45 PM** (31 days from 12 April 2026 – exceeds 30-day maximum schedule window)

Module 5 - Messaging

Test data	Data Type	Size	Descriptions
Message text	String	1 - 5999 characters	Valid: **“hi”**
		0 characters	Invalid: **(empty)**
		≥ 1 space	Invalid: **“ ”**
Sender account	TikTok account	-	Valid: **Use an account that already follow the recipient** (message goes directly to inbox)
		-	Invalid: **Use an account that does not follow the recipient** (first message goes to requests folder and second message is blocked until accepted)

1.6 Testing activities and estimates

The testing process will be based on the standard Software Testing Life Cycle (STLC), which ensures a systematic approach for testing from planning to closure. The Work Breakdown Structure (WBS) technique has been used to decompose the project into manageable tasks and estimate the effort required to minimise the sudden surprises of tasks.

Requirement Analysis

In the requirement analysis phase, the main task is to carefully review the system's specifications as to understand what the TikTok web application must accomplish in terms of its functionality. The first task is to identify the features that can be tested in the system and make sure that all the requirements are clearly defined, specific to avoid confusion, and not abstract before we start any planning. The team will analyze the application's requirements to understand and identify features that need testing and what the expected behavior should be for each function that they test. In this phase, we also define the high-level testing objectives. This phase will produce outputs like the requirement review notes and a list of testable features, which will help guide the later testing activities.

Test Planning

After the analysis phase, we will continue with the test planning phase, where we will define the scope of the testing. First, we will focus on the identification of the testing objectives and then choose which testing strategy we should use to measure those objectives. We can do this by determining suitable testing methods and selecting the most appropriate one to be used. We also identify the project risk and plan to mitigate that risk as much as possible to reduce future chances of these risks becoming a major problem. The roles and responsibilities of each team member are also assigned, and the test schedule and timeline will be made. We will also set the entry criteria and exit criteria that are suitable for the testing of the TikTok Application. The main deliverable from this phase is the Test Plan Document.

Test Case Development

The test case development is the phase where we design detailed test cases for each feature. We only use black box testing as we do not have access to the source code of the TikTok application. We apply techniques like Equivalence partitioning, boundary value analysis, and many more. We will then create positive and negative test cases to show the normal and expected output, and using the negative test case, we will show how the TikTok application reacts to invalid input and user errors. After that, we will define the expected results for each test case and even conduct a peer review of test cases. The main product of these phase is the test case specifications and test data sets.

Test Environment Setup

We will prepare the hardware and software needed for the testing based on the specifications of the TikTok application. We will also install and configure the testing tools we need and the required browser that we will be using. We will need to set up test user accounts, as we need to be able to access the TikTok web application. At the same time, we must have files and media that are relevant and needed during testing. We must make sure we match the environment to real user conditions and have realistic situations. Finally, we will perform an environmental readiness check.

Test Execution

In test execution, we will execute the test cases according to the test plan from the test planning phase. To correctly execute them, we must follow a step-by-step testing procedure to get the results. After getting the results, we must first compare them to the expected results and record them. Usually, the test results are recorded as pass, fail, or pending. If they fail, we should log the defects and must have evidence and attachments to prove that that is the case. With the defects being listed, we also need to assign the defects a severity and priority based on their importance. After bugs are fixed, we will perform regression testing and maintain the test execution logs and defect reports.

Test Cycle Closure

Last but not least, we will conduct the Test cycle closure phase. In this phase, we will first review all the testing activities and make sure and confirm that all planned test cases were executed and none were left out or forgotten. Then we will analyse the test results and review the defects and unresolved issues. With that information, we will prepare a test summary report that includes statistics and the evaluation of the performance and effectiveness of the web application. In this phase, we will also mention the lessons learned and improvements that can be made.

1.7 Staffing

Staffing table displays the team members and their responsibilities,they are solely responsible for the modules that they are given and will be assigned tasks specifically based on their roles.

Team Member	Assigned Module	Role	Responsibilities
Ryan Julian Rajesh	User Profile Module	Functional Test Analyst, Tester	Reviews system requirements, ensures test coverage, and validates functionality for the User Profile module.
Natalie Koa Hao Yee	Video Comment Module	Functional Test Analyst, Tester	Reviews system requirements, ensures test coverage, and validates functionality for the video comment module.
Lester Liau Jiun Lang	TikTok Shop Seller Center Module	Functional Test Analyst, Tester	Tests seller functions including product creation and promo code setup, reports defects for the TikTok Shop Seller Center module.
Tiong Chang Vui	Video Creation Module	Team Leader, Tester	Leads test execution, tracks progress, resolves blockers, ensures quality, and executes tests for the video creation module.
Wong Kai Lit Kerry	Messaging Module	Team Manager, Tester	Oversees the overall test plan, coordinates team tasks, manages timelines, reviews reports, and executes tests for the messaging module.

1.8 Schedule

Task	Date	Week(S)	Duration (Days)	Responsible Member(s)
Requirement Analysis: Understanding project requirements for testing the TikTok application and identifying testable features.	26/01/2026 - 08/02/2026	Week 1 - 2	14	Test Analyst
Test Planning: Creating a test strategy, defining test scope, identifying risks, and assigning roles.	09/02/2026 - 01/03/2026	Week 3 - 5	21	Team Manager, Team Leader
Test Case Development: Design black-box test cases and prepare test data.	02/03/2026 - 29/03/2026	Week 6 - 9	28	All Members
Test Environment Setup: Configure Selenium IDE, and test accounts.	30/03/2026 - 05/04/2026	Week 10	7	Team Manager, Team Leader
Test Execution: Running manual and automated test cases, logging defects, and recording execution logs..	06/04/2026 - 12/04/2026	Week 11-12	7	Test Analyst, All Members

Test Cycle Closure: Final analysis, evaluating test coverage, and creating the Test Execution Summary Report for submission.	13/04/2026 - 20/04/2026	Week 13	8	Team Manager, Team Leader

2. Test Case Specifications

2.1 Overview (about Test Case)

Basically, the test case is a detailed set of instructions that clearly defines what needs to be tested, what steps to take, and what results to expect. For this assignment, we primarily focused on designing test cases with specific inputs and actions since, for us, the end goal is to identify defects in the TikTok web portal before they even arise for a live user. Rather than writing random test cases, our group opted to use a set of dynamic black-box techniques by first designing Cause-Effect Graphs (CEG), where we clearly defined the logical cause-and-effect relationships between input elements termed "causes" and their resultant effects. Next, we created a set of systematic Decision Tables to ensure that all possible combinations of test cases were covered, eliminating any possibility of omitting important test scenarios. For ease of reading, each of the five test cases for each module will follow a standardized template consisting of test case ID, pre-conditions, execution steps, and post-conditions. This will ensure that we can confirm a stable system, one that processes both expected and unexpected user behavior appropriately.

2.2 References

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International Organization for Standardization, International Electrotechnical Commission, & Institute of Electrical and Electronics Engineers. (2013). Software and systems engineering – Software testing – Part 3: Test documentation (ISO/IEC/IEEE 29119-3:2013).

TikTok Help Center. (2024). Account safety and privacy guidelines. https://support.tiktok.com

TikTok Shop Seller Academy. (2024). Product listing and promo code guidelines. https://seller-my.tiktok.com

2.3 Test Cases

Module 1 - User Profile Management

Test Case Template	
Test Case #: UserProfileManagement_Login Account_ TC001	**Test Case Name: Login**
System: TikTok Web Application	**Module: User Profile Management**
Design By: Ryan Julian Rajesh	**Design Date: 28/03/2026**
Executed By: Wong Kai Lit Kerry	**Execution Date: 10/4/2026**
Short Description: **Valid username and password that a user can use to successfully log in .**	

Pre-conditions:
User has a registered TikTok account with valid credentials
TikTok web application is open on Chrome browser
User is on the login page

Step	Action	Expected System Response	Pass/Fail	Comments
1	Enter valid username user2016830387848 in the username field	Username field accepts input and displays entered text	Pass	
2	Enter valid password “CorrectPass123@” in the password field	Password field masks characters with dots	Pass	

3	Click the “Log in” button	System redirects user to the main feed page	Pass	
4	Verify user profile icon is visible	Profile icon appears in the top corner	Pass	

Post-conditions:
System allows users to log into their account and can access all features.

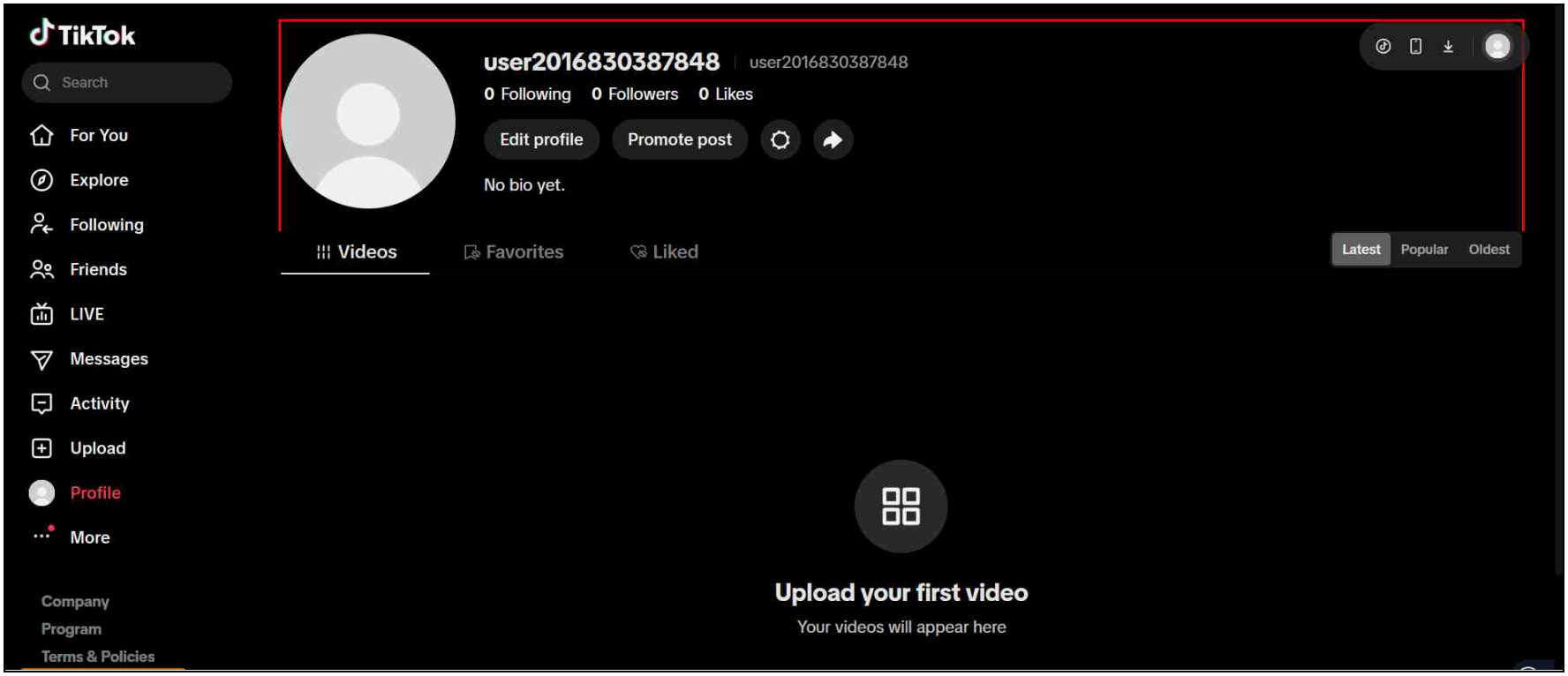


Figure 11: Output for Login Successful

Test Case Template	
Test Case #: UserProfileManagement_Login Account_ TC002	**Test Case Name: Login**
System: TikTok Web Application	**Module: User Profile Management**
Design By: Ryan Julian Rajesh	**Design Date: 28/03/2026**
Executed By: Wong Kai Lit Kerry	**Execution Date: 7/4/2026**
Short Description: **Invalid password is used and will cause the login to fail.**	

Pre-conditions:
User has a registered TikTok account
TikTok web application is open on Chrome browser
User is on the login page

Step	Action	Expected System Response	Pass/Fail	Comments
1	Enter valid username “registered_user” in the username field	Username field accepts input	Pass	
2	Enter incorrect password “WrongPassXYZ” in the password field	Password field masks characters	Pass	
3	Click the “Log in” button	System displays error message: “Username or Password doesn't match our records.Try again”	Pass	
4	Verify user remains on login page	Page does not redirect to feed	Pass	

Post-conditions:

System makes user stay on login page and displays an error message .

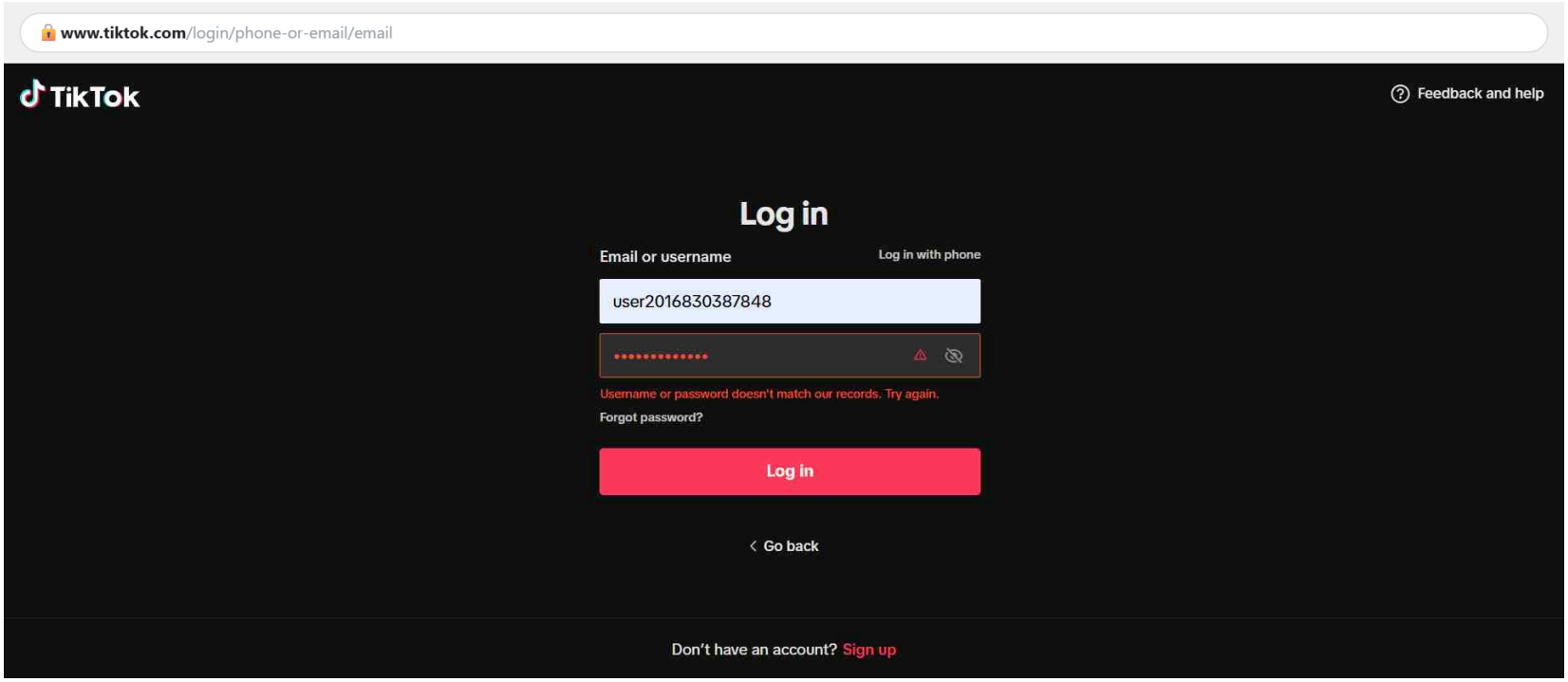


Figure 12: Output for Incorrect Password Error

Test Case Template	
Test Case #: UserProfileManagement_Login Account_ TC003	**Test Case Name: Login**
System: TikTok Web Application	**Module: User Profile Management**
Design By: Ryan Julian Rajesh	**Design Date: 28/03/2026**
Executed By: Wong Kai Lit Kerry	**Execution Date: 7/4/2026**
Short Description: **Invalid username will cause the Login to fail.**	

Pre-conditions: **TikTok web application is open on Chrome browser** **User is on the login page**

Step	Action	Expected System Response	Pass/Fail	Comments
1	Enter unregistered username ghost_user_999 in the username field	Username field accepts input	Pass	
2	Enter any password (e.g., dummy123) in the password field	Password field accepts input	Pass	
3	Click the “Log in” button	System displays error message: “Invalid Username or password you have 4 attempts left.Please try again”	Pass	
4	Verify no redirect occurs	User stays on login page	Pass	

Post-conditions: **System displays error message and the user will remain in the login page.**

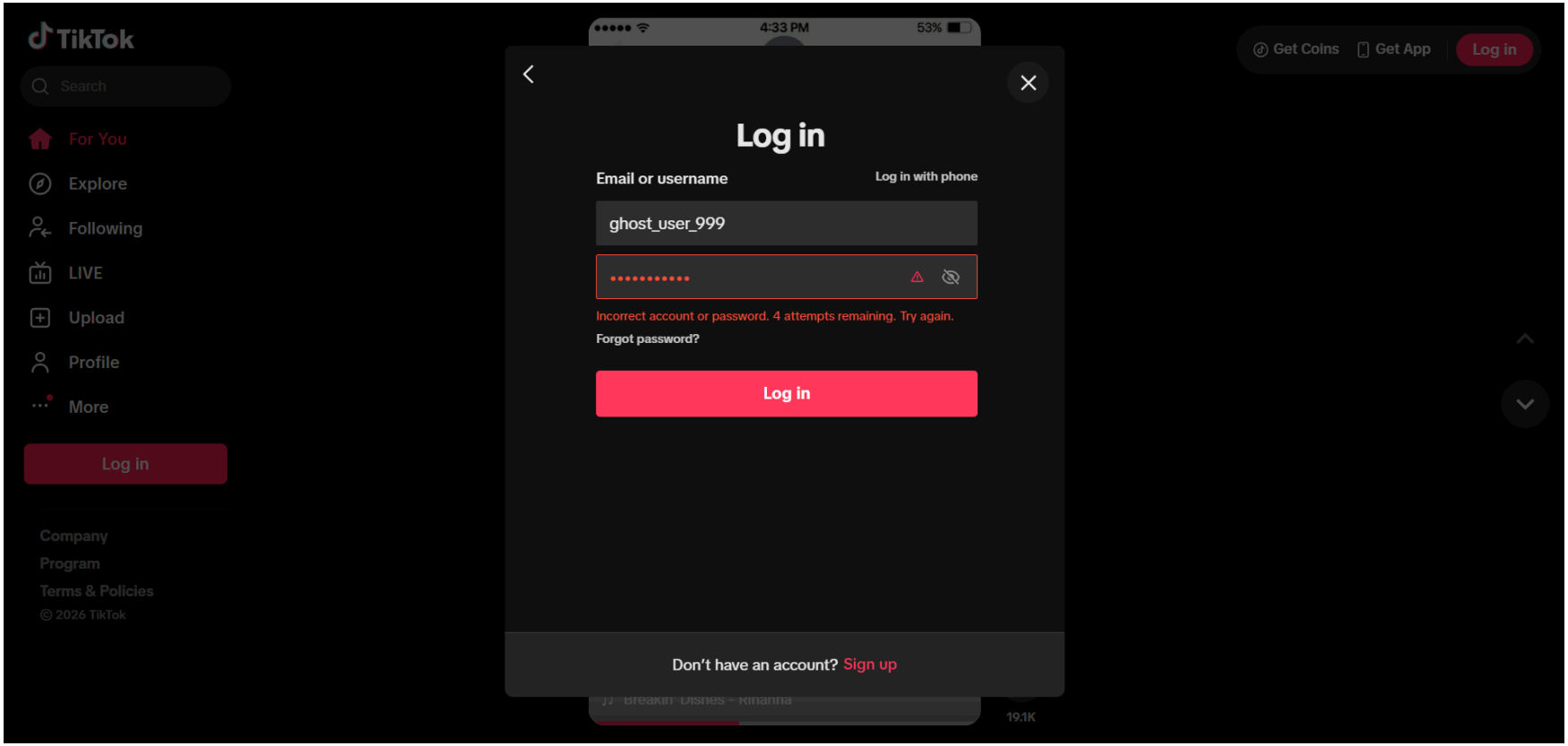


Figure 13: Output for Unregistered Account Error

Test Case Template	
Test Case #: UserProfileManagement_EditProfile_ TC004	**Test Case Name: Edit profile**
System: TikTok Web Application	**Module: User Profile Management**
Design By: Ryan Julian Rajesh	**Design Date: 28/03/2026**
Executed By: Wong Kai Lit Kerry	**Execution Date: 7/4/2026**
Short Description: **Invalid username causes the edit profile function to fail.**	

Pre-conditions:
User is logged into TikTok account
User is on the Edit Profile page

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click on the edit Profile button to edit	Profile becomes editable	Pass	
2	Enter username "a" (1 character)	Text appears in the field	Pass	
3	Click “Save” button	System displays error message: “Include at least 2 characters in your username”	Pass	

Post-conditions:
System will not update username and will display a error message.

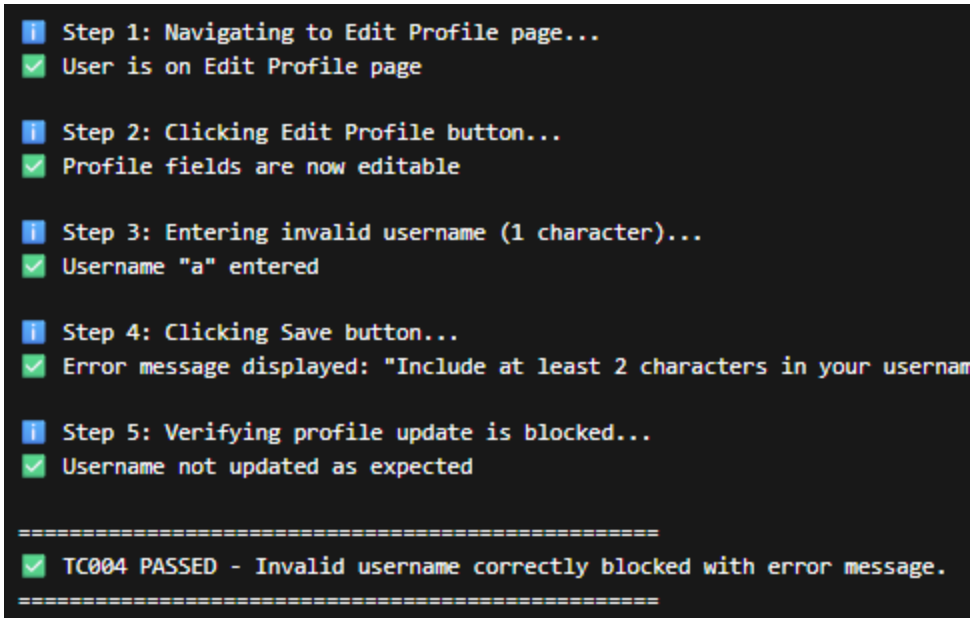


Figure 14: Successful Execution log for Edit Profile TC004

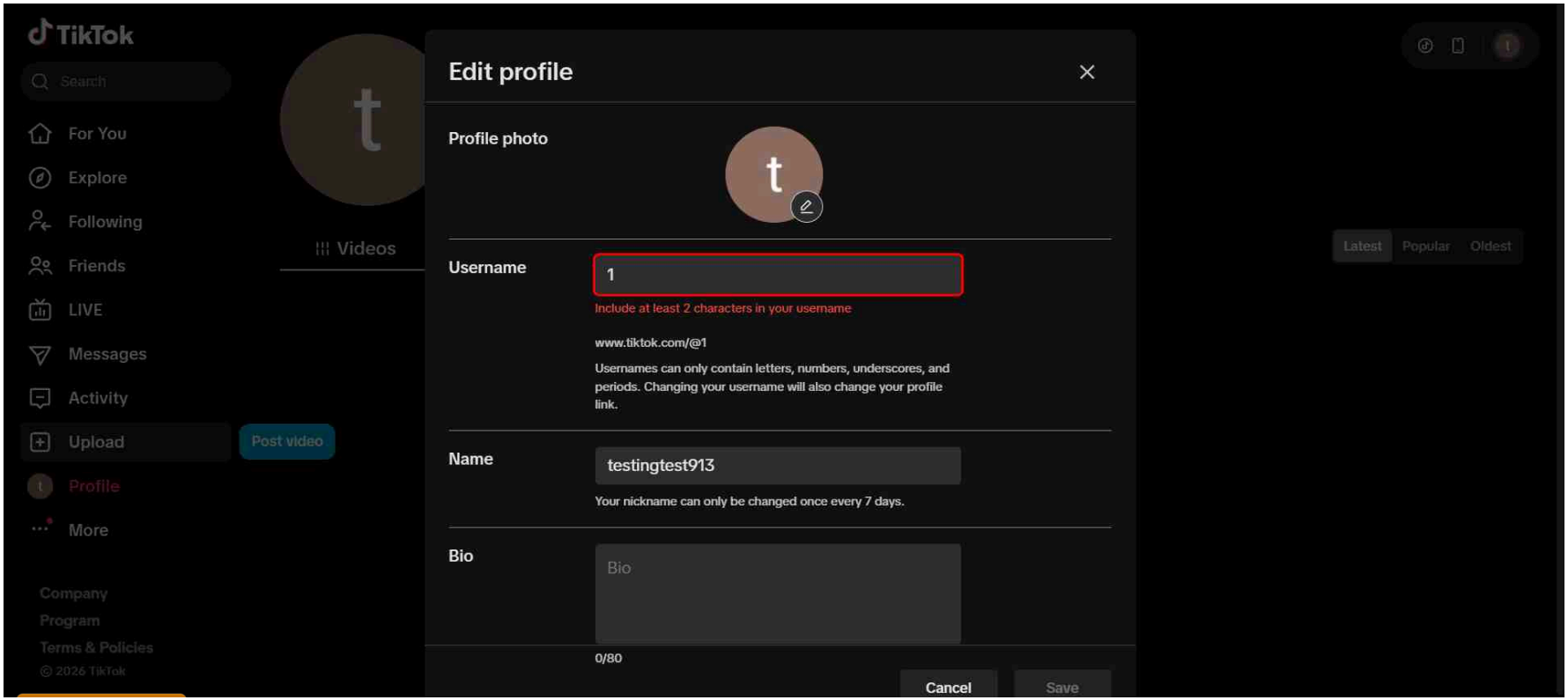


Figure 15: Output for Username Too Short Error

Test Case Template	
Test Case #: UserProfileManagement_EditProfile_ TC005	**Test Case Name: Edit profile**
System: TikTok Web Application	**Module: User Profile Management**
Design By: Ryan Julian Rajesh	**Design Date: 28/03/2026**
Executed By: Wong Kai Lit Kerry	**Execution Date: 7/4/2026**
Short Description: **Invalid bio over 80 characters causes the system to display an error message.**	

Pre-conditions:
User is logged into TikTok account
User is on the Edit Profile page

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click on the edit Profile button to edit	Profile becomes editable	Pass	
2	Enter an 81-character bio (e.g., 123456789012345678901234567890123456789012345678901234567890123456789012345678901)	Bio field may stop accepting input at 80 chars or allow but show error later	Pass	
3	Click “Save” button	System displays error message: “81/80”	Pass	

Post-conditions:
System will not update bio beyond 80 characters and display an error message.

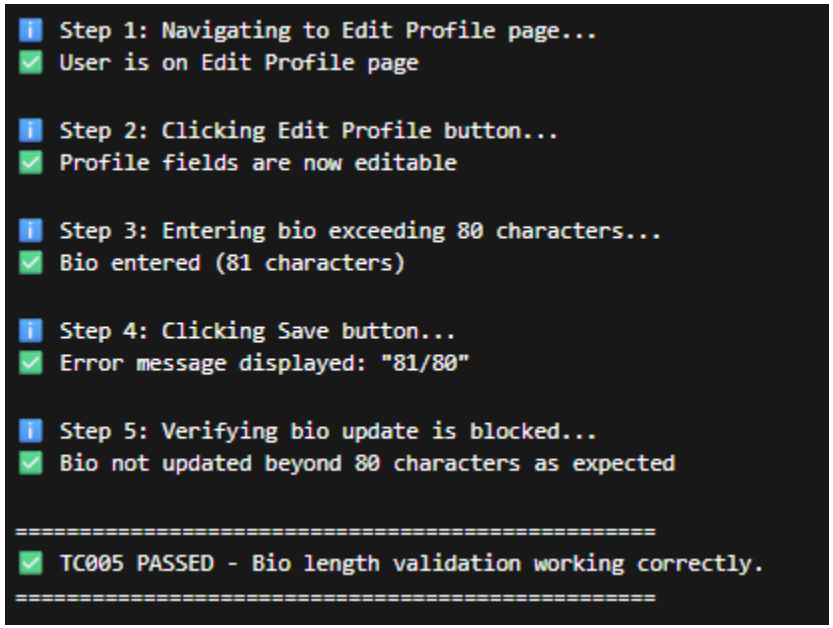


Figure 16: Successful Execution log for Edit Profile TC005

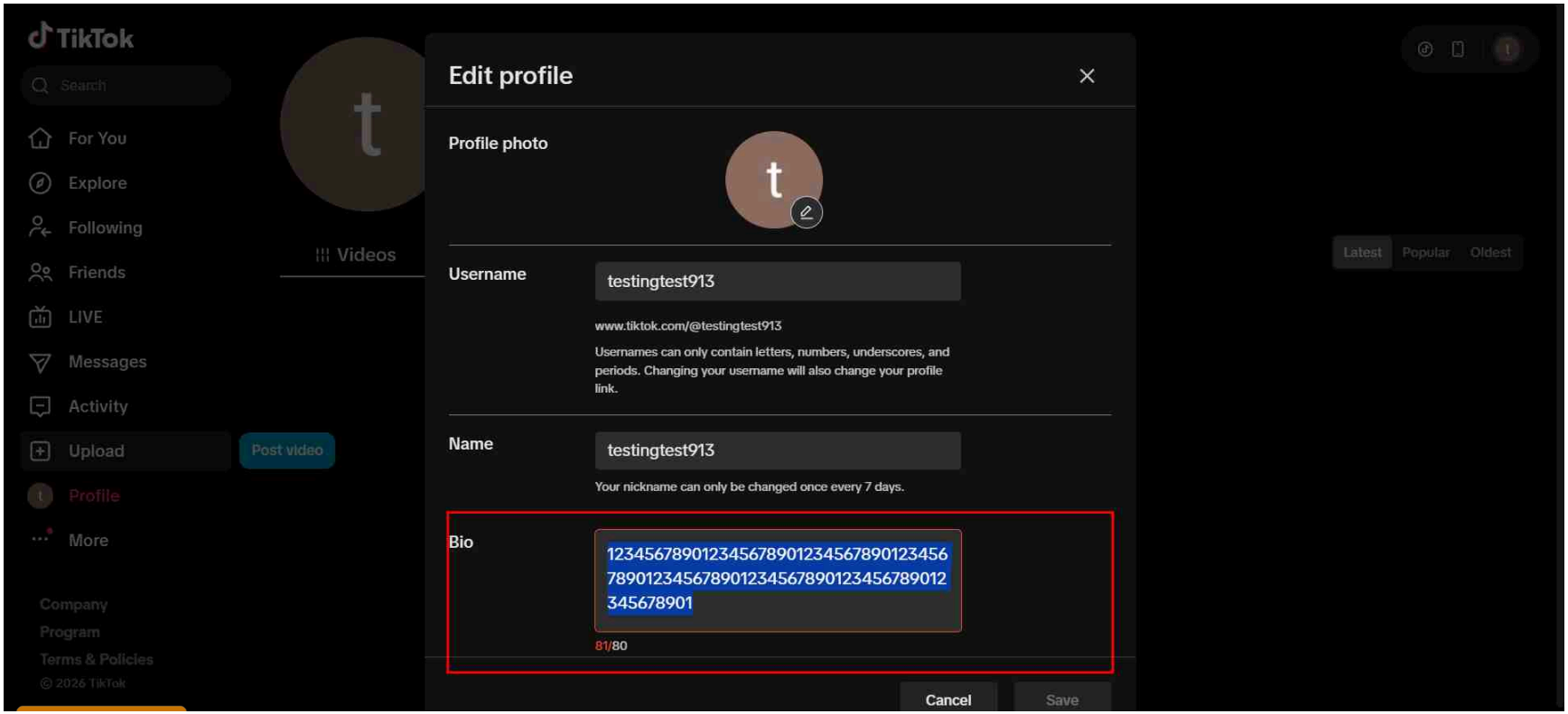


Figure 17: Output for Bio Too Long Error

Module 2 - Video Comment

Test Case Template	
Test Case #: VideoComment_PostComment_TC001	**Test Case Name: Post Comment**
System: TikTok Web Application	**Module: Video Comment**
Design By: Natalie Koa Hao Yee	**Design Date: 28/03/2026**
Executed By: Ryan Julian Rajesh	**Execution Date: 07/04/2026**
Short Description: **Test and verify posting a valid comment within the character limit**	

Pre-conditions:
User is logged into a TikTok account
A video is open with comment section available

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click on the comment field	Field becomes active with cursor blinking	Pass	
2	Type a valid comment: Great Video!	Text field appears	Pass	
3	Click the “post” button	Comment appears in comment section	Pass	
4	Refresh the page	Display videos you may like	Pass	
	Click on “Comments”	Previous comment is displayed		

Post-conditions:

System saves the comment and displays it publicly under the video.

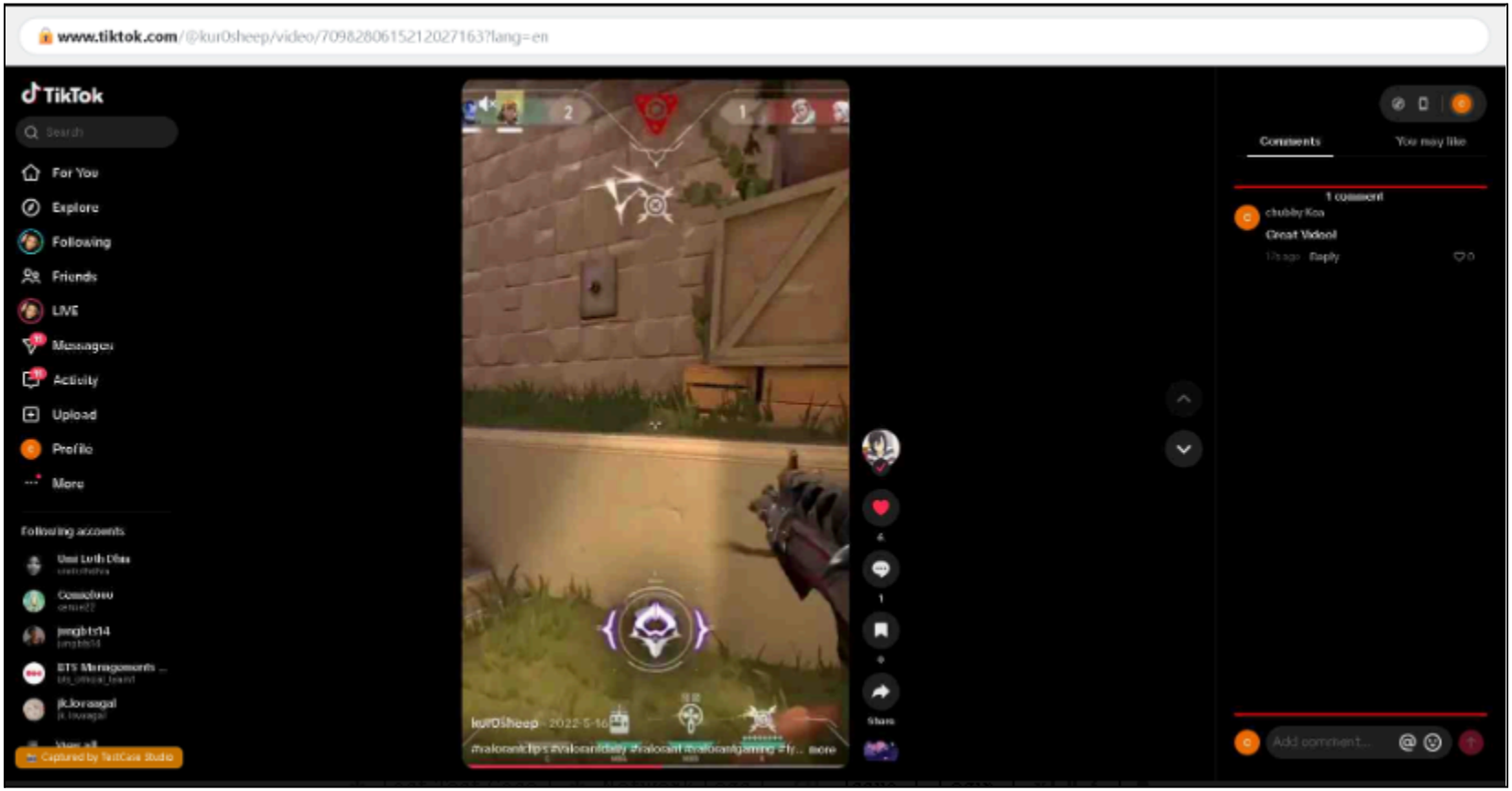


Figure 18: Output for Video Comment Posted Successfully

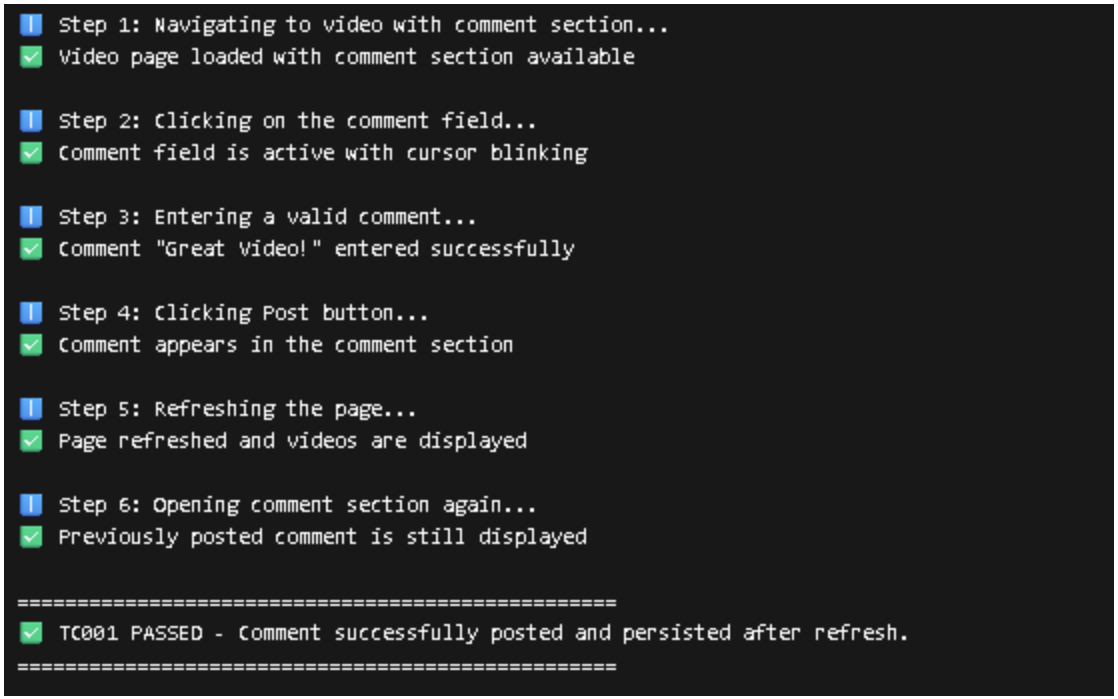


Figure 19: Successful Execution log for Post Comment TC001

Test Case Template	
Test Case #: VideoComment_PostComment_TC003	**Test Case Name: Post comment**
System: TikTok Web Application	**Module: Video Comment**
Design By: Natalie Koa Hao Yee	**Design Date: 28/03/2026**
Executed By: Ryan Julian Rajesh	**Execution Date: 07/04/2026**
Short Description: **Test and verify posting an invalid comment containing only space characters**	

Pre-conditions:
The user is logged into a TikTok account.
A video is open with the comment section visible.

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click on comment field	Field becomes active with cursor blinking	Pass	
2	Type only space characters:	Field appears blank (spaces are invisible) with cursor blinking	Pass	
3	Click the “post” button	Error: "Couldn't post comment. Try Again”.	Pass	
4	Check the comment section	No new comment appears	Pass	

Post-conditions:
System does not post the comment and displays an error message to the user.

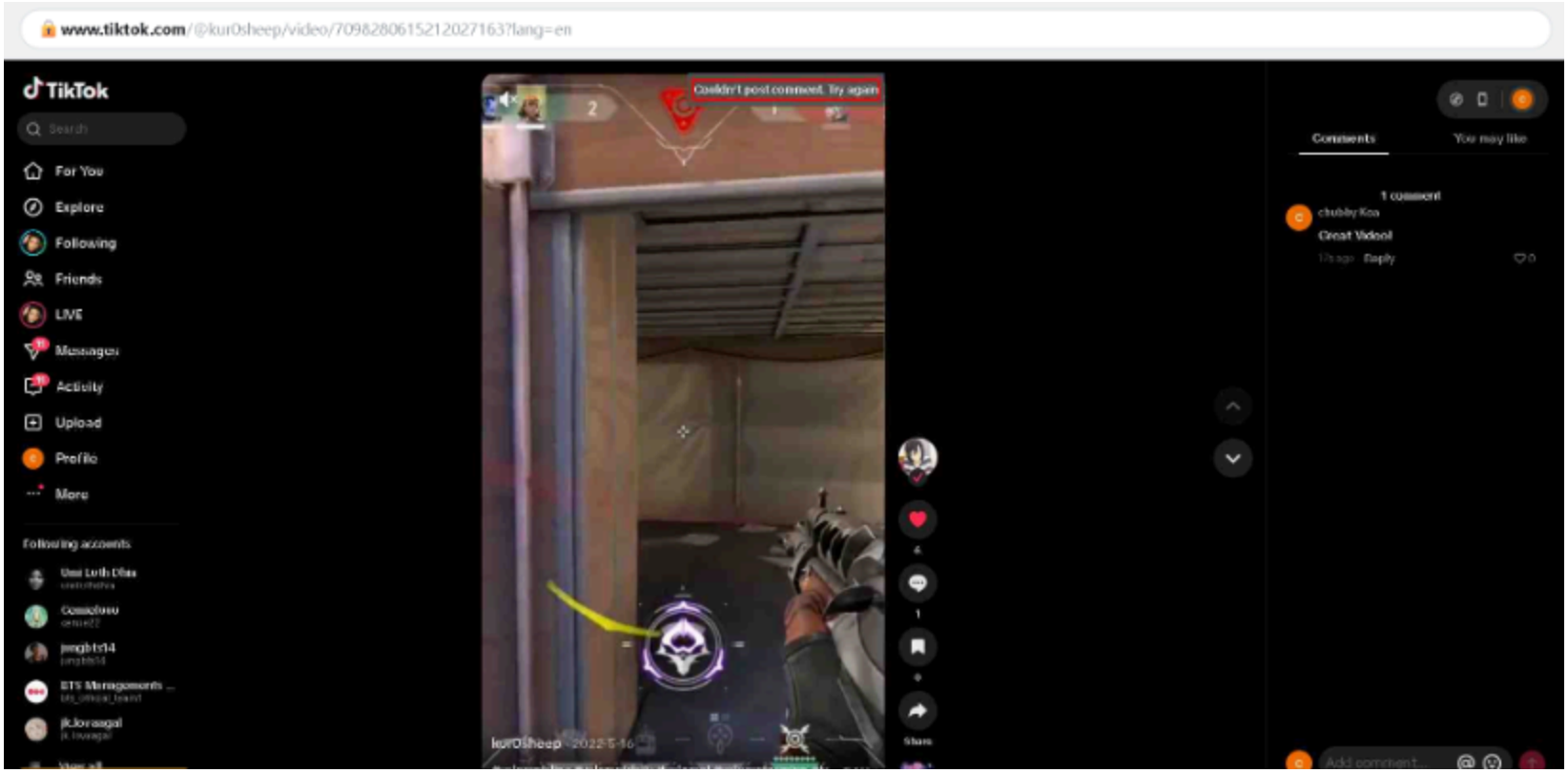


Figure 20: Output for Space-Only Error

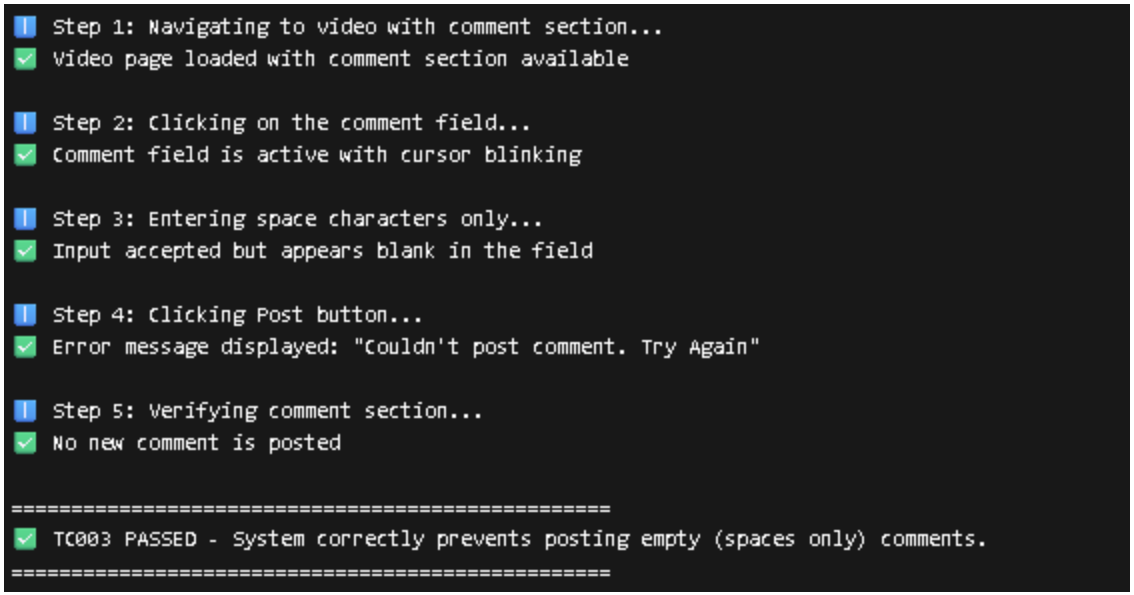


Figure 21: Successful Execution log for Post Comment TC002

Test Case Template	
Test Case #: VideoComment_PostComment_TC004	**Test Case Name: Post Comment**
System: TikTok Web Application	**Module: Video Comment**
Design By: Natalie Koa Hao Yee	**Design Date: 28/03/2026**
Executed By: Ryan Julian Rajesh	**Execution Date: 07/04/2026**
Short Description: **Test and verify posting an invalid comment exceeding 2200 characters**	

Pre-conditions:
The user is logged into a TikTok account.
A video is open with the comment section available.
The maximum character limit for TikTok is 2200 characters for comments.

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click on comment field	Field becomes active with cursor blinking	Pass	
2	Type a comment: HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI	System stops accepting input at 2200 and Error: “Character limit reached” appears.	Pass	

	HIHIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH IHIHIHIHIHIHIHIHIHIHIHI HIHIHIHIHIHIHIHIHIHIHIH			

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3	Check character counter shows limit reached	Counter shows “2200/2200”	Pass	

Post-conditions: **System does not allow input beyond 2200 characters and the character counter shows 2200/2200.**

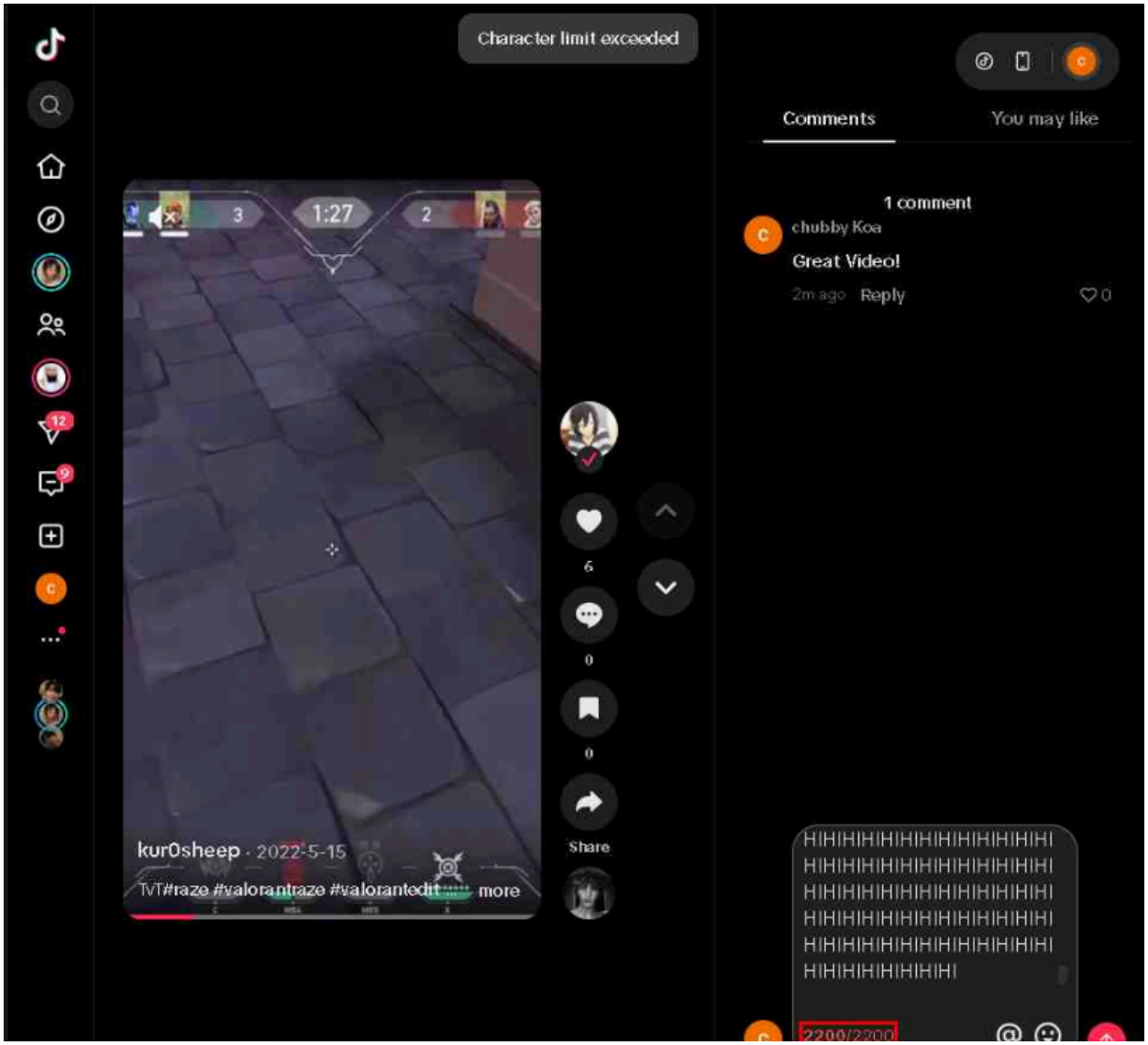


Figure 22: Output for Character Limit Reached

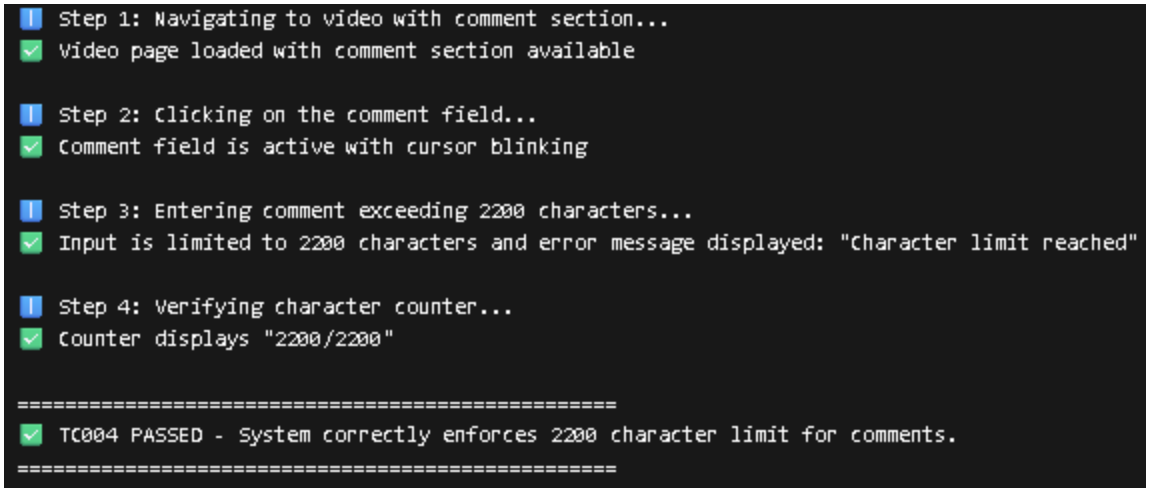


Figure 23: Successful Execution log for Post Comment TC003

Test Case Template	
Test Case #: VideoComment_ReplytoComment_TC006	**Test Case Name: Reply to comment**
System: TikTok Web Application	**Module: Video Comment**
Design By: Natalie Koa Hao Yee	**Design Date: 28/03/2026**
Executed By: Ryan Julian Rajesh	**Execution Date: 07/04/2026**
Short Description: **Test and verify replying with a valid @mention that is clickable.**	

Pre-conditions:
User A (reply sender) is logged into a TikTok account.
A video with an existing comment from another user is open.
User B privacy settings for mention is set to "Everyone" or allows mention from User A.

Step	Action	Expected System Response	Pass/Fail	Comments
1	Locate an existing comment.	Comment is visible	Pass	
2	Click the "Reply" button below the comment.	Reply field appears with cursor blinking	Pass	
3	Type : @	User suggestion list appears.	Pass	
4	Select User B from the list	"@UserB" appears in the reply field.	Pass	
5	Click the "Post" button	Reply posts successfully and appears nested under the original comment.	Pass	
6	Click the mentioned username "@UserB"	User A will be redirected to User B profile.	Pass	

Post-conditions:

System saves the reply, makes @UserB clickable, and redirects to UserB profile when clicked.

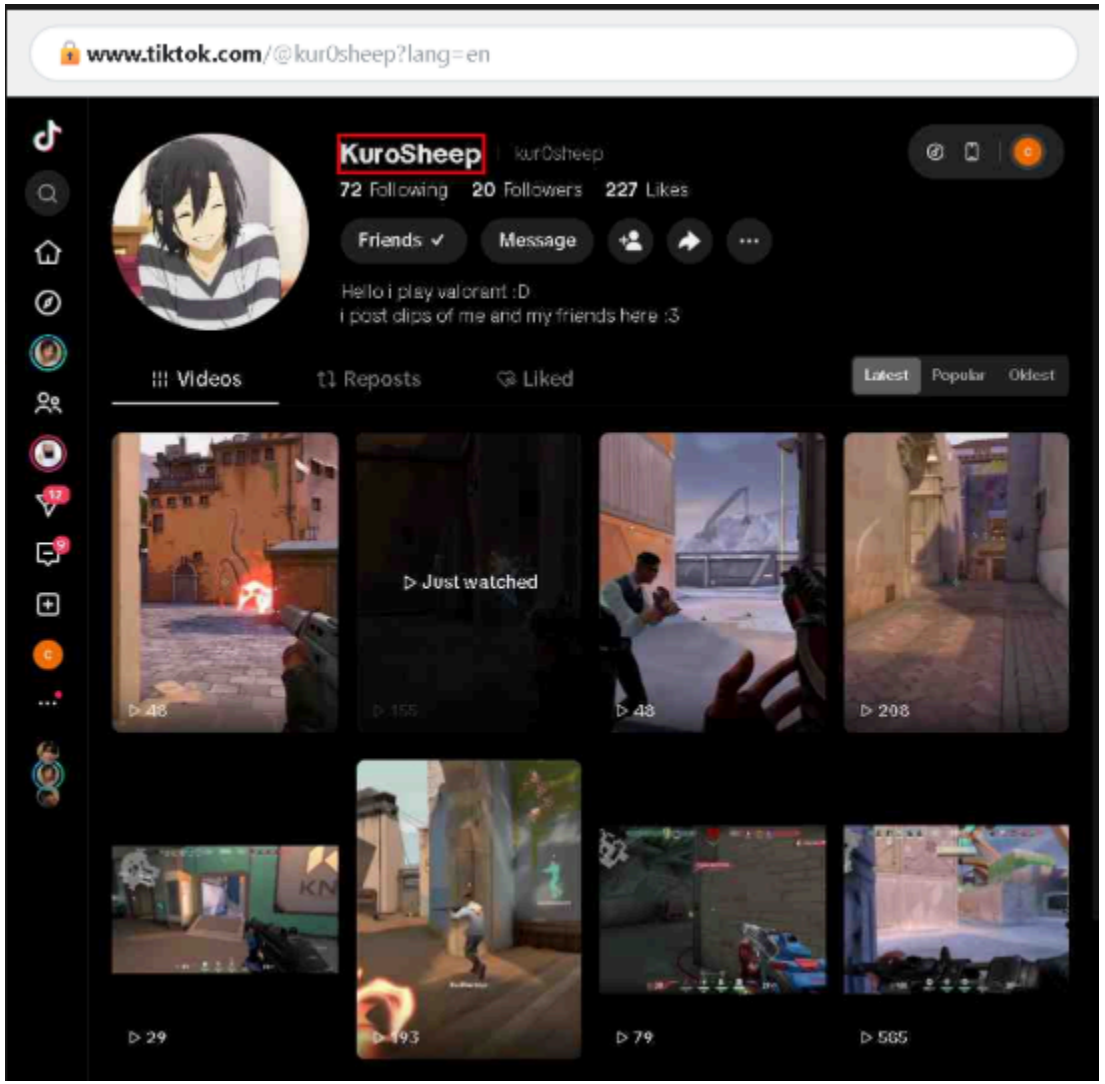


Figure 24: Output for @mention clickable

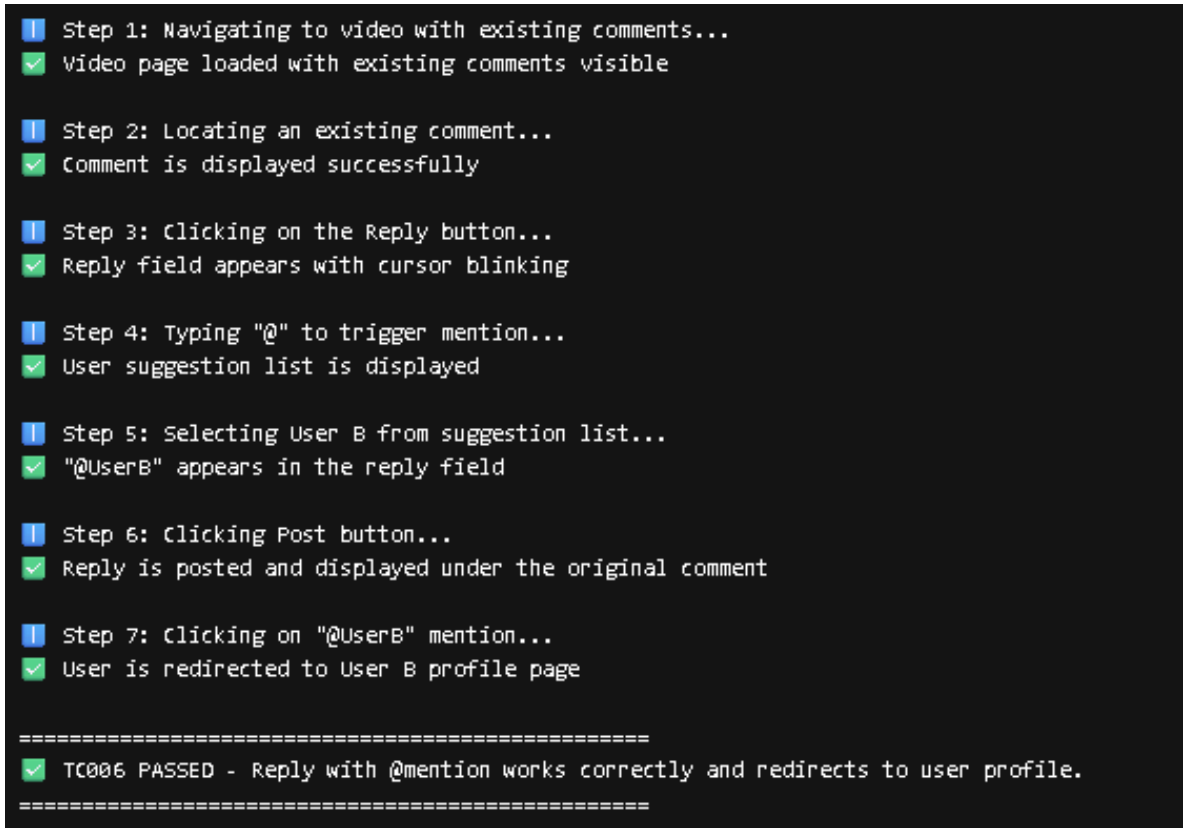


Figure 25: Successful Execution log for Reply To Comment TC004

Test Case Template	
Test Case #: VideoComment_ReplytoComment_TC008	**Test Case Name: Reply to comment**
System: TikTok Web Application	**Module: Video Comment**
Design By: Natalie Koa Hao Yee	**Design Date: 28/03/2026**
Executed By: Ryan Julian Rajesh	**Execution Date: 07/04/2026**
Short Description: **Test and verify replying with an invalid @mention when the user has disabled mentions in privacy settings.**	

Pre-conditions:
User A (reply sender) is logged into TikTok Account
A video with at least one existing comment is open
User C (to be mentioned) has their TikTok account privacy setting for mentions set to "No one"

Step	Action	Expected System Response	Pass/Fail	Comments
1	Locate existing comment	Comment is visible	Pass	
2	Click "Reply" button below the comment	Reply field appears with cursor blinking	Pass	
3	Type : @	User suggestion list appears	Pass	
4	Click on User C	User C is grayed out and Error: "Can't mention this person due to their privacy settings"	Pass	
5	Manually type : @UserC and click "Post"	Comment posts successfully.	Pass	
6	Click on "@UserC"	Not directed to UserC profile page	Pass	

Post-conditions:
System post comment but the @mention is not functional (not clickable, no notification)

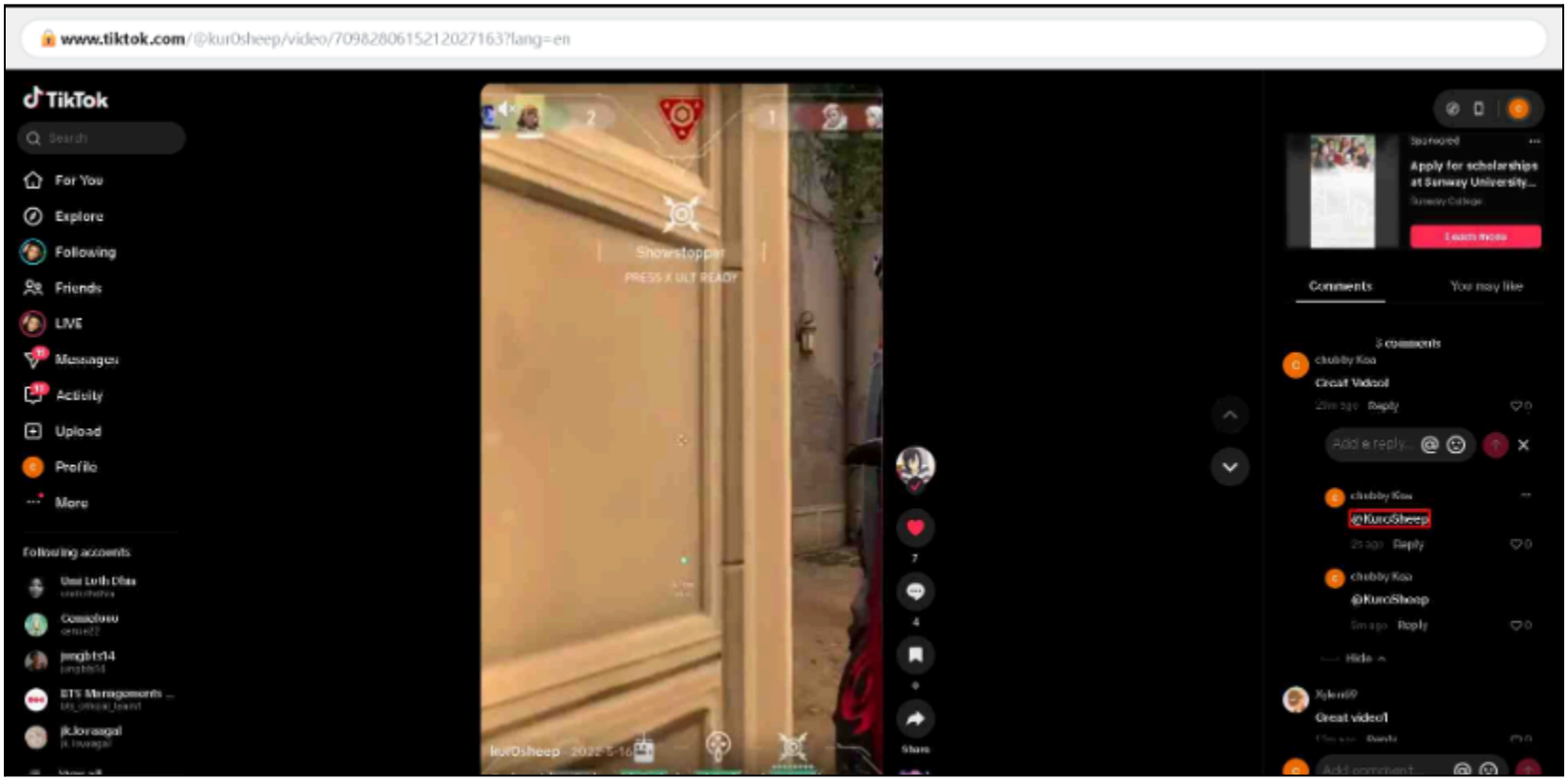


Figure 26: Output for @mention blocked

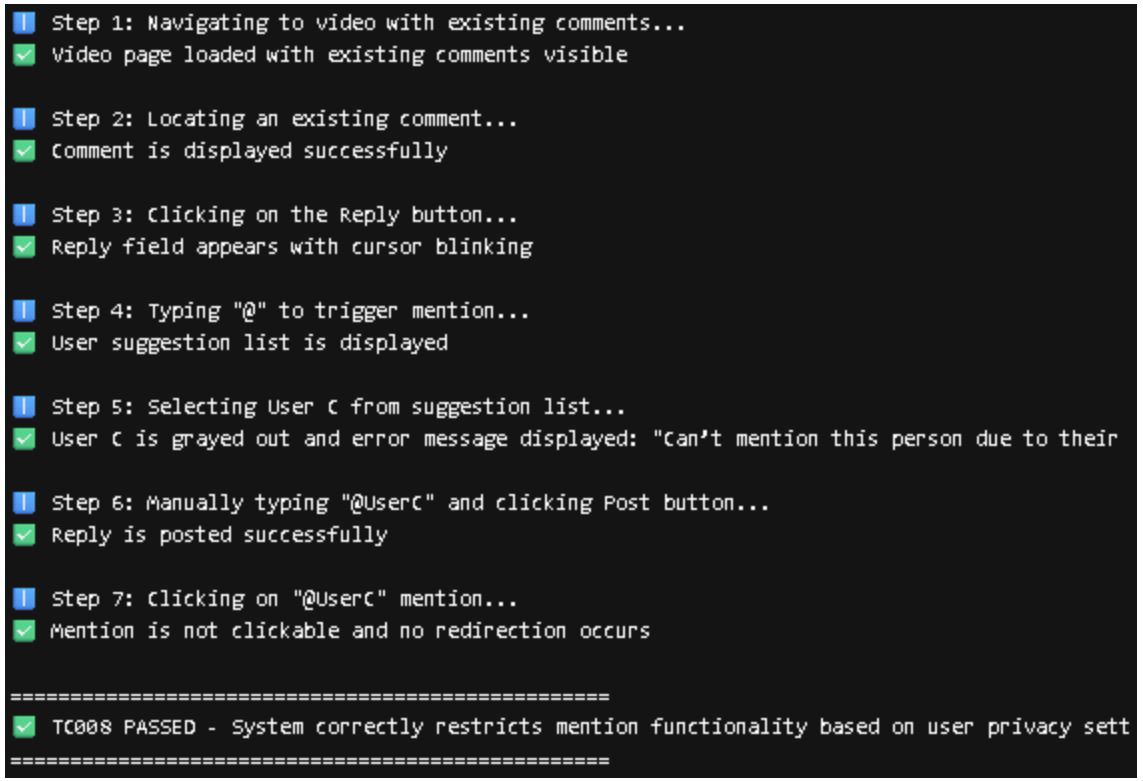


Figure 27: Successful Execution log for Reply To Comment TC005

Module 3 - Tiktok Shop Seller Center

Test Case Template	
Test Case #: **TikTokShopSellerCenter_CreateProductListing_TC001**	**Test Case Name: CreateProductListing**
System: TikTok Web Application	**Module: TikTok Shop Seller Center**
Design By: Lester Liau Jiun Lang	**Design Date: 28/03/2026**
Executed By: Natalie Koa Hao Yee	**Execution Date: 6/04/2026**
Short Description: **Test and verify create product listing with valid details**	

Pre-conditions:
Seller is logged into TikTok Seller Center
Seller has completed shop setup
All required product information is ready

Step	Action	Expected System Response	Pass/Fail	Comments
1	Navigate to "Products" section and click "Add Product"	Product creation form loads	Pass	
2	Enter valid product name, description, and price	Fields accept input correctly	Pass	
3	Upload valid product images	Images upload successfully	Pass	

4	Set inventory quantity to valid number (e.g., 10)	Quantity field accepts input	Pass	
5	Click "Save" or "Publish" button	System shows success message "Product created successfully"	Pass	
6	Verify product appears in product list	New product is visible in the product inventory	Pass	

Post-conditions:
System saves the new product and displays it in the seller's product list

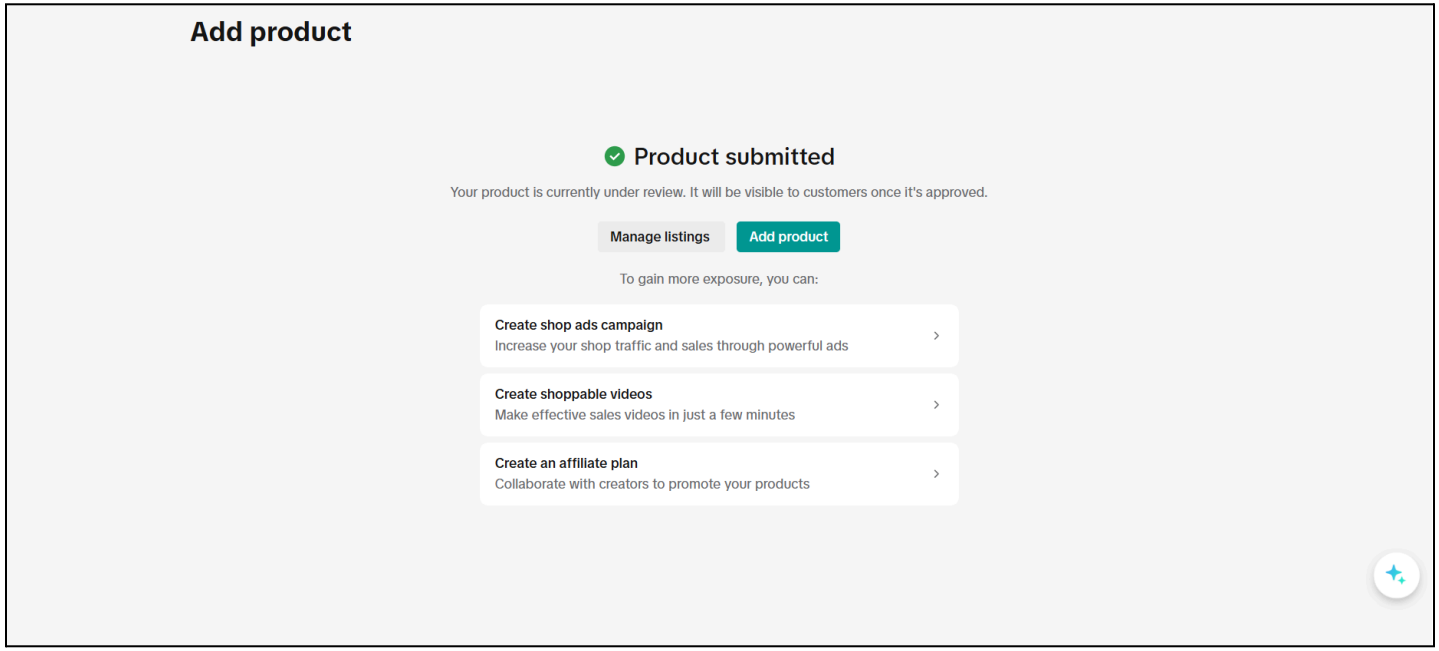


Figure 28: Output for Product Created Successfully

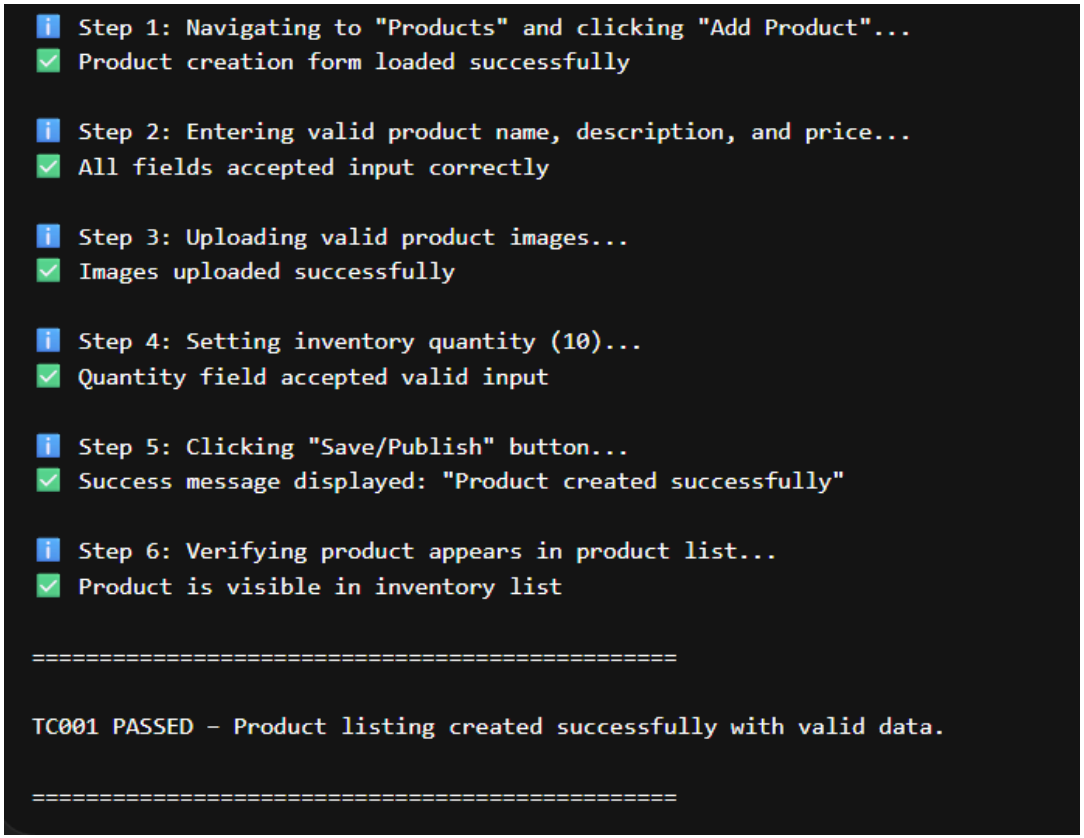


Figure 29:Successful Execution log for Create Product Listing TC001

Test Case Template	
Test Case #: **TikTokShopSellerCenter_CreateProductListing_TC002**	**Test Case Name: CreateProductListing**
System: TikTok Web Application	**Module: TikTok Shop Seller Center**
Design By: Lester Liau Jiun Lang	**Design Date: 10/03/2026**
Executed By: Natalie Koa Hao Yee	**Execution Date: 6/04/2026**
Short Description: **Test and verify create product listing with invalid data (missing required field)**	

Pre-conditions:
Seller is logged into TikTok Seller Center
Product creation form is open

Step	Action	Expected System Response	Pass/Fail	Comments
1	Leave product name field empty	Field may show as required	Pass	
2	Enter valid description and price	Other fields accept input	Pass	
3	Click "Save" or "Publish" button	System displays error message "Product name is required"	Pass	

4	Check if product is saved	Product does not appear in product list	Pass	

Post-conditions:
System does not create the product. Seller remains on the creation form with an error message.

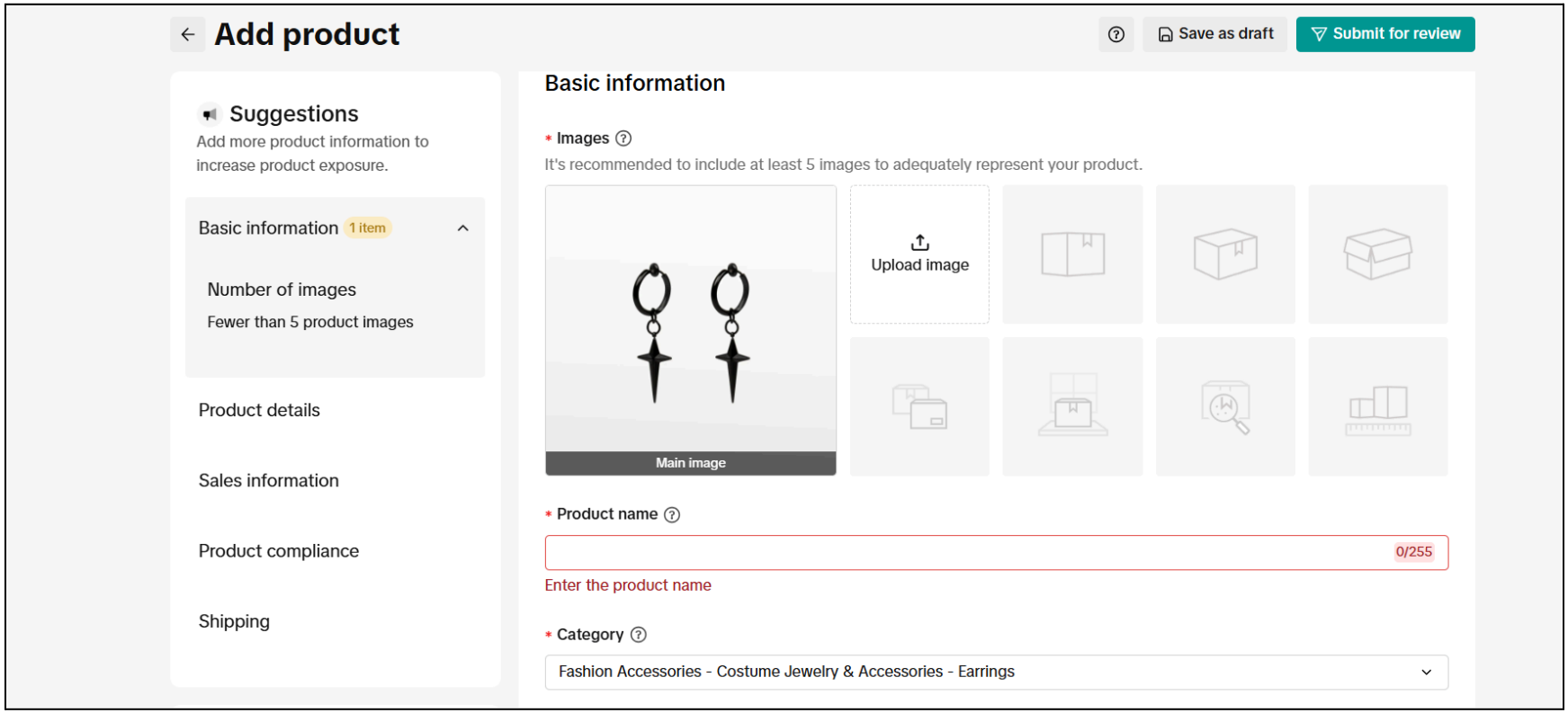


Figure 30: Output for Missing Product Name

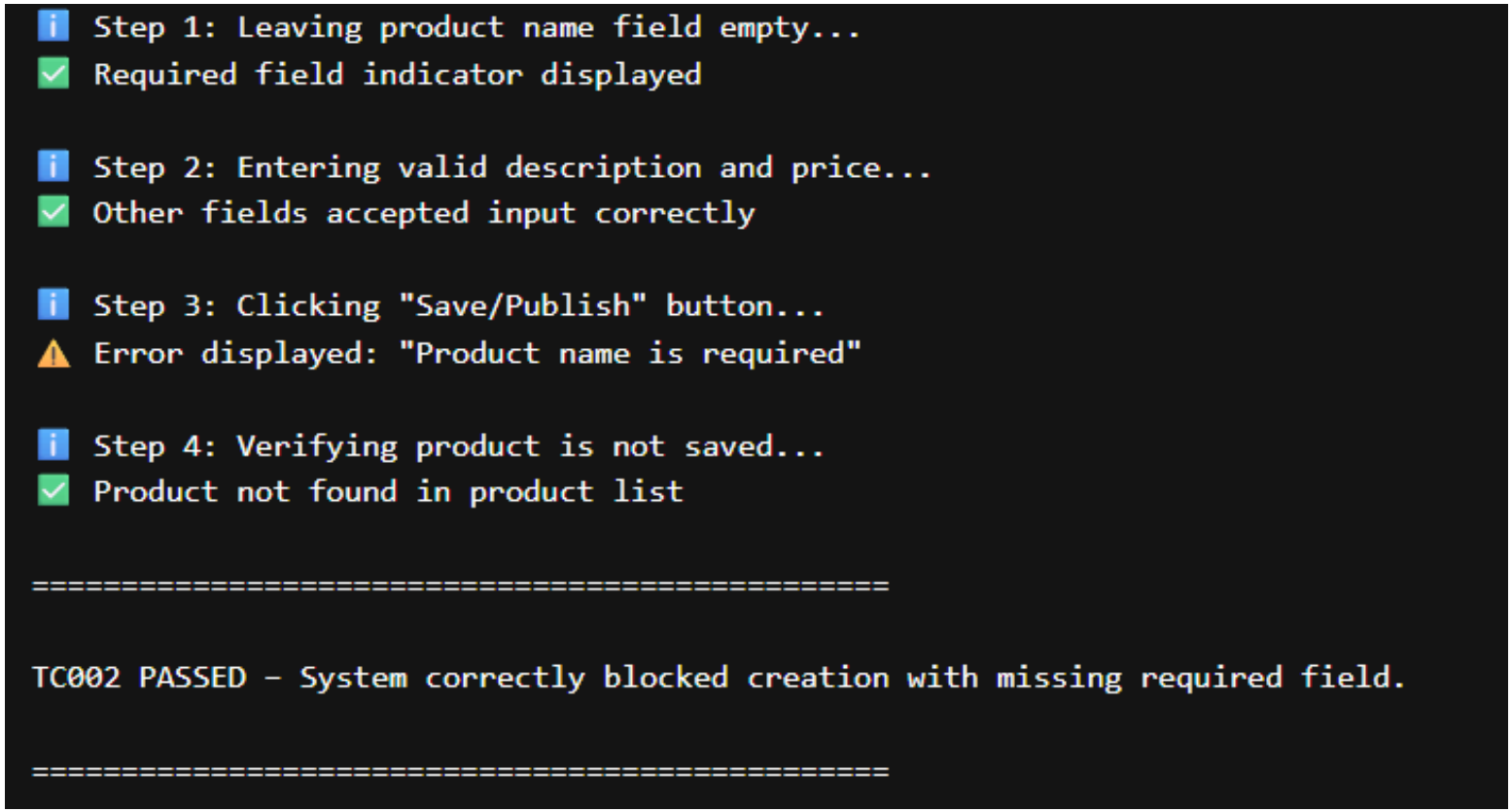


Figure 31: Successful Execution log for Create Product Listing TC002

Test Case Template	
Test Case #: **TikTokShopSellerCenter_CreateProductListing_TC003**	**Test Case Name: CreateProductListing**
System: TikTok Web Application	**Module: TikTok Shop Seller Center**
Design By: Lester Liau Jiun Lang	**Design Date: 10/03/2026**
Executed By: Natalie Koa Hao Yee	**Execution Date: 6/04/2026**
Short Description: **Test and verify create product listing with invalid price format.**	

Pre-conditions: **Seller is logged into TikTok Seller Center** **Product creation form is open**

Step	Action	Expected System Response	Pass/Fail	Comments
1	Enter valid product name and description	Fields accept input	Pass	
2	Enter price with letters or symbols (e.g., "abc" or "$10")	Field may reject non-numeric characters	Pass	
3	If field accepts, click "Save"	Error message appears "Price must be a valid number"	Pass	
4	Verify product not saved	Product does not appear in product list	Pass	

Post-conditions:
System does not create the product due to the invalid price format.

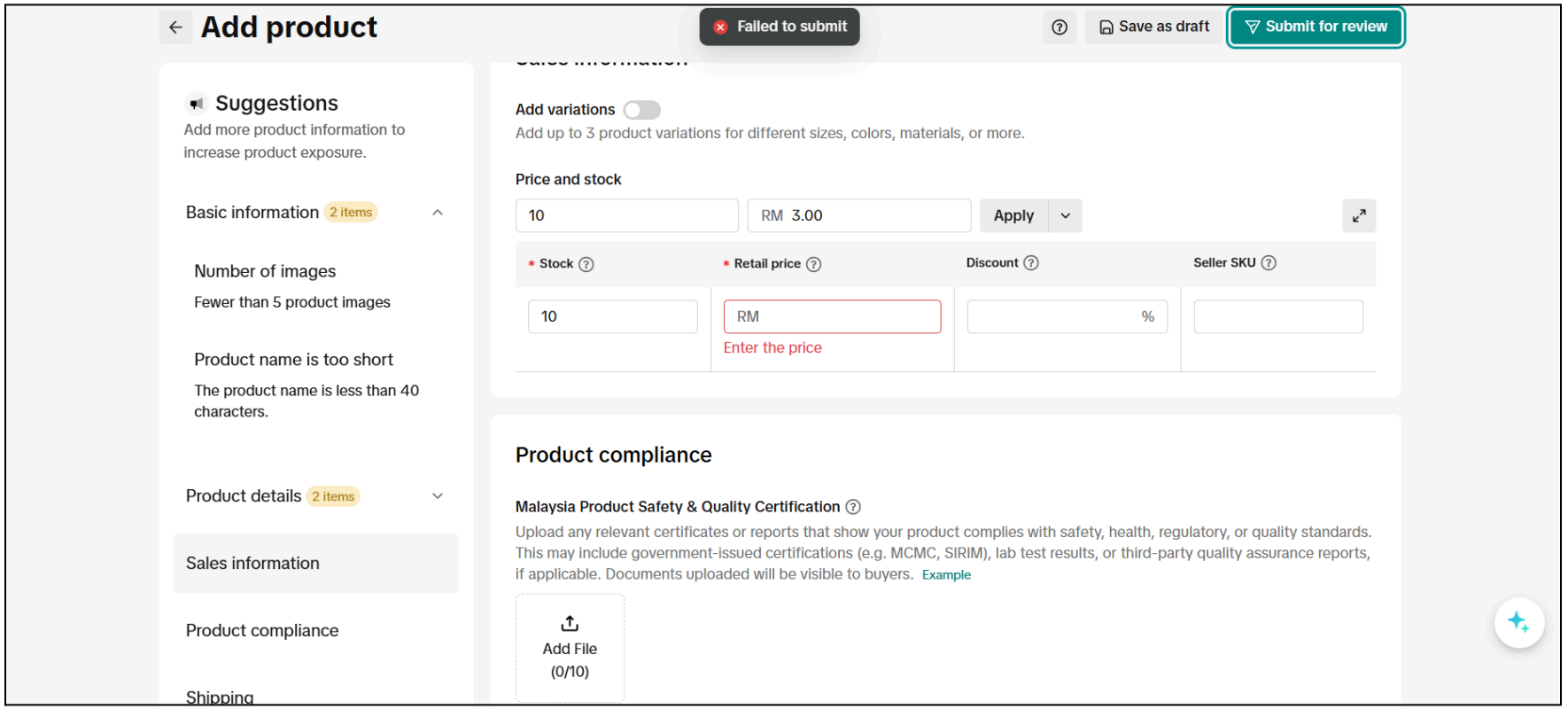


Figure 32: Output for Invalid Price Format

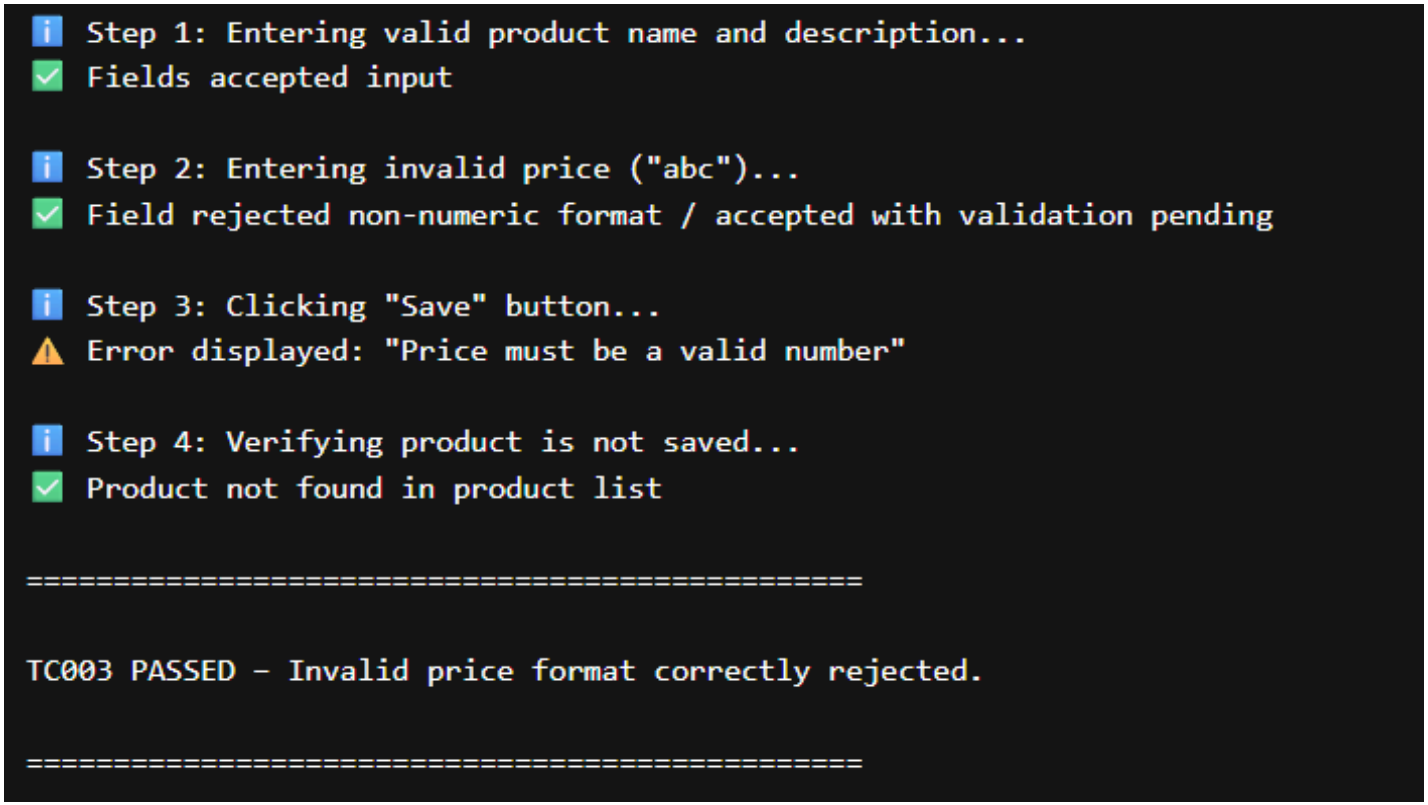


Figure 33: Successful Execution log for Create Product Listing TC003

Test Case Template	
Test Case #: **TikTokShopSellerCenter_CreatePromoCode_TC004**	**Test Case Name: CreatePromoCode**
System: TikTok Web Application	**Module: TikTok Shop Seller Center**
Design By: Lester Liau Jiun Lang	**Design Date: 10/03/2026**
Executed By: Natalie Koa Hao Yee	**Execution Date: 6/04/2026**
Short Description: **Test and verify create valid promo code with minimum spend.**	

Pre-conditions: **Seller is logged into TikTok Seller Center** **At least one product exists in inventory**

Step	Action	Expected System Response	Pass/Fail	Comments
1	Navigate to "Marketing" or "Promo Codes" section	Promo code management page loads	Pass	
2	Click "Create Promo Code"	Promo code creation form opens	Pass	
3	Enter valid promo code name (e.g., "5/5 Sale")	Field accepts input	Pass	
4	Set discount type and amount (e.g., 50% off)	Discount fields work correctly	Pass	

5	Set minimum spend condition (e.g., RM5)	Field accepts minimum spend value	Pass	
6	Set valid date range	Date picker works correctly	Pass	
7	Click "Save" or "Create"	Success message appears "Promo code created successfully"	Pass	
8	Verify promo code appears in list	New promo code visible in promo code list	Pass	

Post-conditions:
System saves the promo code and makes it available for customers to use.

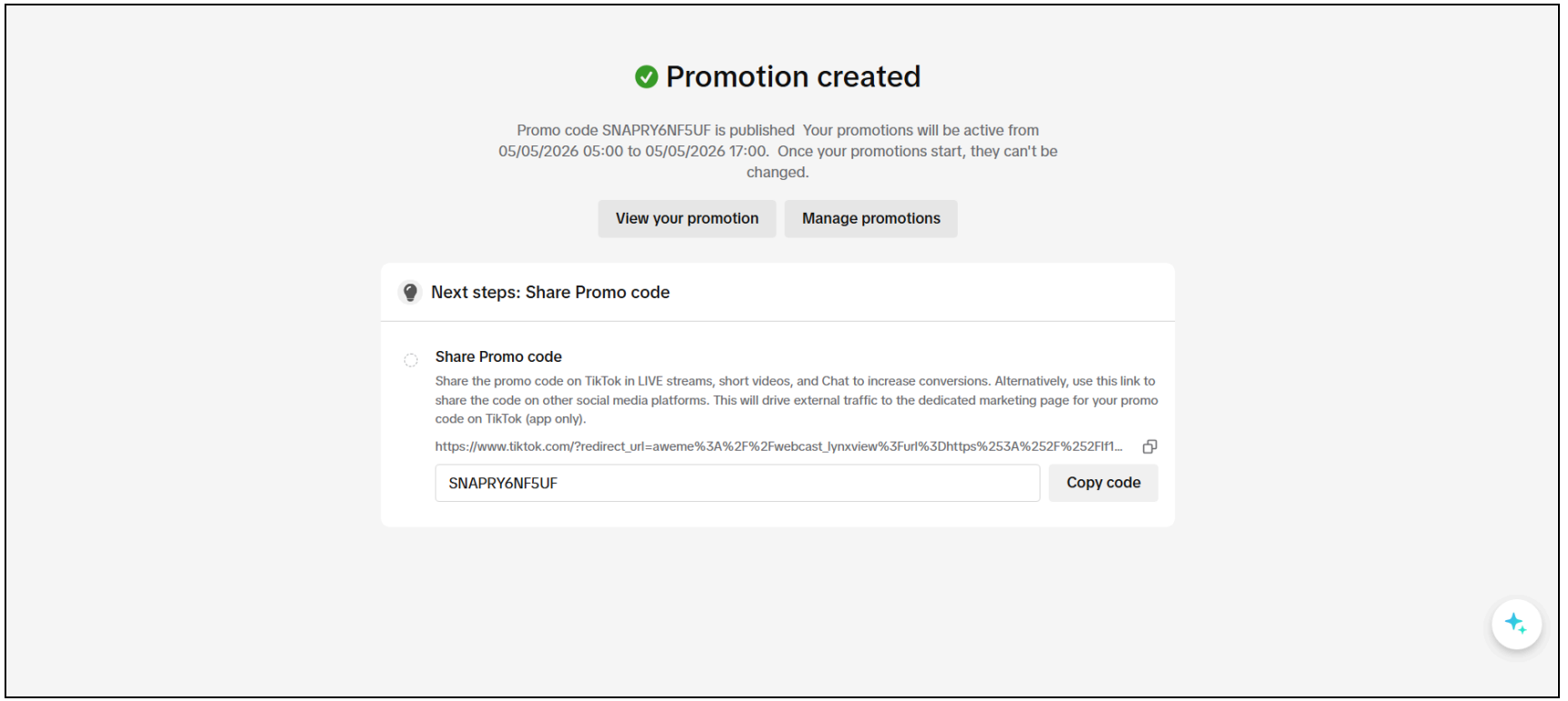


Figure 34: Output for Promotion Code Created Successfully

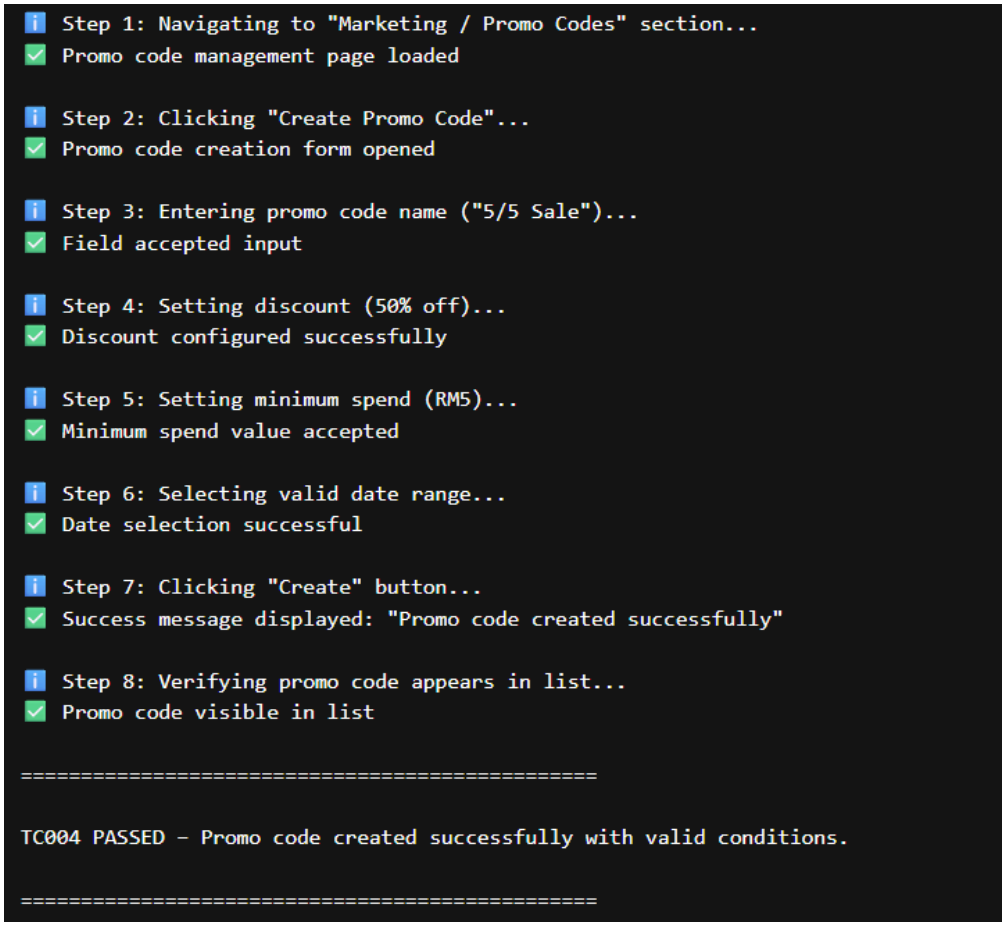


Figure 35: Successful Execution log for Create Promo Code TC004

Test Case Template	
Test Case #: TikTokShopSellerCenter_CreatePromoCode_TC005	**Test Case Name: CreatePromoCode**
System: TikTok Web Application	**Module: TikTok Shop Seller Center**
Design By: Lester Liau Jiun Lang	**Design Date: 10/03/2026**
Executed By: Natalie Koa Hao Yee	**Execution Date: 6/04/2026**
Short Description: **Test and verify create promo code with invalid date range.**	

Pre-conditions: **Seller is logged into TikTok Seller Center** **Promo code creation form is open**

Step	Action	Expected System Response	Pass/Fail	Comments
1	Enter valid promo code name and discount details	Fields accept input	Pass	
2	Set start date to tomorrow	Date selected	Pass	
3	Set end date to next year (more than 365 days)	Date selected	Pass	
4	Click "Save" or "Create"	System displays error message "The promotion must be less than 365 days"	Pass	

5	Verify promo code not saved	Promo code does not appear in promo code list	Pass	

Post-conditions:
System does not create the promo code. Seller remains on the form with an error message.

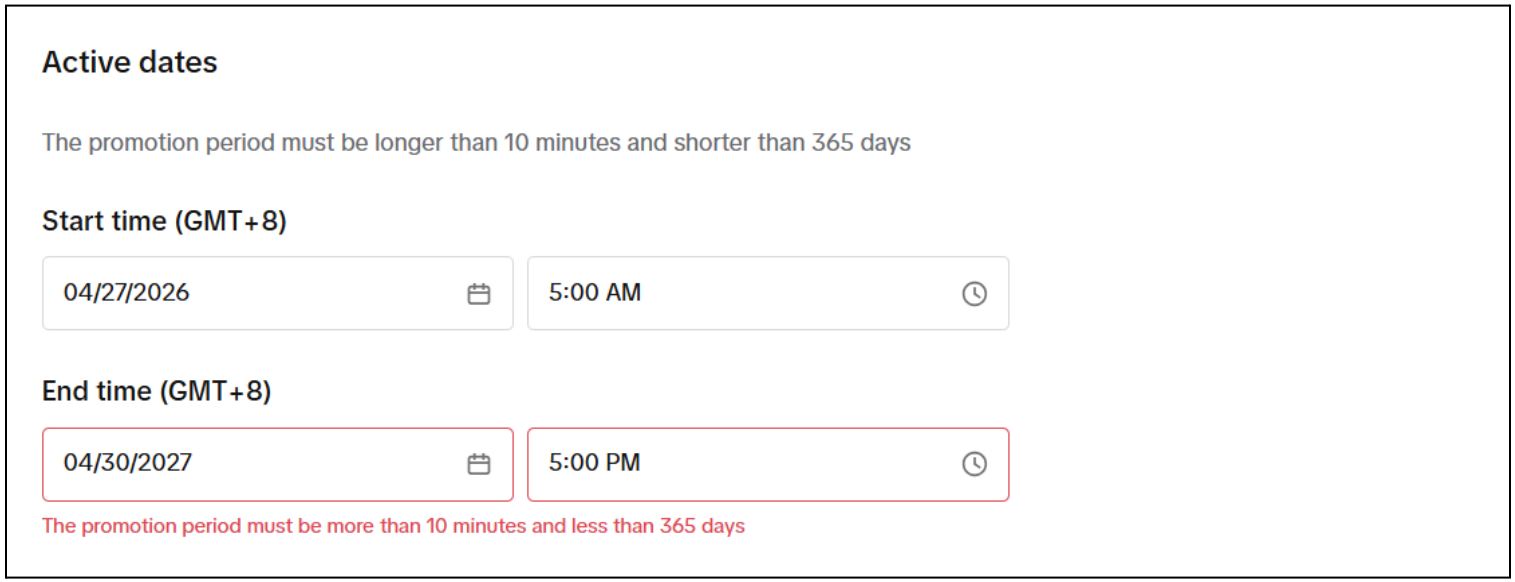


Figure 36: Output for set date more than 365 days

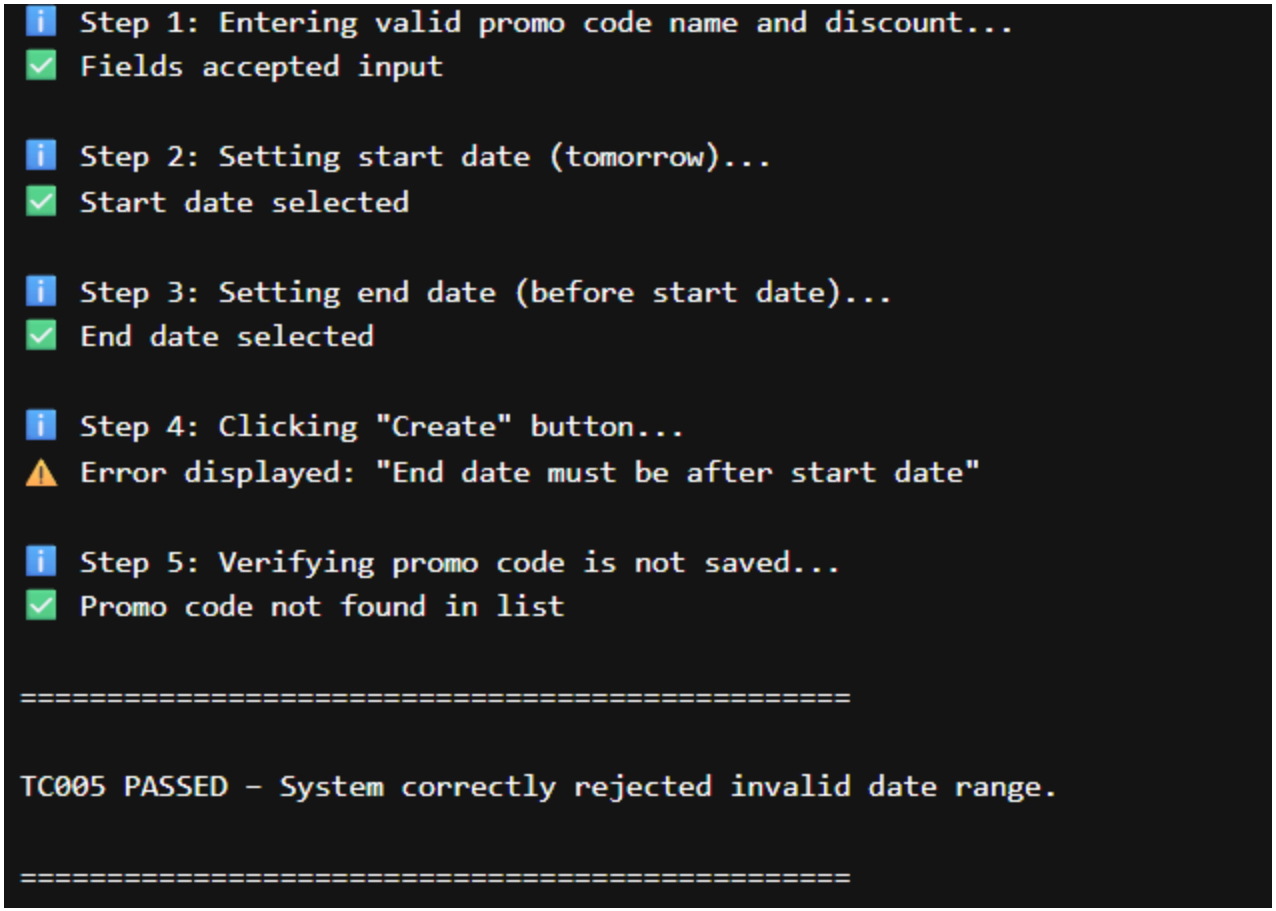


Figure 37: Successful Execution log for Create Promo Code TC005

Module 4 - Video Creation

Test Case Template	
Test Case #: VideoCreation_UploadVideo_TC001	**Test Case Name: Upload Video**
System: TikTok Web Application	**Module: Video Creation**
Design By: Tiong Chang Vui	**Design Date: 28/03/2026**
Executed By: Lester Liau Jiun Lang	**Execution Date: 12/04/2026**
Short Description: **Test and verify the upload with valid video file (MP4) and valid size**	

Pre-conditions: **User is logged into TikTok account** **User is on the video upload page**

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click "Select Video" button	File explorer opens	Pass	
2	Select video file: video.mp4	File name appears, upload progress starts	Pass	
3	Wait for upload to complete	Progress bar reaches 100%, the system responses uploaded	Pass	

Post-conditions:
System uploads the video successfully and displays the video preview.

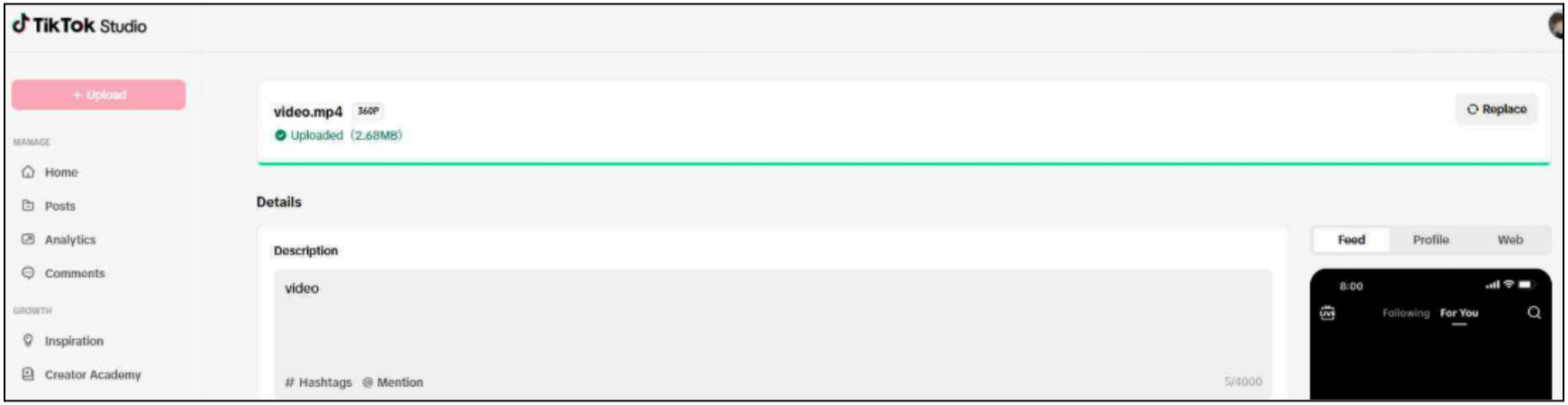


Figure 38: Output for Video Uploaded Successfully

```
ℹ Step 1: Navigating to TikTok Studio upload page...
✅ Navigation successful
✅ Login verified / session active
ℹ Step 2: Locating file input element...
✅ File input element located
ℹ Step 3: Sending video file path to input...
✅ File path sent successfully (OS dialog bypassed)
ℹ Step 4: Waiting for upload to complete...
✅ Upload completed

==================================================
✅ TC001 PASSED - Video uploaded successfully.
==================================================
```

Figure 39: Successful Execution log for Upload Video TC00

Test Case Template	
Test Case #: VideoCreation_UploadVideo_TC002	**Test Case Name: Upload Video**
System: TikTok Web Application	**Module: Video Creation**
Design By: Tiong Chang Vui	**Design Date: 28/03/2026**
Executed By: Lester Liau Jiun Lang	**Execution Date: 12/04/2026**
Short Description: **Test and verify the upload with invalid file format PDF (TikTok Web only accepts MP4/MOV video files**	

Pre-conditions:
User is logged into TikTok account
User is on the video upload page

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click "Select Video" button	File explorer opens	Pass	
2	In the file picker dialog, change the file type filter from “Video Files” to “All Files” , then select video file : video.pdf	Error message appears "Unsupported file format. Please upload MP4 or other video formats."	Pass	
3	Verify no video is uploaded	Upload area remains empty	Pass	

Post-conditions:
System rejects the invalid file, and the user can try again with a valid file.

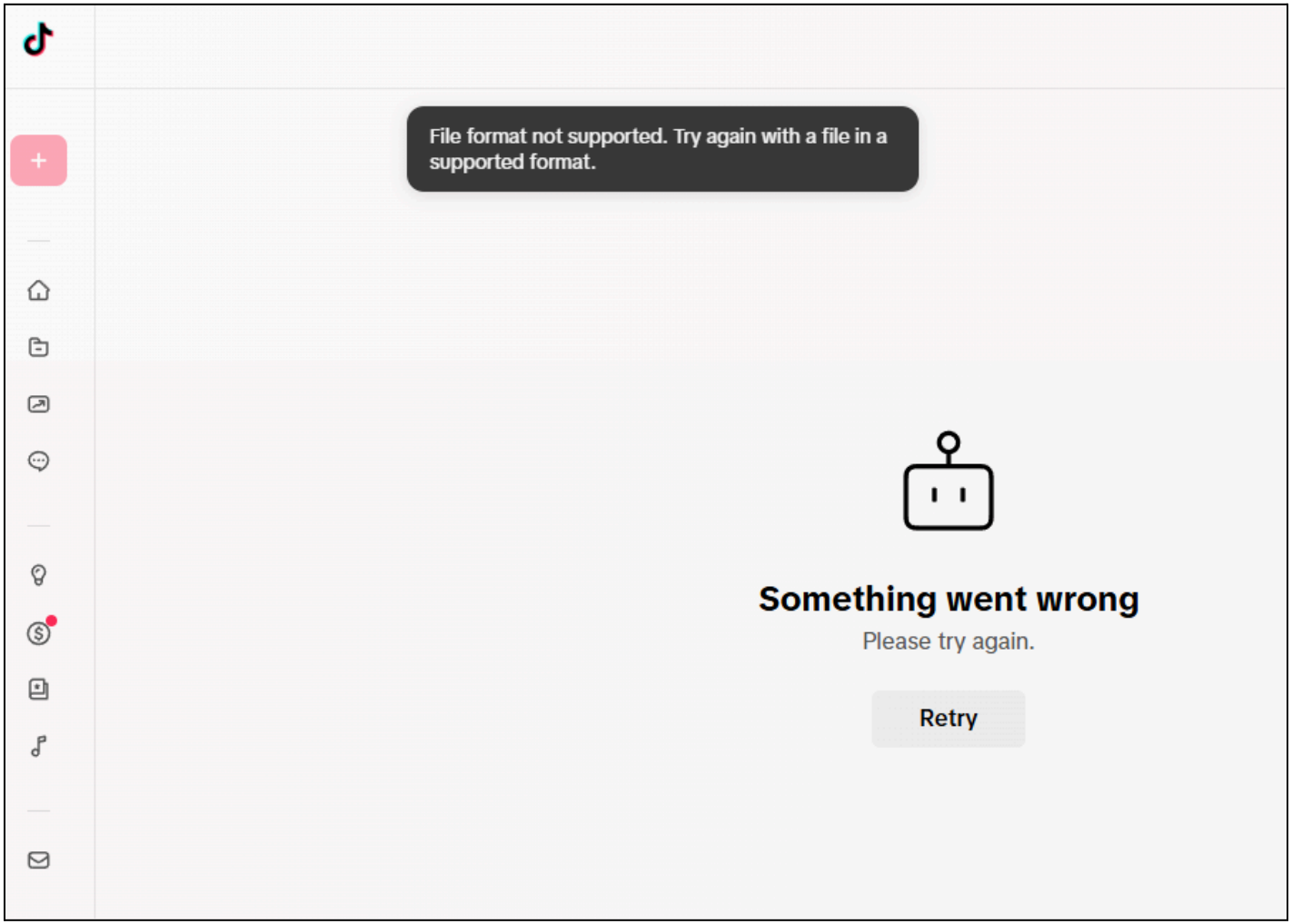


Figure 40: Output for Unsupported Format Error

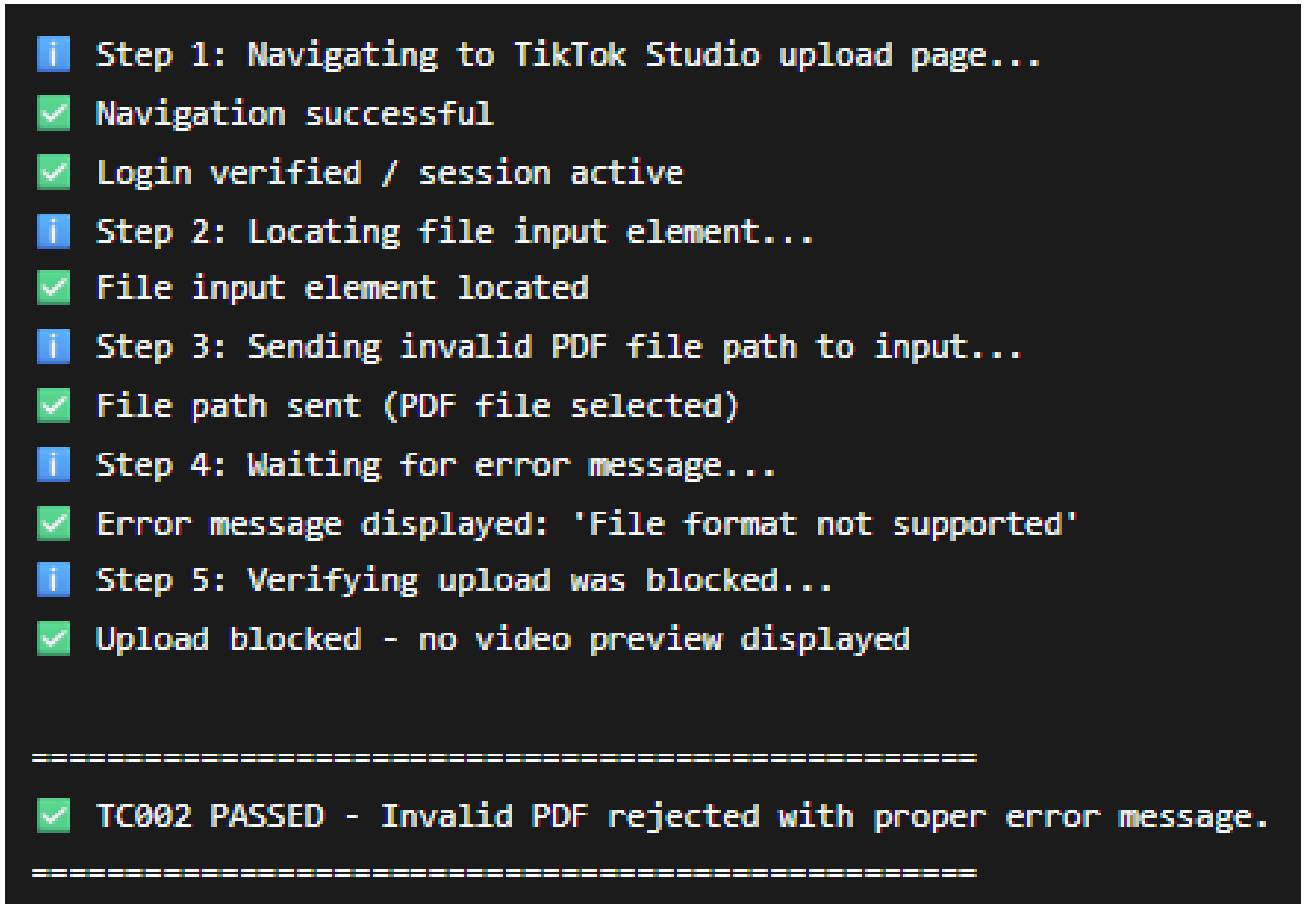


Figure 41: Successful Execution log for Upload Video TC002

Test Case Template	
Test Case #: VideoCreation_ScheduleVideo_TC003	**Test Case Name: Schedule Video**
System: TikTok Web Application	**Module: Video Creation**
Design By: Tiong Chang Vui	**Design Date: 28/03/2026**
Executed By: Lester Liau Jiun Lang	**Execution Date: 12/04/2026**
Short Description: **Schedule video with valid future date and time**	

Pre-conditions:
User is logged into TikTok account
The user is on the video posting page
Valid MP4 video file has been uploaded

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click the “Schedule” radio button.	Date and time picker appears.	Pass	
2	Select Date: 12/04/2026	Dates are accepted.	Pass	
3	Select Time: 12:45 PM	Time is accepted	Pass	
4	Click “Schedule” button	Success message “Video scheduled” appears. The video will appear in	Pass	

		the list with the selected schedule date/time.		

Post-conditions:
System schedules the video and will automatically post it at the selected date and time.

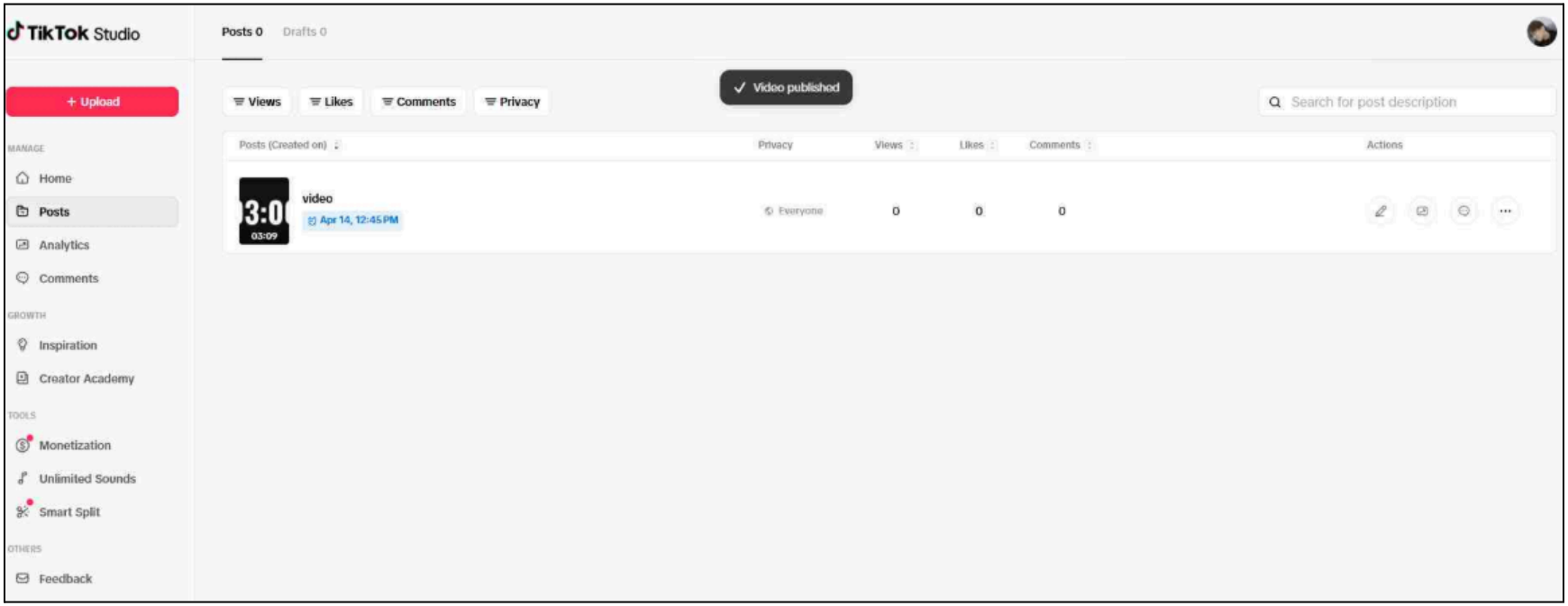


Figure 42: Output for Video Scheduled Successfully

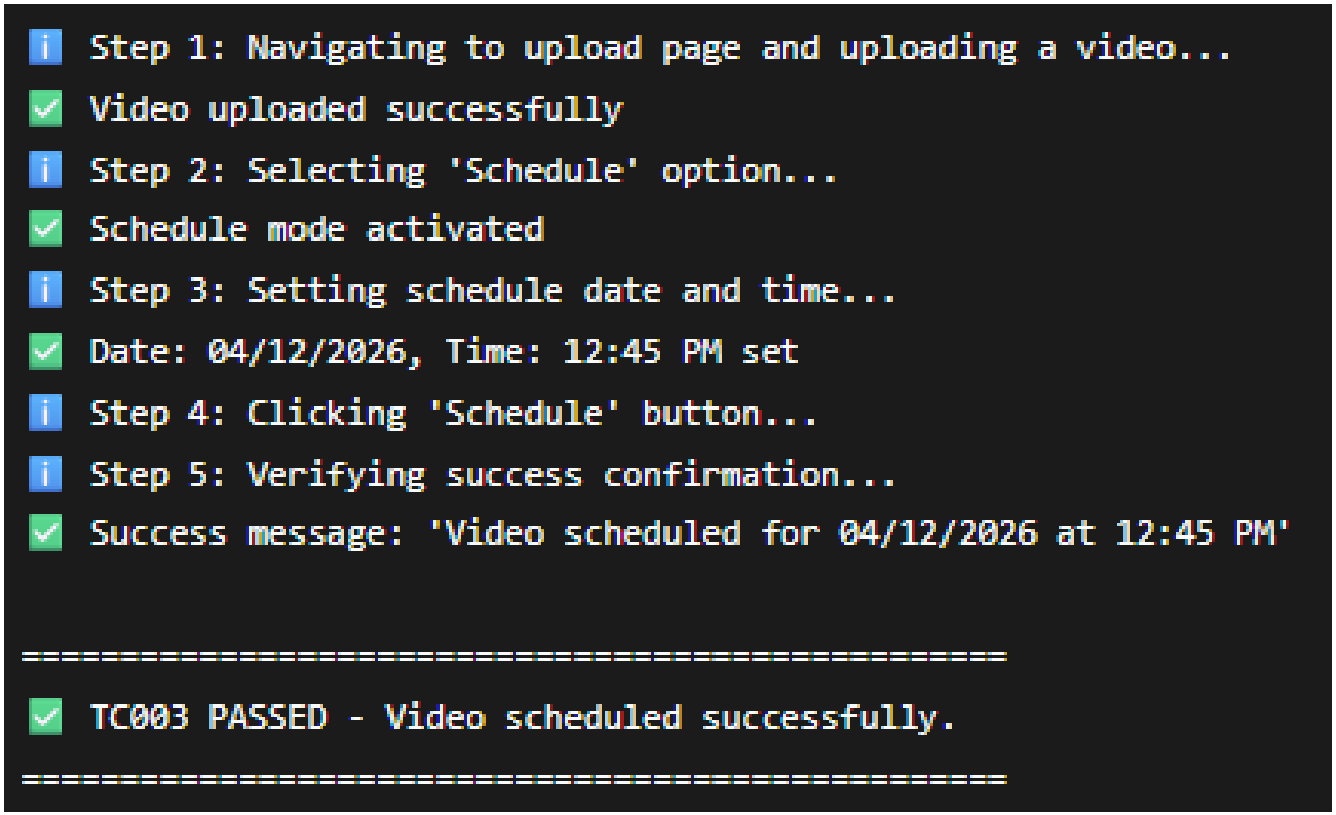


Figure 43: Successful Execution log for Schedule Video TC003

Test Case Template	
Test Case #: VideoCreation_ScheduleVideo_TC004	**Test Case Name: Schedule Video**
System: TikTok Web Application	**Module: Video Creation**
Design By: Tiong Chang Vui	**Design Date: 28/03/2026**
Executed By: Lester Liau Jiun Lang	**Execution Date: 12/04/2026**
Short Description: **Test and verify the schedule video function with invalid time** **(5 minutes from current time, e.g., if current time is 14:00, fails minimum 15-minute advance)**	

Pre-conditions: **User is logged into TikTok account** **The user is on the video posting page** **Valid MP4 video file has been uploaded**

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click the “Schedule” radio button.	Date and time picker appears.	Pass	
2	Select date: 12/04/2026	Date accepted.	Pass	
3	Select time: 12:40 PM	The system prevents selection or displays a warning.	Pass	
4	Verify that the schedule cannot be confirmed.	Warning appears: “Schedule at least 15 minutes in advance” and the	Pass	

		“Schedule” button is disabled or the action is blocked.		

Post-conditions: **System does not schedule the video and warns the user to choose a valid future time.**

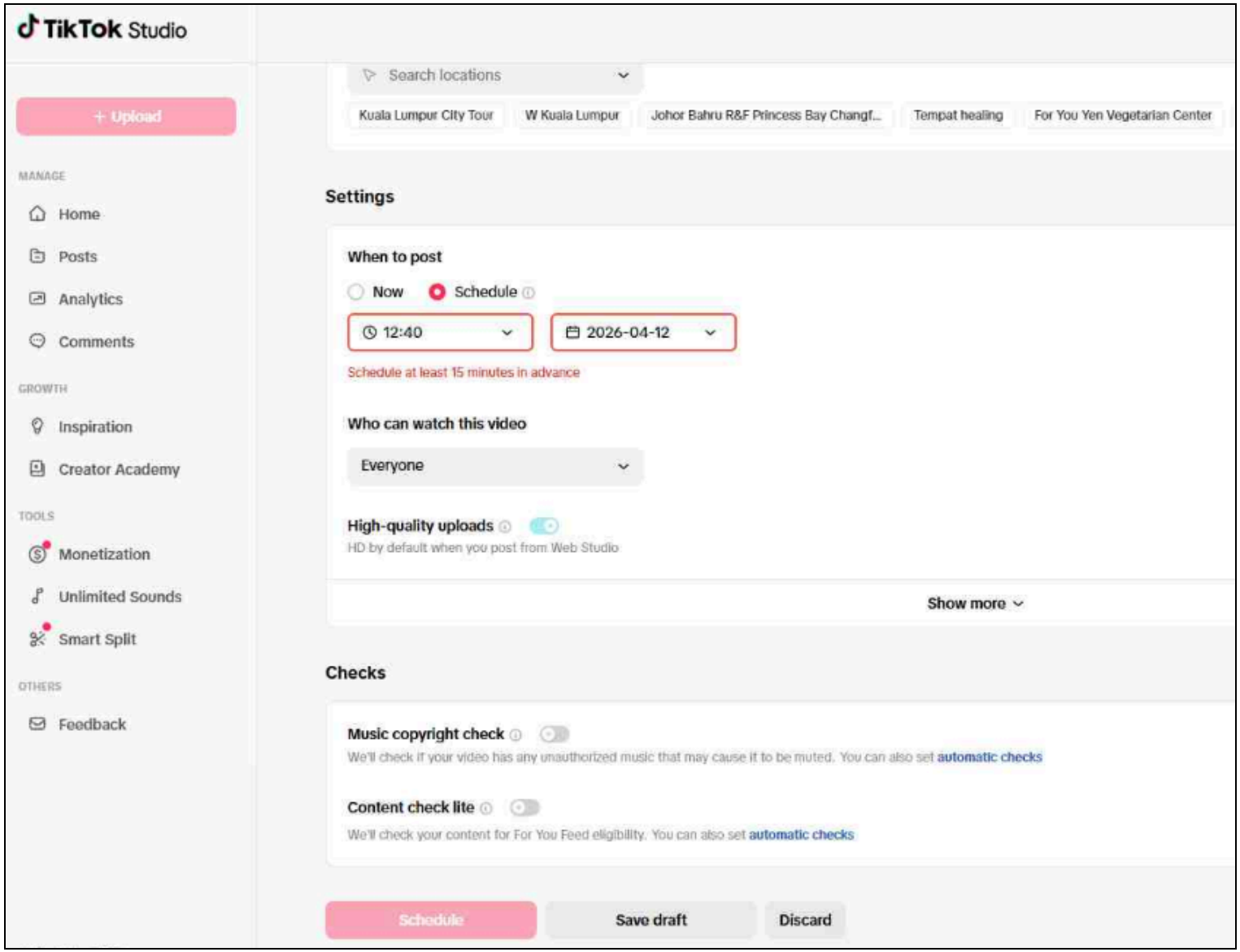


Figure 44: Output for Time too soon warning

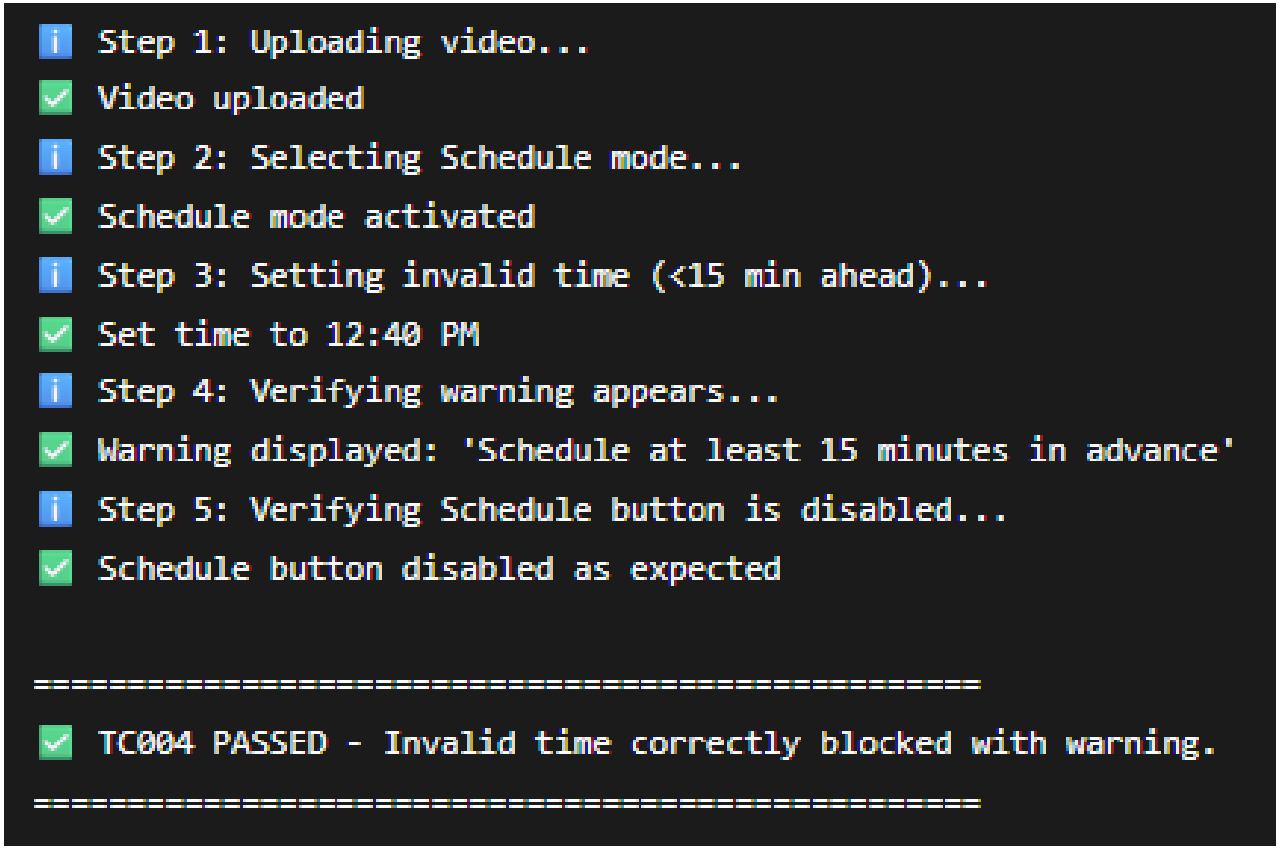


Figure 45: Successful Execution log for Schedule Video TC004

Test Case Template	
Test Case #: VideoCreation_ScheduleVideo_TC005	**Test Case Name: Schedule Video**
System: TikTok Web Application	**Module: Video Creation**
Design By: Tiong Chang Vui	**Design Date: 28/03/2026**
Executed By: Lester Liau Jiun Lang	**Execution Date: 12/04/2026**
Short Description: **Test and verify the schedule video function with invalid date** **(31 days from 7 April 2026 – exceeds 30-day maximum schedule window)**	

Pre-conditions:
User is logged into TikTok account
The user is on the video posting page
Valid MP4 video file has been uploaded

Step	Action	Expected System Response	Pass/Fail	Comments
1	Click the “Schedule” radio button.	Date and time picker appears.	Pass	
2	Select date: 12/05/2026	The system prevents selection or displays an error/warning.	Pass	
3	Verify that the schedule cannot be confirmed.	Warning appears: “Posts can be scheduled up to 30 days ahead” and the “Schedule” button is disabled or the action is blocked.	Pass	

Post-conditions:
System does not schedule the video and warns the user to choose a date within the 30-day.

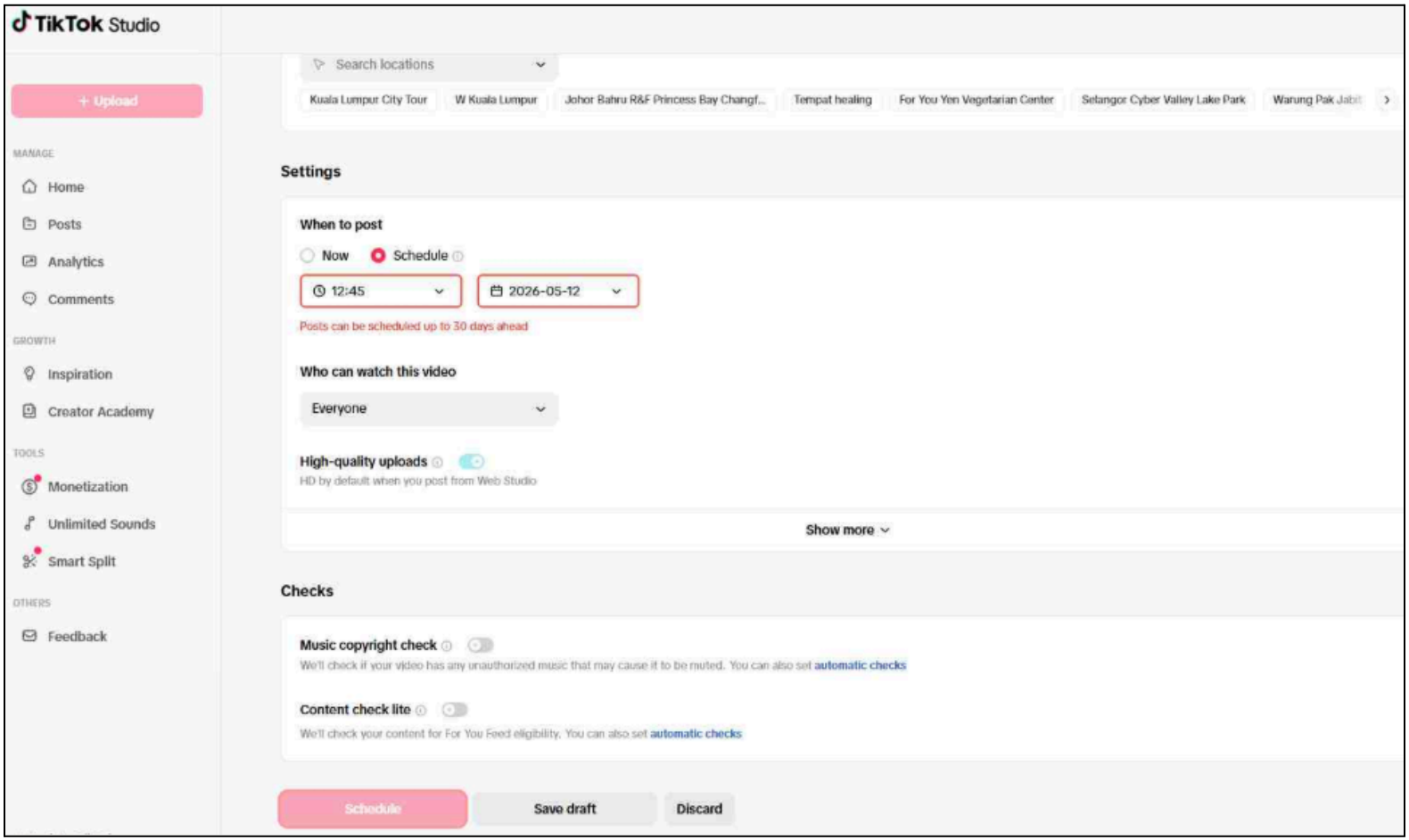


Figure 46: Output for Date Too Far Warning

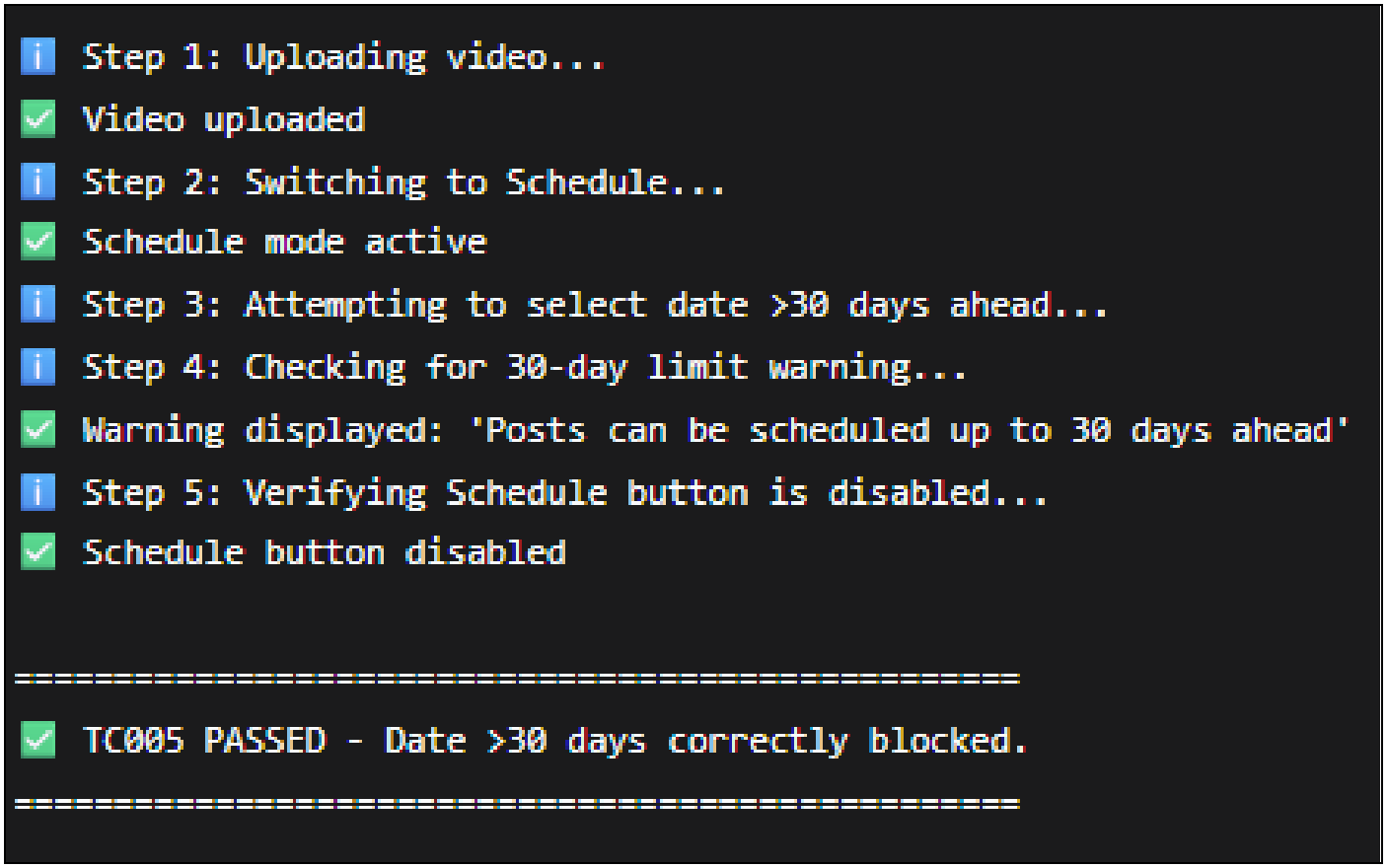


Figure 47: Successful Execution log for Schedule Video TC005

Module 5 - Messaging

Test Case Template	
Test Case #: Messaging_SendMessage_TC001	**Test Case Name: Send Message**
System: TikTok Web Application	**Module: Messaging**
Design By: Wong Kai Lit Kerry	**Design Date: 28/03/2026**
Executed By: Tiong Chang Vui	**Execution Date:**
Short Description: Test and verify send a valid text message to another user	

Pre-conditions:
User A is logged in
User A has an existing chat with User B.

Step	Action	Expected System Response	Pass/Fail	Comments
1	In the chat input field, type a message: Hi	Text appears in the field.	Pass	
2	Press Enter or click the send button.	Message is sent and appears in the chat for User A.	Pass	

Post-conditions:
System displays the message in the chat for both User A and User B.

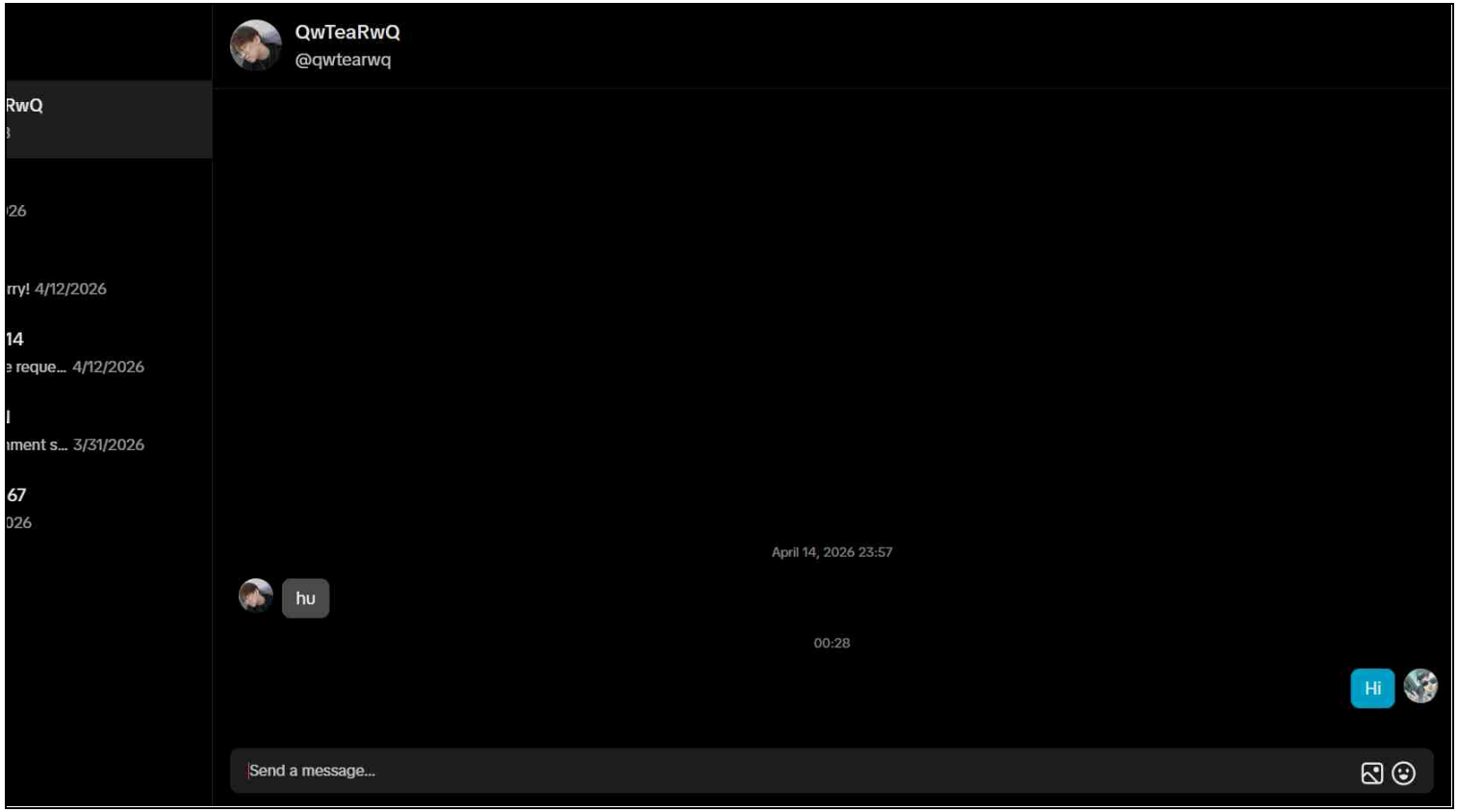


Figure 48: Output for Valid Message Sent

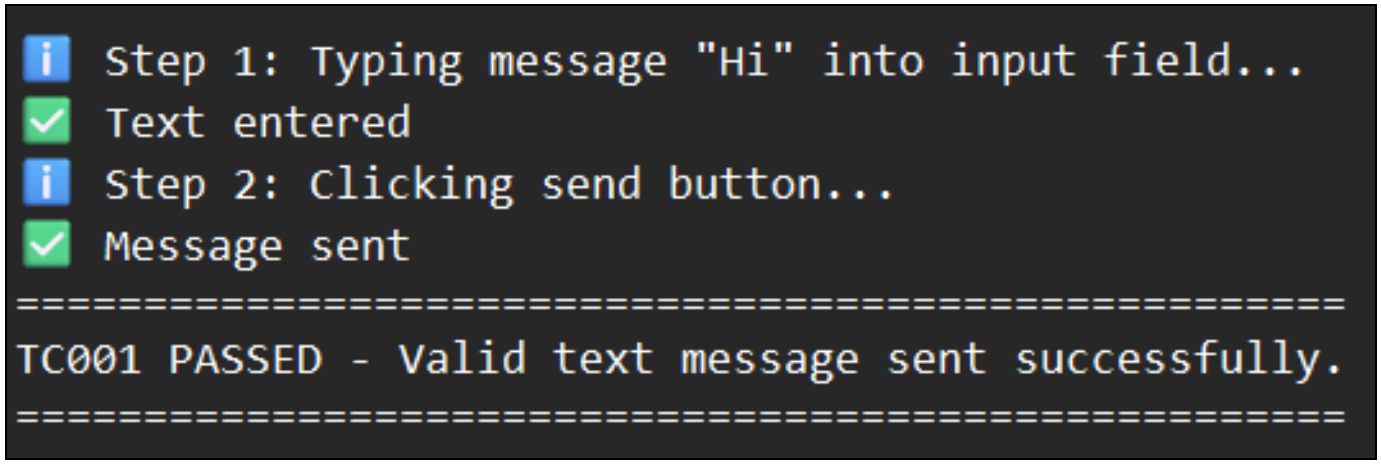


Figure 49: Successful Execution log for Send Message TC001

Test Case Template	
Test Case #: Messaging_SendMessage_TC002	**Test Case Name: Send Message**
System: TikTok Web Application	**Module: Messaging**
Design By: Wong Kai Lit Kerry	**Design Date: 28/03/2026**
Executed By: Tiong Chang Vui	**Execution Date:**
Short Description: Test and verify send an invalid empty message	

Pre-conditions:
The user is logged in and in a chat.

Step	Action	Expected System Response	Pass/Fail	Comments
1	Leave the message input field empty.	The field is empty.	Pass	
2	Attempt to send by pressing Enter or clicking send.	The send button is disabled, or nothing happens, and no message is sent.	Pass	
3	Check chat history.	No empty message appears.	Pass	

Post-conditions:
System does not send any message and the chat remains unchanged.

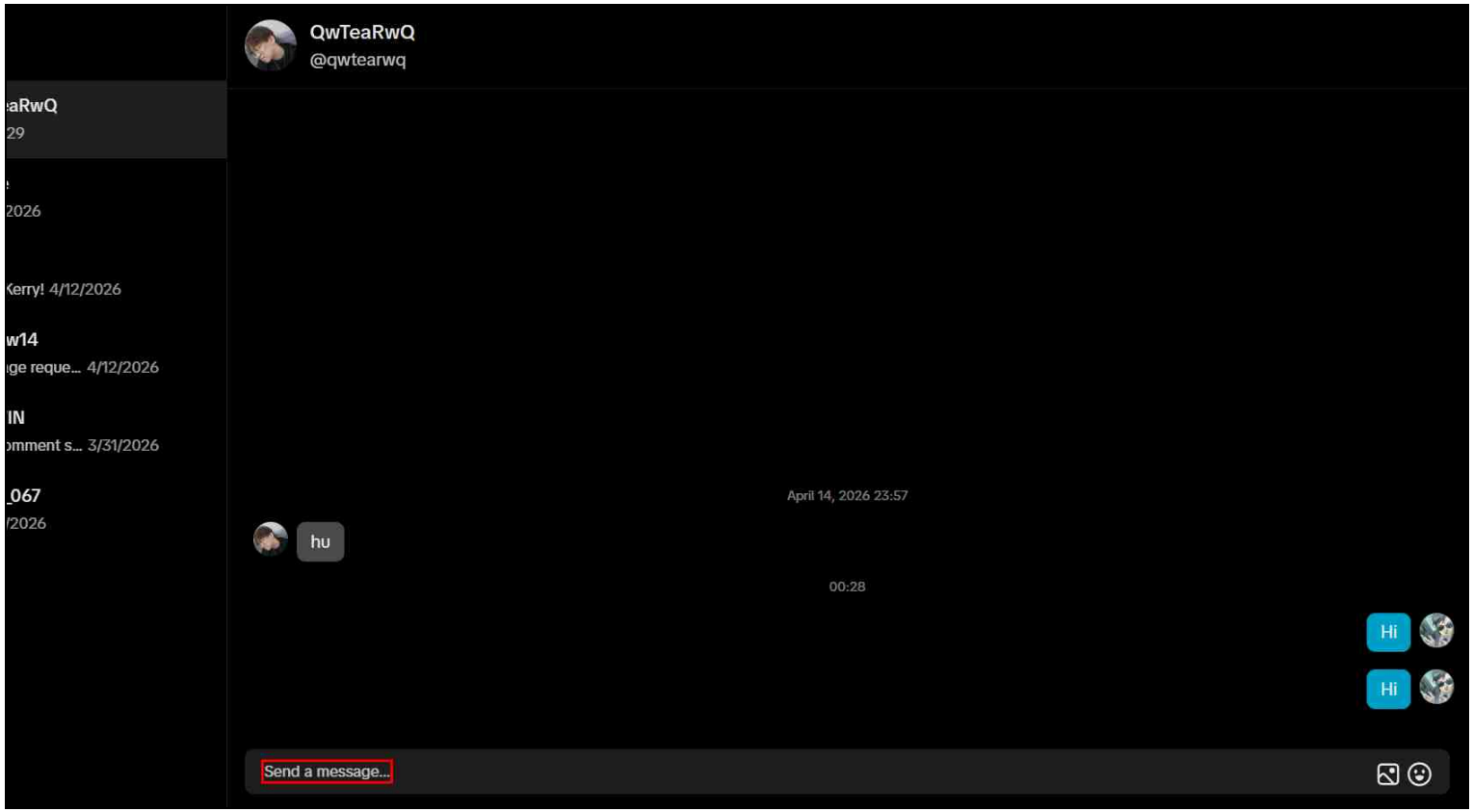


Figure 50: Output for Empty Message

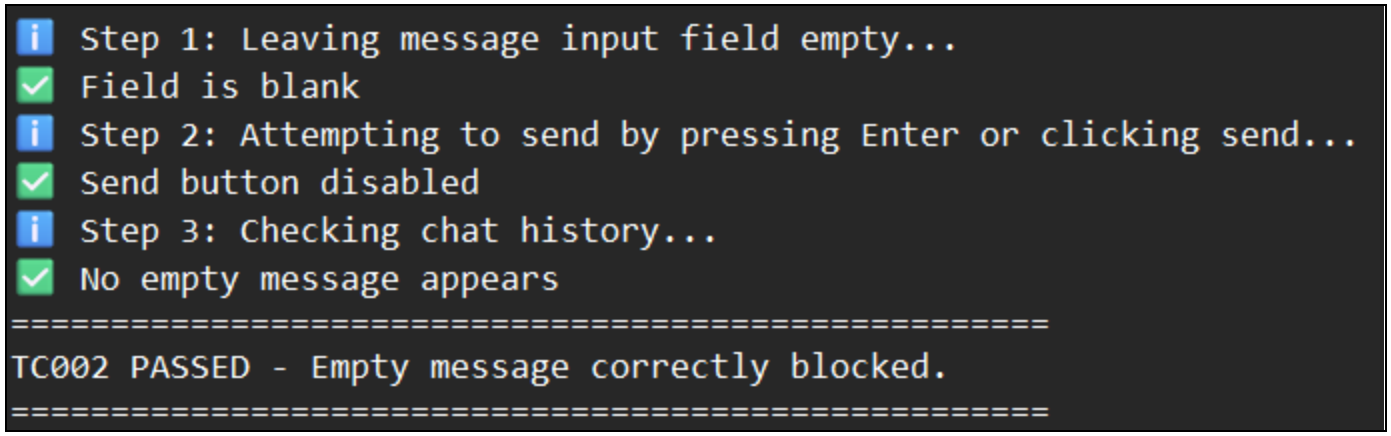


Figure 51: Successful Execution log for Send Message TC002

Test Case Template	
Test Case #: Messaging_SendMessage_TC003	**Test Case Name: Send Message**
System: TikTok Web Application	**Module: Messaging**
Design By: Wong Kai Lit Kerry	**Design Date: 28/03/2026**
Executed By: Tiong Chang Vui	**Execution Date:**
Short Description: Test and verify send an invalid message containing only spaces	

Pre-conditions:
The user is logged in and in a chat.

Step	Action	Expected System Response	Pass/Fail	Comments
1	Type several spaces into the message field.	The field appears blank with no visible characters.	Pass	
2	Attempt to send.	The send button is disabled, or the message is not sent or may be ignored.	Pass	
3	Verify no message appears.	Chat history does not show any blank message.	Pass	

Post-conditions:
System treats the input as empty and does not send any message.

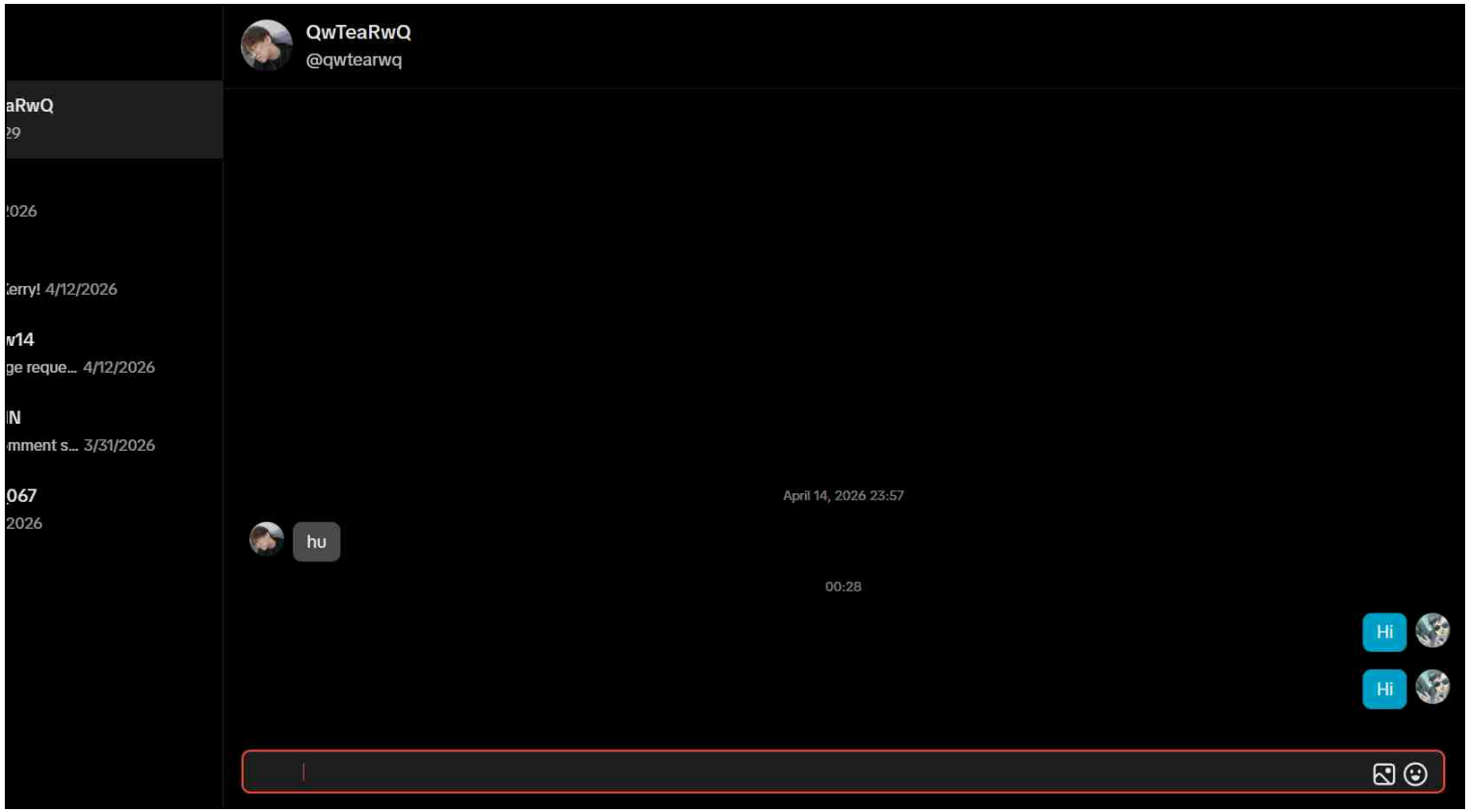


Figure 52: Output for Spaces only

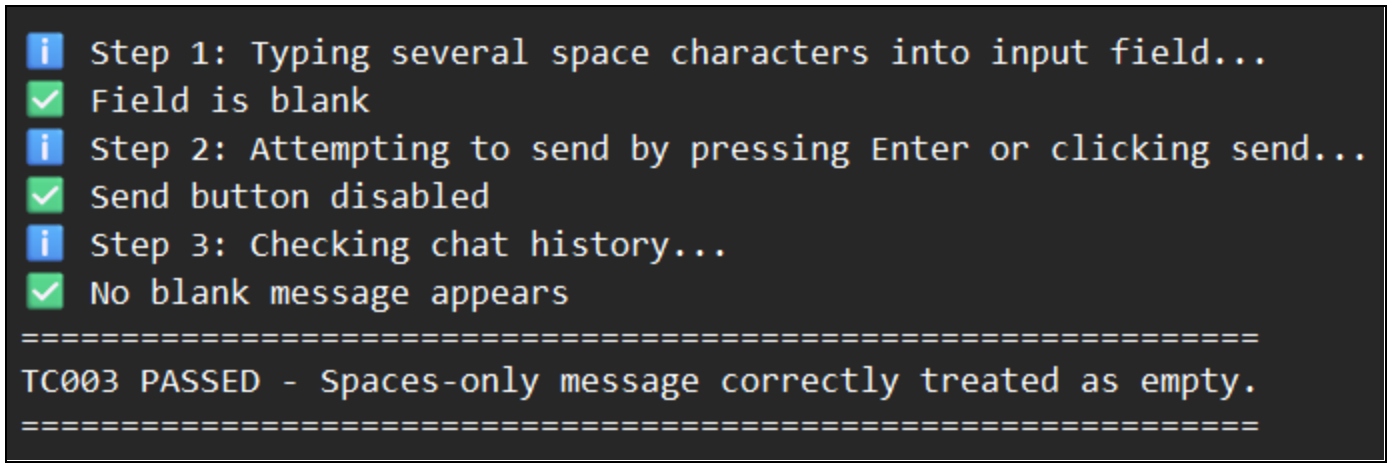


Figure 53: Successful Execution log for Send Message TC003

Test Case Template	
Test Case #: Messaging_ManageMessageRequest_TC004	**Test Case Name: Manage Message Request**
System: TikTok Web Application	**Module: Messaging**
Design By: Wong Kai Lit Kerry	**Design Date: 28/03/2026**
Executed By: Tiong Chang Vui	**Execution Date:**
Short Description: Test and verify that a valid message request from a non-follower can be accepted.	

Pre-conditions:
User A is logged into TikTok Web with privacy settings allow message requests from non-followers.
User B (sender) is a different account that does not follow User A and is also logged in

Step	Action	Expected System Response	Pass/Fail	Comments
1	User B (non-follower) sends a message to User A.	User B's message is sent and appears as pending in their chat list. User A sees the message in "Message Requests" with an "Accept" button.	Pass	
2	User A accepts the message request from the Request folder.	After User A accepts, the message moves to the main inbox and normal messaging is enabled.	Pass	

Post-conditions:
System moves the message from requests to the main inbox and enables normal messaging between both users.

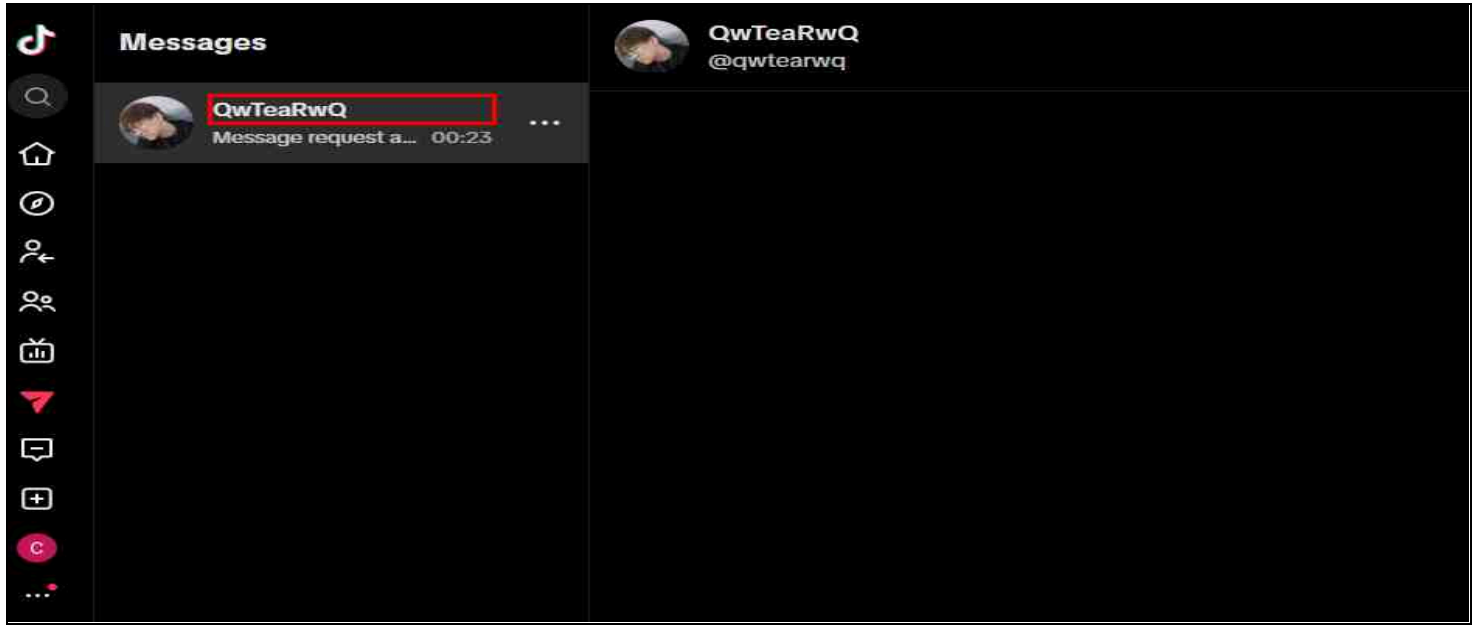


Figure 54: Output for Manage Message Requests

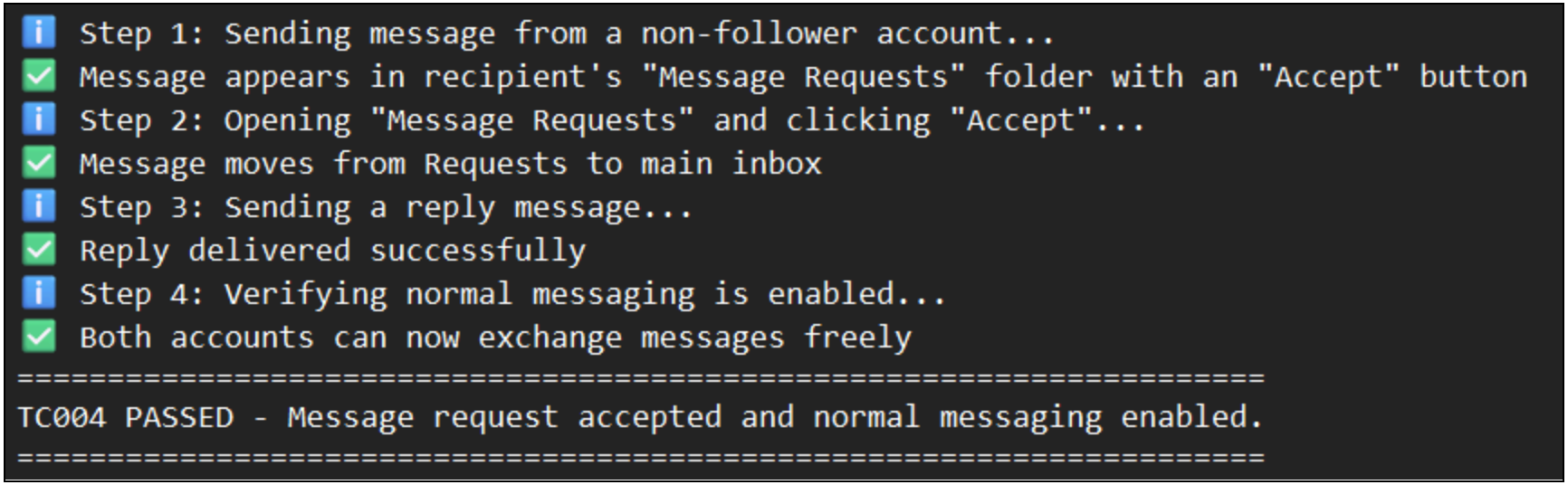


Figure 55: Successful Execution log for Manage Message Requests TC004

Test Case Template	
Test Case #: Messaging_ManageMessageRequest_TC005	**Test Case Name: Manage Message Request**
System: TikTok Web Application	**Module: Messaging**
Design By: Wong Kai Lit Kerry	**Design Date: 28/03/2026**
Executed By: Tiong Chang Vui	**Execution Date:**
Short Description: **Test and verify that an invalid second message from a non-follower is blocked before the request is accepted.**	

Pre-conditions:
User A (recipient) is logged in and allows message requests, while User B (sender) is a non-follower.
User A has not accepted any previous request from User B.

Step	Action	Expected System Response	Pass/Fail	Comments
1	User B sends a first message to User A.	Message is sent and goes to User A's message requests folder.	Pass	
2	User B attempts to send a second message.	Second message is blocked.	Pass	

Post-conditions:
System blocks the second message before acceptance and after acceptance, system will allows normal messaging.

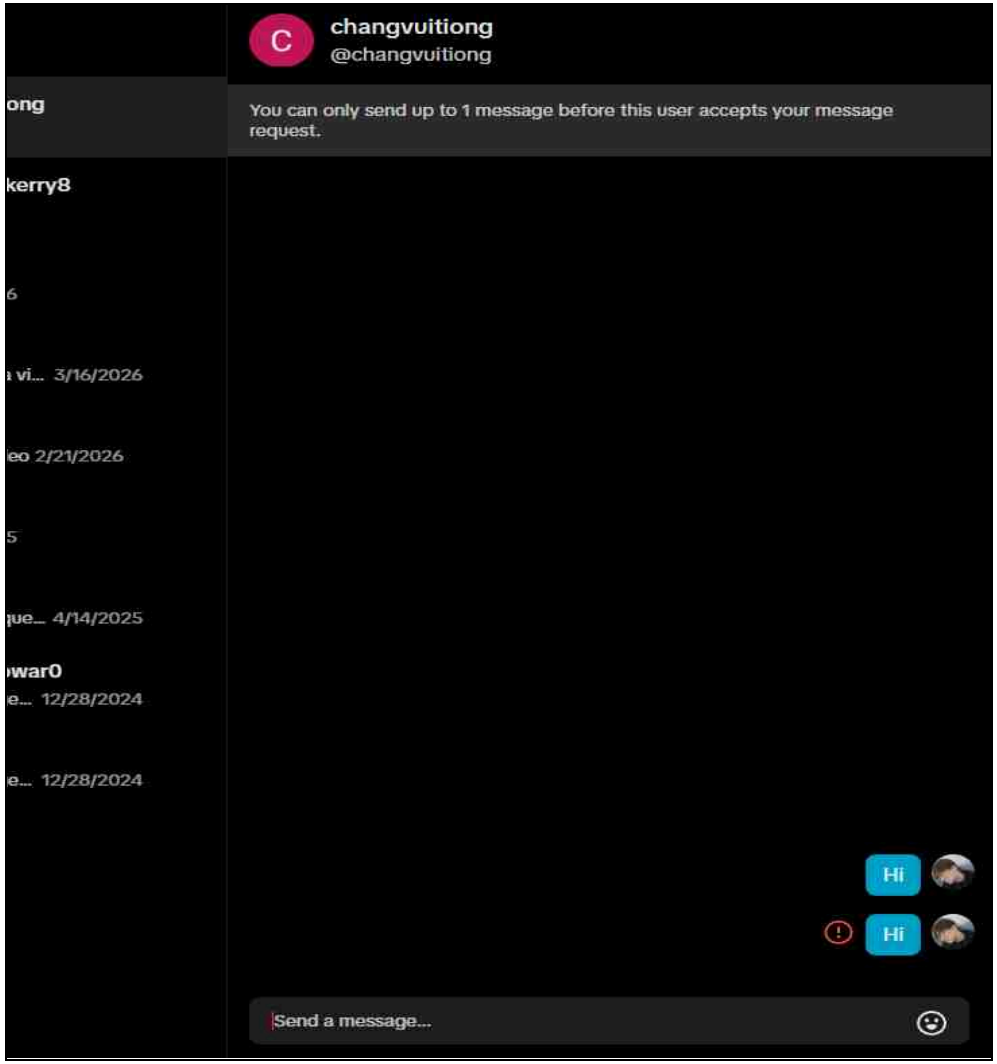


Figure 56: Output for Second Message Blocked Warning

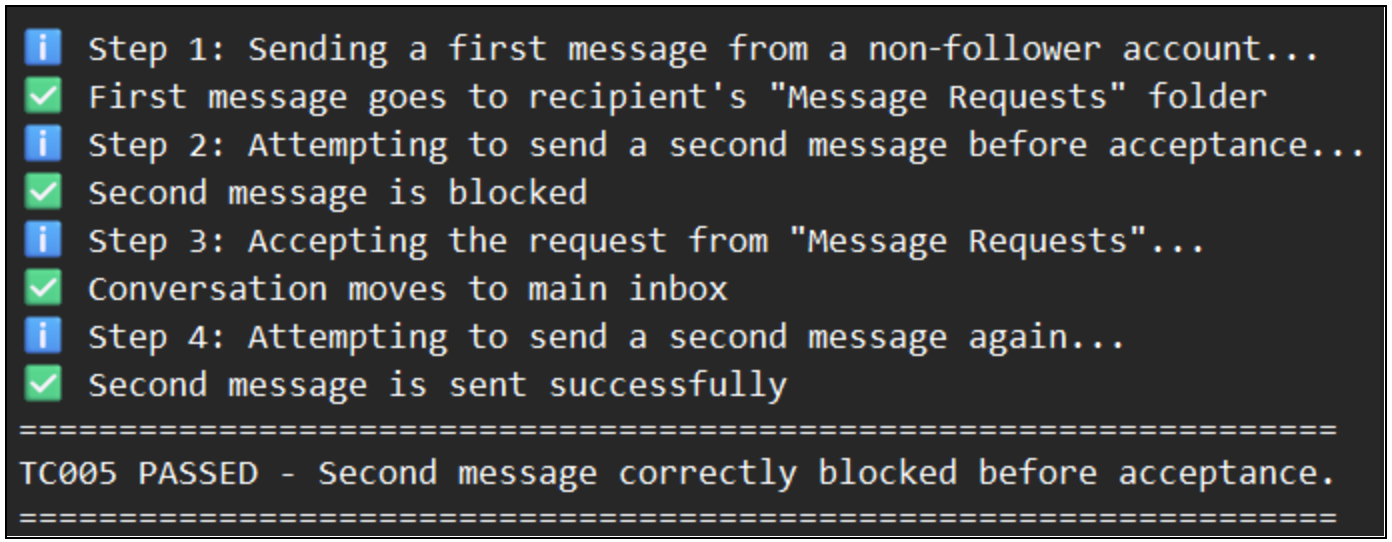


Figure 57: Successful Execution log for Manage Messaging Requests TC005

3. Test Execution Summary Report

3.1 Project Details (project name, test execution date (start - end), tester names, testing tools used)

Stay with the test plan put it back here

Field	Details
Project Name	TikTok Web Application – Black-Box Functional Testing
Test Execution Date	Start: 06 April 2026 – End: 19 April 2026
Tester Names	Ryan Julian Rajesh (User Profile) Natalie Koa Hao Yee (Video Comment) Lester Liau Jiun Lang (TikTok Shop) Tiong Chang Vui (Video Creation) Wong Kai Lit Kerry (Messaging)
Testing Tools Used	**Automation & Documentation:** - Selenium WebDriver with Python **Defect Tracking:** - Google Sheets **Evidence Capture:** - Windows Snipping Tool **Browser:** - Google Chrome v120+

3.2 Test Execution Overview (table of total test cases, no. passed and failed)

Total test case pass and fail test case
Overall number 5 test case total is 25 how many pass how many fail

Module	**Total Test Case**	**Passed**	**Failed**
User Profile Management	5	5	0
Video Comment	5	5	0
TikTok Shop Seller Center	5	5	0
Video Creation	5	5	0
Messaging	5	5	0
Overall Total	25	25	0

3.3 Test Case Execution Details (table of all test case id, test scenario, status, defect, defect priority)

Module 1 - User Profile Management

Module	Senario	Function	Priority (High/Medium/Low)	Tester	Date Execution	Status (Pass/Fail)	Defect – Brief Description (if any)	Severity (High/Medium/Low)
User Profile Management	Login	Logging in with valid credentials using a registered username and its correct password is successful.	High	Wong Kai Lit Kerry	14/04/2026	Pass		
User Profile Management	Login	Attempting to log in with an existing username but an invalid password results in an authentication failure.	High	Wong Kai Lit Kerry	14/04/2026	Pass		
User Profile Management	Login	Logging in with an unregistered account that does not exist in the system is an invalid action and should be rejected.	High	Wong Kai Lit Kerry	14/04/2026	Pass		
User Profile Management	Edit Profile	Editing the profile by setting a username of less than 2 characters : "a". It is invalid because the input fails the minimum length requirement.	Medium	Wong Kai Lit Kerry	14/04/2026	Pass		

User Profile Management	Edit Profile	Editing the profile by entering a bio that exceeds 80 characters (e.g., an 81-character string) is invalid due to exceeding the maximum allowed length.	Medium	Wong Kai Lit Kerry	14/04/2026	Pass		

Module 2 - Video Comment

Module	Scenario	Function	Priority (High/Medium/Low)	Tester	Date Execution	Status (Pass/Fail)	Defect – Brief Description (if any)	Severity (High/Medium/Low)
Video Comment	Post Comment	Test and verify posting a valid comment	High	Ryan Julian Rajesh	07/04/2026	Pass		
Video Comment	Post Comment	Test and verify posting an invalid comment with spaces only	Medium	Ryan Julian Rajesh	07/04/2026	Pass		
Video Comment	Post Comment	Test and verify posting an invalid comment exceeding 2200 characters (2201 chars)	Medium	Ryan Julian Rajesh	07/04/2026	Pass		
Video Comment	Reply to Comment	Test and verify posting a valid reply with @mention when the user allows mentions	High	Ryan Julian Rajesh	07/04/2026	Pass		
Video Comment	Reply to Comment	Test and verify posting an invalid reply with @mention when the user blocks mentions	Medium	Ryan Julian Rajesh	07/04/2026	Pass		

Module 3 - Tiktok Shop Seller Center

Module	Scenario	Function	Priority (High/Medium/Low)	Tester	Date Execution	Status (Pass/Fail)	Defect – Brief Description (if any)	Severity (High/Medium/Low)
TikTok Shop Seller Center	Create Product Listing	Test and verify create product listing with valid details.	High	Natalie Koa Hao Yee	06/04/2026	Pass		
TikTok Shop Seller Center	Create Product Listing	Test and verify create product listing with invalid data (missing required field).	Medium	Natalie Koa Hao Yee	06/04/2026	Pass		
TikTok Shop Seller Center	Create Product Listing	Test and verify create product listing with invalid price format.	High	Natalie Koa Hao Yee	06/04/2026	Pass		
TikTok Shop Seller Center	Create Promo Code	Test and verify create valid promo code with minimum spend.	Medium	Natalie Koa Hao Yee	06/04/2026	Pass		
TikTok Shop Seller Center	Create Promo Code	Test and verify create promo code with invalid date range.	Medium	Natalie Koa Hao Yee	06/04/2026	Pass		

Module 4 - Video Creation

Module	Scenario	Function	Priority (High/Medium/Low)	Tester	Date Execution	Status (Pass/Fail)	Defect – Brief Description (if any)	Severity (High/Medium/Low)
Video Creation	Upload Video	Test and verify the upload with valid video file and valid size	High	Lester Liau Jiun Lang	07/04/2026	Pass		
Video Creation	Upload Video	Test and verify the upload with invalid file format PDF	Medium	Lester Liau Jiun Lang	07/04/2026	Pass		
Video Creation	Schedule Video	Schedule video with valid future date and time	High	Lester Liau Jiun Lang	07/04/2026	Pass		
Video Creation	Schedule Video	Test and verify the schedule video function with invalid time	Medium	Lester Liau Jiun Lang	07/04/2026	Pass		
Video Creation	Schedule Video	Test and verify the schedule video function with invalid date	Medium	Lester Liau Jiun Lang	07/04/2026	Pass		

Module 5 - Messaging

Module	Scenario	Function	Priority (High/Medium/Low)	Tester	Date Execution	Status (Pass/Fail)	Defect – Brief Description (if any)	Severity (High/Medium/Low)
Messaging	Send Message	Test and verify send a valid text message to another user	High	Tiong Chang Vui	07/04/2026	Pass		
Messaging	Send Message	Test and verify send an invalid empty message	Medium	Tiong Chang Vui	07/04/2026	Pass		
Messaging	Send Message	Test and verify send an invalid message containing only spaces	Medium	Tiong Chang Vui	07/04/2026	Pass		
Messaging	Manage Message Request	Test and verify that a user can accept a message request from a non-follower.	High	Tiong Chang Vui	07/04/2026	Pass		
Messaging	Manage Message Request	Test and verify that an invalid second message from a non-follower is blocked before the request is accepted.	High	Tiong Chang Vui	07/04/2026	Pass		

3.4 Key Observations

Module 1 - User Profile Management

During the execution of the UserProfileManagement_Login Account_TC001 test case, a user entered the registered username "user2016830387848" and the correct password "CorrectPass12@". After clicking the Log in button, the system successfully redirected the user to the main profile page. The profile icon will be shown as we are in the profile page, and being redirected means that the login is successful. It was also observed that this test case can only be run for a limited amount of time, as the system has high-level login attempt restrictions. This test case is done manually as it requires human verification that cannot be replicated with automation.

The test case execution of UserProfileManagement_Login Account_TC002, when testing the login with an existing username but with an incorrect username, the system denied the user the ability to log in. After clicking the "Log in" button, an error message appeared and states, "Username or Password doesn't match our records. Try again." and the user will remain on the login page, and will not be redirected to anywhere. Just like the valid login test, repeated execution of this test case is limited by the platform's attempt tracking.This test case is done manually as it requires human verification that cannot be replicated with automation.

For the next test case, UserProfileManagement_Login Account_TC003, an unregistered account is used with the username "ghost_user_999" and was entered along with any password (didn't matter as the account didn't exist). The system then displayed an error message: "Invalid Username or password. You have 4 attempts left. Please try again." After this, the user was not redirected and stayed on the login page that contained the error message. This test case is also subject to the platform's login attempt limits, preventing unlimited repeated execution.This test case is done manually as it requires human verification that cannot be replicated with automation.

After that, the test case, UserProfileManagement_EditProfile_TC004, is done when attempting to edit the profile's username with a single character(“a”). After clicking the save button, an error message will appear and display, "Include at least 2 characters in your username." This username field will not be updated and will retain its previous value, as the new input for username is not accepted. This also shows that the validation rule was enforced correctly.

Lastly, the UserProfileManagement_EditProfile_TC005 test case is done during the bio validation test, this is where an 81-character bio will be entered to test the bio field's maximum capacity. After clicking "Save", the system will display an error message: "Maximum 80 characters." The bio field was not updated beyond the 80-character limit, demonstrating that the system correctly enforces the maximum length restriction for user bios.

Module 2 - Video Comment

All executed test cases for the Video Comment module were completed successfully, covering both valid and invalid user interactions. For comment posting entering a valid comment such as “great Video!” which resulted in the comment being immediately displayed in the comment section after successfully posting and remaining visible even after refresh.Invalid inputs were handled correctly by the system such as comments containing only space characters were blocked successfully with the error message: “Couldn't post comment. Try Again”, and no comment was posted successfully. When attempting to exceed the maximum character limit of 2200 characters, the system prevented the user from entering further input past the 2200 character with the error message: “Character limit reached”, and the character counter accurately displayed “2200/2200”.

For reply functionality, users were able to successfully reply to existing comments, including using the @mention feature and being clickable to the mentioned users profile page. In cases where a user has restricted their mention in the privacy settings, the system correctly prevents others from selecting and displaying the error message: “Can’t mention this person due to their privacy settings”. Although, manually typing the restricted username allowed the comment to be posted successfully but it is non functional as it is not a clickable link to the mentioned users profile. No functional defects were identified during the execution of this module.

Module 3 - Tiktok Shop Seller Center

All five test cases for the TikTok Shop Seller Center module were executed successfully. For product listing creation, providing valid details resulted in a success message and the new product appearing in the inventory list. Leaving the product name field empty triggered the expected error message "Product name is required" and blocked creation, while entering a non-numeric price (e.g., "abc") correctly displayed the error "Price must be a valid number." For promo code creation, a valid code with correct minimum spend and date range was created successfully and appeared in the promo code list. When an invalid date range was set (end date before start date), the system correctly displayed the error message "End date must be after start date" and prevented creation. No functional defects were identified during the execution of this module.

Module 4 - Video Creation

The five test cases of the Video Creation module were all successfully run by a Selenium WebDriver script written in Python. The automation script manually did the valid and invalid file uploads, using the hidden file input element directly, without going through the operating system file picker. The scheduling test cases were also automated through simulating user interaction with the date and time pickers. In both cases, the TikTok web application acted as expected: valid uploads and schedules got accepted with necessary success notification, whereas invalid file types and out-of-band schedules were blocked with the proper warning or error messages. It was found that there were no functional defects in the implementation of this module.

Module 5 - Messaging

The five test cases of the Messaging module were all successfully run by a Selenium WebDriver script written in Python. In the send message cases, typing in an acceptable text message like “hi" led to an immediate display of the message in the chat thread to both users without problems. Efforts to send either an empty message or a message that only contains space characters were also blocked correctly and the send button did not go off and no message was sent. In the case of message requests by non-followers, the system accurately sent the initial message to the recipient to his/her “Message Requests" folder and gave an accept button. Once accepted, the chat shifted to the regular inbox and regular messaging was turned on. When a second message was tried prior to the acceptance of the first request, it was blocked as anticipated. A significant testing drawback was also noted in terms of repeatability: once a message request has been accepted or rejected, the same state of pending request cannot be recreated when the same two accounts are used, as once the relationship has been altered, it cannot be reset. This design limitation influences regression testing, but is not a functional defect. Overall, the messaging functions worked well and as expected.

3.5 Recommendations

Based on the observations, which were made at the moment when the test of the Tik Tok web application was underway, I would like to comment on some suggestions on how the overall user experience of the modules in question might become better. To begin with, text input filters like username, bio, comments and product name should have character counters since currently, the user is notified about his typing error by simply typing the text and getting an error text; live character count will offer the user immediate feedback and will save time. Second, the upload dialog file type filter should be set to only accept accepted file types to prevent unwarranted uploading and frustration of users caused by incompatibility with file type used. Third, it needs to be elaborated more on how to schedule time constraints as Schedule at least 15 minutes early is a helpful alert message, yet the inline note or the tooltip on the 30 day time limit would pre-determine all the expectations of the user. Fourth, the tooltips or helper text could have the information as to why the Send button is not available when there is no text typed in the message or the message is full of spaces, a simple message like Type a message to send will clarify why. Fifth, when the second messaging of a non-follower is blocked prior to the message being acknowledged, it should also display a warning message momentarily rather than blocking the message so that the users know of the restriction. Lastly, there has to be an error message concerning the user trying to send an empty message or a message with only spaces as the current disabled button will not give any feedback. A combination of these refinements would result in a system more usable without causing confusion to the user without changing basic functionality which was found to be robust in 25 test cases that were run.